MAgent Theory 2.0

Gauge-Covariant Agents
and Inherited Organization

Robert C. Dennis

Independent Researcher

October 2026

Abstract

An agent carries a local probability law and mechanisms for receiving records, making predictions, and changing that law. We formulate these objects on associated statistical bundles, with independent directed interactions that may depend on the current belief and model laws. On regular statistical families, the joint-law Fisher metric and a connection induce a covariant pullback tensor on the contextual base. A collective least-squares construction selects a connection class from declared comparison data. Exact identities distinguish information lost under a channel from changes caused by incompatible connections, and characterize when a parent retention preserves the collective connection.

A declared interaction potential generates coupled Fisher dynamics. Conditional-information identities describe communication, common recovery, and the cost of discarding record history. Parent dynamics and boundary responses must descend through the retained configuration. Exact constructions separate state completion, public compression, and response closure. An atomless profile construction preserves coupled identity-channel dynamics and a declared response interface through arbitrary finite binary depth on a supplied family, while its discarded partition information has no Fisher tangent.

A conditional four-agent construction under the original smooth KL potential selects two parents and then their union. It uses non-Gaussian matrix laws with prepared common precision, moment-dependent channels, a fixed membership threshold, and fresh extensive responses. Every join discards continuous public model-law distinctions while preserving original-time dynamics, admitted adaptive response histories, and curved rank-two contextual geometry. The children remain distinct at every finite time. The quotient architecture is supplied; formation from less structured preparations and derivation of a preferred membership or interface rule remain open. A separate nonsmooth extension gives finite-time fusion in a lognormal family.

Reading the construction

The mathematical objects are introduced in the order in which they are needed: agent laws and records, interactions, statistical and contextual geometry, dynamics, predictive compression, and inherited organization. The main results include their decisive proofs. The appendices give the internal history construction, directional links and base connections, an optional determinant-metric and circle example, and notation. Secondary calculations are collected in a separate supplement.

Definitions specify the model's objects; constitutive choices specify how those objects interact. A theorem states a consequence of explicit hypotheses. An open problem identifies a further construction that is not supplied by those hypotheses. General probability measures are used whenever measurability suffices. Finite alphabets, smooth statistical families, and vector fibers enter only where an example or a differential calculation requires them. The general definitions assume no Gaussian closure; the fusion example states its lognormal restriction explicitly.

Contents

Abstract i

1 Agents, Possibilities, and Records 1
1.1 Statistical sections . . . 1
1.2 Presentation and probability . . . 2
1.3 Internal records . . . 3
1.4 What correlations describe . . . 4

2 Law-Dependent Interactions 5
2.1 Two sectors and their comparison spaces . . . 5
2.2 Sending, receiving, and exponential links . . . 6
2.3 An explicit constitutive family for general laws . . . 7
2.4 Dependence, direction, and inherited responses . . . 8

3 Geometry of Statistical Sections 9
3.1 Regular law families . . . 9
3.2 Covariant statistical change . . . 10
3.3 Null directions and quotients . . . 11
3.4 A collective choice of connection . . . 12
3.5 Information retained by a channel . . . 14
3.6 Geometry across a retention . . . 15

4 Coupled inference on agent laws 17
4.1 A potential for the two interacting sectors . . . 17
4.2 Source selection and reduction . . . 18
4.3 The derivative includes every interaction role . . . 19
4.4 Fisher dynamics with equal mobilities . . . 19
4.5 What a fixed record model changes . . . 20

5 Prediction and Selective Communication 22
5.1 Model presentations and their evaluators . . . 22
5.2 What the receiver reconstructs . . . 23
5.3 A predictive coding criterion . . . 25

6 Inherited Parents and Recursive Formation 26
6.1 Two objects that a parent must retain . . . 26
6.2 Exact deterministic inheritance . . . 26
6.3 Exact stochastic inheritance . . . 28
6.4 Boundary responses and composition . . . 29
6.5 A finite two-level construction . . . 30
6.6 Averaged laws under smooth peer dynamics . . . 31
6.7 Memory and formation . . . 34
6.8 Statistical partitions and their limits . . . 35
6.9 The formation obligation . . . 37

7 Finite Fusion with Exact Statistical Inheritance **38**
7.1 A law family closed under geometric averaging . . . 38
7.2 Messages and state-dependent scalar links . . . 39
7.3 The continuous subgradient dynamics . . . 40
7.4 An invariant class with explicit global solutions . . . 41
7.5 Two formation stages without a supplied partition . . . 42
7.6 The inherited handoff and its scope . . . 43

8 Sufficient States for Smooth KL Evolution **45**
8.1 The force that survives marginalization . . . 45
8.2 Conditional reconstruction within the original potential . . . 46
8.3 An exact law-valued completion . . . 48
8.4 Boundary records and feedback . . . 49
8.5 What is still required for formation . . . 51

9 State-Controlled Erasure and Persistent Groups **52**
9.1 A homogeneous population with internal gate coordinates . . . 52
9.2 Finite entry into an exactly erasing region . . . 53
9.3 A change of preferred groups without synchronization . . . 54
9.4 The exact inherited state . . . 56

10 Public compression and inherited responses **58**
10.1 An exact collective phase for general laws . . . 58
10.2 An ordinary response that the phase does not preserve . . . 60
10.3 What unrestricted Gibbs probes retain . . . 61
10.4 The remaining recursive requirement . . . 62

11 Boundary Closure and Recursive Composition **63**
11.1 A general-law family on a supplied tree . . . 63
11.2 The exact inherited profile equations . . . 64
11.3 Recursive retention and the response process . . . 66
11.4 Public information lost at every depth . . . 67
11.5 Why ordinary response completion can remove all compression . . . 68

12 Conditional Recursive Formation **70**
12.1 Matrix laws and the declared architecture . . . 70
12.2 The complete smooth Fisher flow . . . 71
12.3 Parent laws and exact joining . . . 73
12.4 Selected membership and persistent distinctions . . . 74
12.5 Exact joint response histories . . . 75
12.6 Inherited collective geometry . . . 76

13 Formation Beyond the Prepared Examples **78**

A A Normalized Internal Record Process **79**
A.1 Prediction followed by assimilation . . . 79

B From Directional Interaction to Base Transport **81**

C A Conditional Determinant Geometry **83**

D Notation **85**

Chapter 1

Agents, Possibilities, and Records

The purpose of the theory is to describe agents whose interactions can produce further agents. The first distinction is between an agent's probability law and the possibilities represented by that law. The second is between comparing two descriptions and specifying an interaction that changes them. A parent construction must preserve enough of both to continue the dynamics.

1.1 Statistical sections

Let $\mathcal{C}$ be a smooth base of contexts and let $P \to \mathcal{C}$ be a principal G-bundle. A context labels where a description applies; it is not identified with physical position or time. Let K and M be standard Borel spaces, with measurable G-actions by bimeasurable bijections. Their elements k and m represent state contents and model presentations. The actions may be different representations of the same group. They act on probability measures by pushforward.

Definition 1.1 (Statistical agent section). Set $\mathsf{Y} = \mathsf{K} \times \mathsf{M}$, with the product action, and give $\mathcal{P}(\mathsf{Y})$ the sigma-algebra generated by evaluation on measurable sets. The associated statistical bundle is

$$\mathcal{B} = P \times_G \mathcal{P}(\mathsf{Y}), \qquad (ug, \zeta) \sim (u, g_{\#}\zeta). \tag{1.1}$$

The statistical component of agent i is a section ζ_i over a domain $\mathcal{U}_i \subseteq \mathcal{C}$. Its full state also includes the internal variables and access mechanisms read by its update rule.

At a fixed context, disintegration gives

$$\zeta_i(dk, dm) = q_i(dk \mid m) s_i(dm), \qquad q_i(dk) = \int q_i(dk \mid m) s_i(dm). \tag{1.2}$$

Here q_i is the state marginal and s_i is the law over model presentations. Regular conditional laws exist on the stated spaces, with versions defined up to the appropriate marginal null sets [13]. A model presentation acquires predictive content through a normalized kernel; its name or coordinate value alone supplies no prediction.

The product restriction $\zeta_i = q_i \otimes s_i$ removes dependence between the two sectors within an agent. A product over agents makes a further restriction on represented contents. Neither restriction prevents the update of one law from depending on other laws. Coupling is a property of the dynamics; factorization is a property of a chosen probability description.

Smoothness is an additional hypothesis. When derivatives of laws are required, use a declared regular statistical family stable under the relevant coordinate changes. The whole space $\mathcal{P}(\mathsf{Y})$ is not assigned a smooth manifold structure by notation. Chapter 3 equips an admitted family with the Fisher metric of the full joint law and pulls that metric back to contextual directions through a

covariant derivative. The product restriction has a simpler metric, but marginal metrics alone do not measure changes of dependence in a general joint family. A collection of pointwise marginals also does not determine a probability law on fields over $\mathcal{C}$.

The population at a context consists of the agents whose domains contain that context. Most of the dynamics below is formulated at a common c_*. This retains the ambient bundle and local sections without requiring parallel transport between distinct base points. Independent interactions between agents at c_* are defined in chapter 2. Contextual statistical geometry is developed in chapter 3; appendix B gives an explicit directional-transport realization of its connection.

1.2 Presentation and probability

In local vector coordinates, write $U_i^q : V_q \to E_i^q$ and $U_i^s : V_s \to E_i^s$ for the sector frames. The spaces E_i^q, E_i^s are the represented state and model fibers. A common principal frame can induce both maps through different representations. Reference laws are

$$\bar{q}_i = ((U_i^q)^{-1})_{\#} q_i, \qquad \bar{s}_i = ((U_i^s)^{-1})_{\#} s_i. \tag{1.3}$$

A passive change sends q_i to $(g_i^q)_{\#} q_i$ and U_i^q to $g_i^q U_i^q$, with the corresponding model-sector rule. The reference laws are unchanged. This is a change of description, not an inference step.

For probability measures μ, ν on one measurable space, relative entropy is

$$D_{\mathrm{KL}}(\mu\|\nu) = \int \log \frac{d\mu}{d\nu} d\mu, \tag{1.4}$$

when $\mu \ll \nu$, and is $+\infty$ otherwise. In this formula and below, a displayed integral is understood with its declared measure; no common Lebesgue density is required. Relative entropy compares distinguishability of laws [15].

Proposition 1.2 (Coordinate invariance). For a bimeasurable bijection f, including divergences with value $+\infty$,

$$D_{\mathrm{KL}}(f_{\#}\mu\|f_{\#}\nu) = D_{\mathrm{KL}}(\mu\|\nu). \tag{1.5}$$

Proof. Absolute continuity is preserved in both directions. If $h = d\mu/d\nu$, the transformed density is $h \circ f^{-1}$. Integrating its logarithm against $f_{\#}\mu$ gives the original integral. If absolute continuity fails, a set of zero ν-measure and positive μ-measure is carried to a set with the same property. □

Definition 1.3 (Normalized channel). A Markov kernel $K : X \rightsquigarrow Y$ assigns a probability measure $K(x, \cdot)$ to each x, measurably in x. Its action on a law is

$$(\mu K)(A) = \int K(x, A)\mu(dx). \tag{1.6}$$

A deterministic measurable map $f : X \to Y$ is represented by $K_f(x, \cdot) = \delta_{f(x)}$. Then $\mu K_f = f_{\#}\mu$.

An invertible frame change preserves all distinctions. A selective or noisy channel need not. Both are well-defined operations on general probability laws, but their roles in an agent interaction differ.

1.3 Internal records

At a fixed context, let S_i be the admitted internal-state space of agent i, and let S be the complete configuration space. It contains the statistical laws, source rows, interfaces, and memory variables that the dynamics reads. Write $\mathrm{pr}_i : \mathsf{S} \to \mathsf{S}_i$ for the local state projection. For the measurable constructions, take these spaces to be standard Borel. A finite collection of probability-law spaces and finite or Euclidean memory spaces supplies such a realization.

Definition 1.4 (Record and readout). A record readout is a measurable map

$$\rho_i : \mathsf{S}_i \longrightarrow \mathsf{O}_i, \tag{1.7}$$

where O_i is the alphabet of internally distinguishable outcomes. If an interaction with source j produces configuration Z_{t+1}, its record variable is

$$O_{ij,t} = \rho_i(\mathrm{pr}_i\, Z_{t+1}). \tag{1.8}$$

For stochastic dynamics this is a measurable function from the probability space of realizations to O_i. A realized value is denoted $o_{ij,t}$. The source label is part of the record only when the internal state retains it.

A record can be transient. Persistence means that the distinction remains accessible through a specified later readout. A list kept outside the agent does not supply this accessibility. A memory coordinate is therefore an explicit part of the model unless the required distinction is already represented in its state.

Proposition 1.5 (Induced record law). Let $\Gamma_{ij} : \mathsf{S} \rightsquigarrow \mathsf{S}$ be a normalized interaction transition. Then

$$\mathcal{R}_{ij}(x, A) = \Gamma_{ij}\left(x, \{x' : \rho_i(\mathrm{pr}_i\, x') \in A\}\right), \tag{1.9}$$

is a normalized record kernel. If $\Gamma_{ij}(x, \cdot) = \delta_{F_{ij}(x)}$, the record is the deterministic function $\rho_i \circ \mathrm{pr}_i \circ F_{ij}$.

Proof. The right side is the pushforward of a probability measure through the measurable map $\rho_i \circ \mathrm{pr}_i$. Its total mass is one, and measurability in x follows from the defining property of a kernel. The deterministic case follows by evaluating a Dirac measure. □

This definition introduces no observing agent outside the population. It specifies where an outcome is available and how it can enter later dynamics. It does not derive the transition Γ_{ij} or decide whether its stochasticity represents objective chance. Those are further modeling choices.

For example, a vector comparison can be represented by

$$r_{ij} : E_i^q \times E_j^q \times \mathbb{R}^r \to \mathbb{R}^r, \qquad r_{ij}(k_i, k_j, \epsilon) = A_i k_i - B_{ij} k_j + \epsilon. \tag{1.10}$$

A noise law turns this map into a record kernel on represented contents. A joint content law determines its predicted outcomes. A transition on agent states must still specify how an outcome is registered and how the agent changes. A predictive likelihood and an internal update thus have different domains.

1.4 What correlations describe

Suppose two represented bits have the independent uniform law. Register the event that they agree. Their conditional joint law assigns probability $1/2$ to each of $(0,0)$ and $(1,1)$. Each marginal remains uniform, while the conditional expectation of their product changes from $1/4$ to $1/2$. An exact joint posterior has acquired dependence that its two marginals cannot report.

This example identifies a concrete need for a joint law: predicting an observable that reads both bits after the shared record. It does not imply that every admissible agent dynamics must propagate that posterior. A coupled product-family dynamics is another precise model, provided it is declared. Similarly, two routes that inspect the same source must share its identity; replacing it by independent redraws specifies a different experiment.

The configuration x of law-valued agents, a sample tuple $(k_i, m_i)_i$ used in a probabilistic model, and an ensemble of possible configurations are three different objects. A deterministic dynamics on x needs no probability law over configurations. A preparation ensemble can be introduced to assess predictions or compression, but it is then stated explicitly. These distinctions will matter when a parent retains a law over contents while discarding information needed to evolve that law.

Chapter 2

Law-Dependent Interactions

An interaction compares or transmits information between agents. Its law may depend on the agents' current statistical states. Gauge covariance specifies how this dependence is represented when coordinates change; it does not require the interaction to be a frame identification or select a unique mechanism. We first construct these interactions at one context. Transport between different contexts is a further construction, developed in appendix B.

2.1 Two sectors and their comparison spaces

Fix a context c and a finite set of agents defined there. Write

$$\nu_i^q = q_i, \qquad \nu_i^s = s_i, \tag{2.1}$$

for the belief and model marginals of agent i. The sector index a takes values in $\{q, s\}$. These laws live on standard Borel spaces E_i^a. No product assumption between the two sectors, or between different agents, is required to define an interaction between their marginals. Such a product is a separate choice when a joint recognition law is needed.

In a vector realization choose reference spaces V_q, V_s and invertible presentation maps

$$U_i^a : V_a \longrightarrow E_i^a, \qquad \bar{q}_i = (U_i^q)_\#^{-1} q_i, \qquad \bar{s}_i = (U_i^s)_\#^{-1} s_i. \tag{2.2}$$

We also write $\bar{\nu}_i^a = (U_i^a)_\#^{-1} \nu_i^a$. The two presentations can come from different representations of the same principal frame. This does not identify the belief and model interaction maps. Their domains, parameters, and dependence on the laws remain separately specified.

A passive re-expression $y_i' = g_i^a y_i$ sends ν_i^a to $(g_i^a)_\# \nu_i^a$ and U_i^a to $g_i^a U_i^a$. Therefore the reference laws in eq. (2.2) are unchanged. A common reference change $U_i^a \mapsto U_i^a h_a$ instead sends every reference law to $(h_a^{-1})_\# \bar{\nu}_i^a$. The first operation changes an agent's displayed coordinates; the second changes the common coordinates in which interaction parameters are written.

Definition 2.1 (Normalized interaction channel). A sector channel from source j to receiver i is a Markov kernel

$$\mathcal{K}_{ij}^a : E_j^a \rightsquigarrow E_i^a. \tag{2.3}$$

It sends a law to $(\nu_j^a \mathcal{K}_{ij}^a)(A) = \int \mathcal{K}_{ij}^a(y, A) \nu_j^a(dy)$. When the channel depends on the agent configuration, this formula is evaluated at that configuration. Covariance requires

$$\mathcal{K}_{ij}^{a\prime}(g_j^a y, A) = \mathcal{K}_{ij}^a(y, (g_i^a)^{-1} A). \tag{2.4}$$

Composition of sending, transmission, and receiving kernels produces another normalized channel by iterated integration. Invertibility is not required. Alternatively, two channels can send the receiver and source laws into a common message space, where their communicated predictions are compared.

Proposition 2.2 (Covariant law comparison). Under eq. (2.4), the divergence

$$D_{ij}^{a} = D_{\mathrm{KL}}(\nu_i^a \| \nu_j^a \mathcal{K}_{ij}^a), \tag{2.5}$$

is unchanged by passive bimeasurable coordinate changes, including when its value is infinite.

Proof. Changing the integration variable in the kernel gives $\nu_j^{a\prime}\mathcal{K}_{ij}^{a\prime} = (g_i^a)_{\#}(\nu_j^a \mathcal{K}_{ij}^a)$. For any bimeasurable bijection g, absolute continuity is preserved and the transformed Radon–Nikodym derivative is $(d\mu/d\eta) \circ g^{-1}$. Its logarithmic integral is therefore unchanged. Failure of absolute continuity is also preserved, giving infinity on both sides. □

This is the coordinate invariance of relative entropy [15]. It makes the comparison intrinsic without asserting that either law is a posterior under a fixed population model.

2.2 Sending, receiving, and exponential links

Choose a message space H_a and reference interfaces

$$T_j^a : V_a \longrightarrow H_a, \qquad O_i^a : H_a \longrightarrow V_a. \tag{2.6}$$

Their direct content map is $M_{ij}^a = U_i^a O_i^a T_j^a (U_j^a)^{-1}$. A deterministic channel can push the entire law through this map even when the image is lower dimensional. A density formula for a nonsingular law cannot then be used on the original ambient space without accounting for that change of support.

Definition 2.3 (Law-dependent reference generator). Let $\mathcal{L}_{ij}^a$ be a measurable function of the reference laws, and of any explicitly declared interface data, with values in $\mathrm{End}(V_a)$. Define

$$L_{ij}^a = \mathcal{L}_{ij}^a(\bar{\nu}_i^a, \bar{\nu}_j^a), \qquad \Theta_{ij}^a = U_i^a \exp(L_{ij}^a)(U_j^a)^{-1}. \tag{2.7}$$

The primitive self-comparison is $L_{ii}^a = 0$. For other edges a receiving–sending factorization is $L_{ij}^a = O_i^a(\bar{\nu}_i^a) T_j^a(\bar{\nu}_j^a)$, provided both maps refer to the same H_a and V_a.

The exponential is taken on the reference endomorphism, not on a map with different domain and codomain. These are $\mathrm{GL}(V_a)$ interaction maps. A smaller group acting on the frames can still re-express them covariantly. Interpreting them as parallel transports of a connection for that smaller group would additionally require generators in its represented Lie algebra.

Proposition 2.4 (Endpoint covariance and reference consistency). The links in eq. (2.7) obey

$$\Theta_{ij}^{a\prime} = g_i^a \Theta_{ij}^a (g_j^a)^{-1}. \tag{2.8}$$

They are also independent of a common reference rechoice if

$$\mathcal{L}^{a\prime}_{ij}((h_a^{-1})_{\#}\bar{\nu}^a_i, (h_a^{-1})_{\#}\bar{\nu}^a_j) = h_a^{-1} L^a_{ij} h_a, \tag{2.9}$$

where the prime includes the re-expression of any supplied interface data.

Proof. Passive site changes leave the reference laws and generator unchanged, so multiplication by the transformed endpoint frames proves eq. (2.8). Under a reference rechoice, the exponential series gives $e^{h_a^{-1} L h_a} = h_a^{-1} e^L h_a$. Substitution cancels both occurrences of h_a in eq. (2.7). □

For factored generators, $O^a_i \mapsto h_a^{-1} O^a_i$ and $T^a_j \mapsto T^a_j h_a$ suffice. Message-coordinate changes $T^a_j \mapsto r_a T^a_j$, $O^a_i \mapsto O^a_i r_a^{-1}$ cancel within the product.

Proposition 2.5 (Selective generator and invertible transport). For finite matrices $O : H \to V$ and $T : V \to H$, the map OT has rank at most $\dim H$, whereas e^{OT} is invertible. Moreover,

$$e^{OT} v = v \quad (v \in \ker T), \qquad \operatorname{im}(e^{OT} - I) \subseteq \operatorname{im} O. \tag{2.10}$$

Proof. The rank statement follows from factorization. The exponential series and $e^{OT} e^{-OT} = I$ prove invertibility. Every positive power of OT vanishes on $\ker T$, and every such power has image in $\operatorname{im} O$. Summing the convergent series proves the remaining assertions. □

Thus the exponential link retains an identity action in unread directions. It is a different operation from sending only the selective content OTv. The identity self-link also differs from a coarse internal loop: a parent may retain nonidentity loop information inherited from several child interactions.

2.3 An explicit constitutive family for general laws

The preceding definition does not select the functional $\mathcal{L}$. The following choice shows that non-constant, covariant generators can depend directly on laws without restricting them to a Gaussian family.

Definition 2.6 (Moment-dependent receiving and sending). For each sector let $V_a = \mathbb{R}^{d_a}$ and admit laws with finite second moments and positive-definite covariance. In reference coordinates write their means and covariances as μ^a_i and M^a_i. For a supplied scalar λ^a_{ij} on each nonself directed edge, put

$$L^a_{ij} = \lambda^a_{ij} \mu^a_i (\mu^a_j)^T (M^a_j)^{-1}, \qquad O^a_{ij}(t) = \lambda^a_{ij} \mu^a_i t, \qquad T^a_j(v) = (\mu^a_j)^T (M^a_j)^{-1} v. \tag{2.11}$$

Here $H_a = \mathbb{R}$, and $L^a_{ij} = O^a_{ij} T^a_j$ has rank at most one. Set $L^a_{ii} = 0$ separately. The edge coefficient is constitutive data, not a consequence of covariance.

Proposition 2.7 (Equivariance of the moment family). The family eq. (2.11) satisfies eq. (2.9) under every invertible linear reference change.

Proof. Under $U^a_i \mapsto U^a_i h_a$, the moments become $\mu^{a\prime}_i = h_a^{-1} \mu^a_i$ and $M^{a\prime}_j = h_a^{-1} M^a_j h_a^{-T}$. Consequently

$$(\mu^{a\prime}_j)^T (M^{a\prime}_j)^{-1} = (\mu^a_j)^T (M^a_j)^{-1} h_a. \tag{2.12}$$

Multiplication with the transformed receiving column gives $L_{ij}^{a\prime} = h_a^{-1} L_{ij}^a h_a$. □

No fixed Euclidean metric was inserted into this comparison: the source covariance supplies the inverse bilinear form and transforms with the law. Finite moments and invertible covariance are assumptions of this example alone. Centered laws give zero generators in this family, so it does not encode all distinctions between laws.

For a non-Gaussian example in two dimensions, take independent centered Laplace coordinates of variance one and translate their joint law to means e_1 and e_2 for two agents. Both covariances equal I. With identity frames and unit edge coefficients,

$$L_{12} = E_{12}, \qquad L_{21} = E_{21}, \qquad \Theta_{12}\Theta_{21} = (I + E_{12})(I + E_{21}) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{2.13}$$

The loop is nontrivial even though both interfaces have rank one. This is a communication loop at one context, not yet the curvature of a base connection.

2.4 Dependence, direction, and inherited responses

Covariance constrains which law-only generators are possible. If a reference transformation fixes both input laws and all supplied data, eq. (2.9) requires its matrix to commute with the resulting generator. Identical centered laws invariant under every orthogonal transformation therefore cannot select a distinguished off-diagonal direction through a fully orthogonally equivariant law-only rule. Directional interfaces can carry additional information, but that information must be declared.

Opposite attending edges are independent constitutive choices. In particular, Θ_{ji} need not equal Θ_{ij}^{-1}, as eq. (2.13) demonstrates. Geometric reversal of the *same* transport is instead its inverse. Endpoint covariance and conjugation of loop products are the familiar transformation laws of lattice link variables [24]; neither imposes an inverse relation between two independently modeled interactions.

When links depend on the current laws, differentiation must include that dependence. At fixed frames a variation of an objective has the form

$$d\mathcal{E} = d_\nu \mathcal{E} + \sum_{a,i,j} \langle d_{L_{ij}^a} \mathcal{E}, D\mathcal{L}_{ij}^a[d\bar{\nu}_i^a, d\bar{\nu}_j^a] \rangle. \tag{2.14}$$

This statement uses a differentiable statistical family and differentiable constitutive maps. It includes both the sender and receiver occurrences. Freezing the link during an update defines another declared dynamics; it is not generally the derivative of the substituted objective. Nor does substituting live recognition laws into a likelihood make that likelihood part of a fixed generative model during recognition variation.

An inherited parent's state must determine its future boundary response, including the law dependence of its generators. If two child configurations with identical retained data give different future responses, those data do not support autonomous closure.

Chapter 3

Geometry of Statistical Sections

A section's statistical change induces contextual geometry relative to a connection. We construct that tensor, a collective rule for choosing the connection, and conditions for their inheritance.

3.1 Regular law families

Definition 3.1 (Fisher geometry on an admitted family). Let $\mathcal{N}$ be a finite-dimensional smooth manifold of probability laws on a standard Borel space Y. In each parameter chart $\theta \mapsto P_\theta$, assume differentiability in quadratic mean with centered square-integrable score $\ell_\theta : \mathsf{Y} \to \mathbb{R}^d$. At a fixed θ, this means that the Lebesgue decomposition $P_{\theta+\delta} = p_\delta P_\theta + P_\delta^\perp$ satisfies

$$P_\delta^\perp(\mathsf{Y}) = o(|\delta|^2), \qquad \left\| \sqrt{p_\delta} - 1 - \tfrac{1}{2}\delta^\top \ell_\theta \right\|_{L^2(P_\theta)} = o(|\delta|). \tag{3.1}$$

For a tangent vector v, its scalar score ℓ_v is the corresponding directional derivative in this expansion. Require the Fisher tensor

$$g_P^F(v,w) = \mathbb{E}_P[\ell_v \ell_w] \tag{3.2}$$

to be smooth and positive definite on the admitted tangent space. Redundant parameter directions must be removed or treated as null directions before imposing this last requirement.

These family-specific hypotheses allow parameter-dependent supports satisfying eq. (3.1), without a common dominating measure [3].

Suppose the finite-dimensional Lie group G acts smoothly on $\mathcal{N}$ by the sample-space pushforwards in definition 1.1. A fixed bimeasurable bijection carries a score ℓ_v to $\ell_v \circ g^{-1}$. Change of variables in eq. (3.2) therefore makes the action a Fisher isometry. Its infinitesimal action is

$$J_P \xi = \left. \frac{d}{dt} \right|_{t=0} (\exp(t\xi))_\# P, \qquad \xi \in \mathfrak{g}. \tag{3.3}$$

The regular family must be stable under every group element used in these comparisons.

Proposition 3.2 (Fisher metric of the joint agent law). Let $\zeta_\theta(dk, dm) = q_\theta(dk \mid m) s_\theta(dm)$ be a regular family. Assume the marginal and conditional derivatives obey the product rule, that the joint score is $\ell_v^\zeta(k,m) = \ell_v^s(m) + \ell_v^{q|m}(k)$, and that these scores are square-integrable. Assume differentiation of conditional normalization is valid, so $\int \ell_v^{q|m}(k) q_\theta(dk \mid m) = 0$ for s_θ-almost every m. Then

$$g_\zeta^F(v,w) = g_s^F(v,w) + \int g_{q(\cdot|m)}^F(v,w) s(dm). \tag{3.4}$$

Proof. Expand $\mathbb{E}_\zeta[(\ell_v^s + \ell_v^{q|m})(\ell_w^s + \ell_w^{q|m})]$. Each mixed term vanishes after conditioning on m. The two remaining terms are exactly those displayed. □

For the actual product family $q_\theta \otimes s_\theta$, eq. (3.4) reduces to $g_q^F + g_s^F$. The two marginal metrics do not in general recover the joint metric. For example, the law

$$\zeta_\rho(k,m) = \frac{1+\rho km}{4}, \qquad k,m \in \{-1,1\}, \qquad |\rho| < 1, \tag{3.5}$$

has two fixed uniform marginals, whereas its joint score is $km/(1+\rho km)$. Summing its squared score over the four outcomes gives

$$g_{\zeta_\rho}^F(\partial_\rho, \partial_\rho) = \frac{1}{1-\rho^2}. \tag{3.6}$$

A declared marginal metric omits this dependence direction. We use the joint law unless a restriction is stated.

3.2 Covariant statistical change

Restrict the statistical bundle of eq. (1.1) to the admitted smooth fiber $\mathcal{N}$, writing $E = P \times_G \mathcal{N}$ and $\pi : E \to \mathcal{C}$. A principal connection ω induces a horizontal lift $H_e^\omega : T_c\mathcal{C} \to T_eE$ and vertical projection $\mathrm{ver}^\omega : T_eE \to V_eE$, where $V_eE = \ker T_e\pi$ and $c = \pi(e)$. The vertical Fisher metric is well defined because the transition action is isometric.

Definition 3.3 (Covariant pullback). For a smooth statistical section $\zeta : \mathcal{U} \to E$, define

$$D^\omega\zeta = \mathrm{ver}^\omega \circ T\zeta, \qquad h_\zeta^\omega(X,Y) = g_\zeta^F(D^\omega\zeta X, D^\omega\zeta Y). \tag{3.7}$$

The vertical projection removes the base component of $T\zeta$ forced by $\pi \circ \zeta = \mathrm{id}$. The resulting base tensor is positive semidefinite.

In a local frame write A for the connection form and $p(c)$ for the law coordinates. With the convention that the horizontal lift is $(X, -J_pA(X))$,

$$D^Ap(X) = dp(X) + J_pA(X). \tag{3.8}$$

Here $J_pA(X)$ is a statistical tangent vector.

Proposition 3.4 (Gauge covariance and connection dependence). Under a passive change $p' = g_\# p$ and $A' = \mathrm{Ad}_g A - (dg)g^{-1}$,

$$D^{A'}p' = T(g_\#)D^Ap, \qquad h_{p'}^{A'} = h_p^A. \tag{3.9}$$

For two connections on the same bundle, let $\alpha = \omega' - \omega$ denote their adjoint-bundle-valued difference. Along one fixed section put $b_X = J_\zeta \alpha(X)$ and $u_X = D^\omega \zeta X$. Then

$$D^{\omega'} \zeta X = u_X + b_X, \tag{3.10}$$

$$h_\zeta^{\omega'}(X,Y) = h_\zeta^\omega(X,Y) + g^F(u_X, b_Y) + g^F(b_X, u_Y) + g^F(b_X, b_Y). \tag{3.11}$$

Proof. Differentiation of $p' = g_\# p$ gives $dp' = T(g_\#)dp + J_{p'}((dg)g^{-1})$. Equivariance of the fundamental action gives $J_{p'} \operatorname{Ad}_g = T(g_\#) J_p$. Substitution in eq. (3.8) cancels the inhomogeneous terms. Fisher invariance proves eq. (3.9). Replacing A by $A + \alpha$ in that same derivative adds b; expanding the bilinear pairing gives eq. (3.11). □

Changing the connection changes the comparison. Appendix B constructs compatible connection data from directional links.

Proposition 3.5 (Transported relative entropy). Fix a smooth curve γ through c with $\dot\gamma(0) = X$, and transport its laws back to the fiber at c:

$$\mu_t = (T^\omega_{\gamma(t)\to c})_\# \zeta(\gamma(t)). \tag{3.12}$$

Assume, in addition to definition 3.1, that μ_t has positive densities f_t with respect to one measure near $t = 0$, that $\log f_t$ is twice differentiable, and that both derivatives can be interchanged with the integrals of f_t and $f_0 \log f_t$. Assume their second derivatives are continuous at zero. Then

$$D_{\mathrm{KL}}(\zeta(c) \| \mu_t) = \frac{t^2}{2} h_\zeta^\omega(X,X) + o(t^2). \tag{3.13}$$

Proof. The derivative of the transported law is $D^\omega \zeta X$ by the horizontal-lift convention. Write $\ell = \partial_t \log f_t|_0$ for its score. Differentiating normalization once gives $\mathbb{E}_{\mu_0} \ell = 0$. Differentiating twice gives $\mathbb{E}_{\mu_0}[\partial_t^2 \log f_t|_0 + \ell^2] = 0$. The first derivative of the displayed relative entropy is therefore zero and its second derivative is $\mathbb{E}_{\mu_0} \ell^2 = h_\zeta^\omega(X,X)$. Taylor's theorem proves the claim. □

Quadratic-mean differentiability alone does not justify this second-order KL expansion.

3.3 Null directions and quotients

Proposition 3.6 (Rank and quotient geometry). For the positive fiber Fisher metric in definition 3.1,

$$\operatorname{rad} h_\zeta^\omega = \ker D^\omega \zeta, \qquad \operatorname{rank} h_\zeta^\omega = \operatorname{rank} D^\omega \zeta. \tag{3.14}$$

If the latter rank is constant, $K = \ker D^\omega \zeta$ is a smooth subbundle and $T\mathcal{U}/K$ carries the positive-definite metric $\bar h([X],[Y]) = h_\zeta^\omega(X,Y)$.

Proof. The diagonal value of h_ζ^ω is the squared Fisher norm of $D^\omega \zeta X$, so it vanishes exactly on the derivative's kernel. For a positive semidefinite form, its diagonal null set is its radical. Rank-nullity gives the second identity, and the constant-rank theorem for bundle maps makes K a smooth subbundle. The pairing is independent of representatives because it annihilates K, and it is positive on every nonzero quotient class. □

A quotient vector bundle is not automatically the tangent bundle of a quotient manifold. Suppose a smooth Hausdorff manifold $\bar{\mathcal{U}}$ and a surjective submersion $q : \mathcal{U} \to \bar{\mathcal{U}}$ with connected fibers satisfy $\ker Tq = K$. The tensor descends to $\bar{\mathcal{U}}$ exactly when it is basic: besides annihilating K, it obeys $\mathcal{L}_Z h^\omega_\zeta = 0$ for every vertical field $Z \in \Gamma(K)$. Necessity follows by differentiating a pulled-back tensor along a fiber. For sufficiency, use submersion coordinates (y, z): the annihilation condition leaves only $dy^a dy^b$ components, and the Lie derivatives in the z directions make their coefficients independent of z. Connected fibers make these local definitions agree across each fiber. The resulting tensor is positive definite by eq. (3.14).

Even involutivity is an additional condition. On the trivial translation bundle over $\mathbb{R}^3$, use the unit-variance normal location family, the constant section of mean zero, and $A = dz - x dy$. Its location Fisher metric is one, so

$$D^A p = dz - x dy, \qquad h^A_p = (dz - x dy)^2. \tag{3.15}$$

This has constant rank one, but its null fields ∂_x and $\partial_y + x\partial_z$ have bracket ∂_z, which is not null. Hence its radical is not a foliation. This counterexample concerns the quotient inference; Gaussian laws are not required for the preceding results.

A contextual tangent quotient differs from a group-orbit quotient or measurable parent retention, which may discard information having no Fisher tangent.

3.4 A collective choice of connection

Several law sections can determine a common comparison rule by minimizing their shared frame motion. Choose smooth comparisons S_i placing the participating laws in compatible associated fibers, and write $p_i = (\mathsf{S}_i)_\# \zeta_i$. For bimeasurable comparisons, independent source changes g_i and a common target change g act by

$$\zeta_i' = (g_i)_\# \zeta_i, \qquad \mathsf{S}_i' = g \mathsf{S}_i g_i^{-1}, \qquad p_i' = g_\# p_i. \tag{3.16}$$

The compositions in this formula mean composition of represented sample maps. More general comparisons may be declared, but their target laws must still form an admitted smooth equivariant family.

Definition 3.7 (Collective Fisher data). Let $\boldsymbol{p} = (p_i)_i$ lie in a finite product of regular statistical manifolds with diagonal G-action. Choose positive constants w_i and define

$$\mathsf{G}_{\boldsymbol{p}}(v, w) = \sum_i w_i g^F_{p_i}(v_i, w_i), \tag{3.17}$$

$$J_{\boldsymbol{p}} \xi = (J_{p_i} \xi)_i, \qquad \mathfrak{h}_{\boldsymbol{p}} = \ker J_{\boldsymbol{p}}, \qquad H_{\boldsymbol{p}} = (\operatorname{im} J_{\boldsymbol{p}})^{\perp_{\mathsf{G}}}, \tag{3.18}$$

$$\mathfrak{k}_{\boldsymbol{p}}(\xi, \eta) = \mathsf{G}_{\boldsymbol{p}}(J_{\boldsymbol{p}} \xi, J_{\boldsymbol{p}} \eta). \tag{3.19}$$

The weights and comparisons are constitutive data. A different declared joint statistical family may replace the product if its metric and group action satisfy the same hypotheses.

Proposition 3.8 (Collective minimizing connection class). Suppose $J_{\boldsymbol{p}(c)}$ has constant rank on $\mathcal{U}$. For each contextual direction X, the minimizers of

$$a \longmapsto \|d\boldsymbol{p}(X) + J_{\boldsymbol{p}} a\|_{\mathsf{G}}^2 \tag{3.20}$$

form an affine class $A^F(X) + \mathfrak{h}_{\boldsymbol{p}}$, characterized by

$$\mathfrak{k}(A^F(X), \xi) = -\mathsf{G}(d\boldsymbol{p}(X), J\xi), \tag{3.21}$$

$$r_X := d\boldsymbol{p}(X) + JA^F(X) = P_H d\boldsymbol{p}(X). \tag{3.22}$$

Smooth local representatives can be chosen, linear in X. With $h^{\min}(X,Y) = \mathsf{G}(r_X, r_Y)$, any other local connection form A satisfies

$$h^A(X,Y) = h^{\min}(X,Y) + \mathfrak{k}((A - A^F)(X), (A - A^F)(Y)). \tag{3.23}$$

The minimizing class transforms by the affine connection law, and r transforms by the tangent action.

Proof. Project $-d\boldsymbol{p}(X)$ orthogonally onto $\operatorname{im} J$. Its preimages under J give all minimizing a, and two preimages differ exactly by $\ker J$. Orthogonality to every $J\xi$ gives the normal equation. At constant rank, choose a smooth local complement to $\ker J$; the positive form $\mathfrak{k}$ restricted to it has a smooth inverse, providing the stated representative. Writing $d\boldsymbol{p} + JA = r + J(A - A^F)$ and using orthogonality gives eq. (3.23). Under a passive target change, the derivative and connection terms cancel exactly as in eq. (3.9). Isometry of the action carries minimizers to minimizers and residuals to residuals. □

If infinitesimal isotropy is nonzero, the construction leaves stabilizer-valued components undetermined. An arbitrarily chosen pseudoinverse representative need not obey the affine law. If $\ker J = 0$, the unique local forms do obey that law and therefore glue on the given principal bundle; discrete stabilizers do not obstruct this conclusion. Separately, a free proper isometric action on a statistical manifold gives a smooth orbit quotient whose orthogonal horizontal distribution is the usual mechanical connection. These geometric antecedents are described by Guichardet [9], Lewis et al. [17]. A single transitive family has $H = 0$ and zero minimizing residual; a collective family need not.

Proposition 3.9 (Aggregate covariant derivative). Let $\mathcal{M}$ be a smooth equivariant map from the collective statistical family to a finite-dimensional representation W of G. For $M = \mathcal{M}(\boldsymbol{p})$ define

$$e(X) = d\mathcal{M}_{\boldsymbol{p}} r_X = dM(X) + \rho_*(A^F(X))M. \tag{3.24}$$

This W-valued one-form is independent of the representative of the minimizing class. If $\dim \mathcal{C} = \dim W$, it is an isomorphism exactly when the actual map $d\mathcal{M}_{\boldsymbol{p}} \circ r$ has full rank.

Proof. Differentiating equivariance gives $d\mathcal{M}_{\boldsymbol{p}} J\xi = \rho_*(\xi)M$. Applying this identity to eq. (3.22) proves eq. (3.24). A change in A^F belonging to $\ker J$ has zero image under $d\mathcal{M}$, proving representative independence. The rank statement is the criterion for a linear map between equal-dimensional spaces to be invertible. □

Surjectivity of $d\mathcal{M}|_H$ establishes that a full-rank first jet can be chosen at the law tuple: choose horizontal preimages of a basis of W and realize those as derivatives in a local statistical chart. It does not establish full rank for the supplied section or its preservation under inference. The determinant specialization in appendix C uses an additional form on W; the Fisher metric above remains positive.

3.5 Information retained by a channel

Theorem 3.10 (Score projection and exact Fisher loss). Let $\{P_\theta\}$ satisfy eq. (3.1) at θ_0, and let $K : \mathsf{Y} \rightsquigarrow \bar{\mathsf{Y}}$ be one normalized Markov kernel independent of θ. Under the joint law $P_{\theta_0}(dy)K(y, d\bar{y})$, the output family is differentiable in quadratic mean with score and Fisher matrix

$$\bar{\ell}(\bar{Y}) = \mathbb{E}[\ell(Y) \mid \bar{Y}], \tag{3.25}$$

$$I(P_{\theta_0}) - I(P_{\theta_0}K) = \mathbb{E}\,\mathrm{Cov}(\ell(Y) \mid \bar{Y}) \succeq 0. \tag{3.26}$$

Equality in direction v holds exactly when $v^\top \ell(Y)$ is measurable from $\bar{Y}$ up to a joint null set.

Proof. Lift the decomposition in eq. (3.1) to the joint space. Its absolutely continuous square root is $V_\delta = \sqrt{p_\delta(Y)}$ with the same L^2 remainder, and its singular mass is still $o(|\delta|^2)$. For its output, put $m_\delta = \mathbb{E}[V_\delta \mid \bar{Y}]$ and $v_\delta = \mathbb{E}[(V_\delta - m_\delta)^2 \mid \bar{Y}]$. The projected density of this absolutely continuous part is $q_\delta = m_\delta^2 + v_\delta$. Conditional expectation is an L^2 contraction, so $m_\delta = 1 + \frac{1}{2}\delta^\top \mathbb{E}[\ell \mid \bar{Y}] + o_{L^2}(|\delta|)$.

To replace m_δ by $\sqrt{q_\delta}$, observe that $m_\delta \geq 0$ and

$$(\sqrt{q_\delta} - m_\delta)^2 = v_\delta \frac{v_\delta}{(\sqrt{q_\delta} + m_\delta)^2} \leq v_\delta. \tag{3.27}$$

Set the fraction to zero where its numerator and denominator both vanish. It tends to zero in probability. Moreover $v_\delta/|\delta|^2$ is uniformly integrable as $\delta \to 0$: the square-root expansion expresses $(V_\delta - 1)/|\delta|$ as a bounded finite-dimensional combination of the fixed L^2 scores plus an L^2 remainder tending to zero, and conditional variance preserves the resulting uniform integrability. Thus the last display has expectation $o(|\delta|^2)$. The projected singular part may acquire an absolutely continuous density a_δ, but its total mass is $o(|\delta|^2)$. The inequality $(\sqrt{q_\delta + a_\delta} - \sqrt{q_\delta})^2 \leq a_\delta$ shows that this change does not affect the score; any remaining singular output mass has the same small bound. This proves output differentiability and its score formula. Total covariance gives the matrix identity. Its quadratic form in v is the expected conditional variance of $v^\top \ell$, which vanishes exactly under the stated measurability condition. □

This score-projection mechanism is the one in Pollard [18, Theorem 7]; the Markov formulation is also treated in Ay et al. [3]. It identifies local score sufficiency. It does not identify a kernel recovering every law in the experiment. For example, take independent Bernoulli variables with probabilities $1/2 + \theta/4$ and $1/2 + \theta^2/4$, and discard the second. At $\theta = 0$ its score vanishes, so no Fisher information is lost. A fixed recovery kernel applied to the first Bernoulli law makes every reconstructed event probability affine in θ, whereas the required second-coordinate probability is quadratic. Such a kernel cannot recover the family on a neighborhood.

Proposition 3.11 (Moving-channel correction). Let the channel instead have density $k_\theta(\bar{y} \mid y)$ with respect to a fixed output measure. Assume the joint family is differentiable in quadratic mean with square-integrable score $\ell_\theta(Y) + b_\theta(Y, \bar{Y})$, where $b_\theta = \partial_\theta \log k_\theta$, and assume differentiation of channel normalization gives $\mathbb{E}[b_\theta \mid Y] = 0$. Then

$$\bar{\ell}_\theta(\bar{Y}) = \mathbb{E}_\theta[\ell_\theta(Y) + b_\theta(Y, \bar{Y}) \mid \bar{Y}], \tag{3.28}$$

$$I(\bar{Y}) = I(Y) + \mathbb{E}_\theta I(K_\theta(Y, \cdot)) - \mathbb{E}_\theta \operatorname{Cov}(\ell_\theta + b_\theta \mid \bar{Y}). \tag{3.29}$$

Proof. Apply theorem 3.10 to the fixed output projection of the joint experiment. Its score is the stated sum. Conditional centering of b_θ eliminates the mixed terms in its Fisher matrix, leaving the input matrix plus the average conditional channel matrix. Subtracting the conditional covariance proves the formula. □

Consequently input-to-output Fisher change has no fixed sign for a moving channel. A constant input followed by a Bernoulli output with success probability $(1 + e^{-\theta})^{-1}$ has zero input Fisher information and output information $e^{-\theta}/(1 + e^{-\theta})^2 > 0$. A singular moving deterministic statistic need not have a channel log density at all. Its actual weak derivative, schematically $d(PK) = (dP)K + P(dK)$ when these derivatives exist, must replace that expression. These distinctions apply to the law-dependent channels used in the interaction potential.

3.6 Geometry across a retention

Let $\Psi : E \to \bar{E}$ be a smooth bundle map covering $f : \mathcal{C} \to \bar{\mathcal{C}}$, where both statistical bundles carry positive vertical Fisher metrics and connections $\omega, \bar{\omega}$. Assume the sections are related,

$$\Psi \circ \zeta = \bar{\zeta} \circ f. \tag{3.30}$$

This is a hypothesis: a map over f does not automatically make every fine section descend through f.

Definition 3.12 (Horizontal transport defect). For $X \in T_c\mathcal{C}$, define

$$a_\Psi(X) = T\Psi(H^\omega_{\zeta(c)} X) - H^{\bar{\omega}}_{\Psi(\zeta(c))}(Tf X). \tag{3.31}$$

Both terms project to TfX, so their difference lies in the same coarse vertical tangent space. Set $u_X = D^\omega \zeta X$, $\Psi_* = T^V \Psi$, and $\Delta_F = g^F - \Psi_*^* \bar{g}^F$ along the section.

Proposition 3.13 (Signed comparison of base tensors). Under eq. (3.30),

$$D^{\bar{\omega}} \bar{\zeta}(Tf X) = \Psi_* u_X + a_\Psi(X), \tag{3.32}$$

$$h^\omega_\zeta - f^* h^{\bar{\omega}}_{\bar{\zeta}} = u^* \Delta_F - \mathcal{X}_\Psi - \mathcal{Q}_\Psi, \tag{3.33}$$

where

$$\mathcal{X}_\Psi(X, Y) = \bar{g}^F(\Psi_* u_X, a_\Psi(Y)) + \bar{g}^F(a_\Psi(X), \Psi_* u_Y), \tag{3.34}$$

$$\mathcal{Q}_\Psi(X, Y) = \bar{g}^F(a_\Psi(X), a_\Psi(Y)). \tag{3.35}$$

If each fiber map is realized by a fixed parameter-independent Markov kernel, then $\Delta_F \succeq 0$. In that case base contraction holds exactly when, for every X,

$$2\bar{g}^F(\Psi_* u_X, a_\Psi(X)) + \|a_\Psi(X)\|^2_{\bar{g}^F} \leq \Delta_F(u_X, u_X). \tag{3.36}$$

Proof. Differentiate eq. (3.30), decompose $T\zeta X = H^\omega X + u_X$, and take coarse vertical parts. This proves the first identity. Expand the coarse Fisher pairing of $\Psi_* u + a_\Psi$ and subtract it from the fine pairing to obtain the second. For a fixed Markov fiber map, theorem 3.10 gives $\Delta_F \succeq 0$. Evaluating the signed identity on the diagonal proves the exact contraction criterion. □

Zero horizontal defect is sufficient, but is not necessary, for eq. (3.36). Information discarded in the fiber and mismatch of horizontal transports enter different terms. For two successive smooth statistical maps R_{01}, R_{12}, the vertical defects also compose exactly:

$$\Delta_{02} = \Delta_{01} + R^*_{01}\Delta_{12}, \qquad \Delta_{ab} = g_a - R^*_{ab} g_b. \tag{3.37}$$

This follows by inserting and subtracting $R^*_{01} g_1$ and using contravariance of pullback. Both terms are nonnegative when the two maps are fixed Markov morphisms.

Proposition 3.14 (Retention of the collective connection). Let $R : \mathcal{N} \to \bar{\mathcal{N}}$ be a smooth G-equivariant map between positive-metric collective statistical families, independent of the contextual coordinate. Denote their metrics by $g, \bar{g}$, their infinitesimal orbit maps by $J, \bar{J}$, and put $\Delta = g - R^* \bar{g}$. Let $\boldsymbol{p}(c)$ be a smooth fine section, and let A^F and $r_X = d\boldsymbol{p}(X) + JA^F(X)$ be a fine minimizing representative and residual. The same representative minimizes the cost at $\bar{\boldsymbol{p}} = R(\boldsymbol{p})$ exactly when

$$\Delta(r_X, J\xi) = 0 \quad \text{for all } X, \xi. \tag{3.38}$$

Under this condition, equality of the entire fine and coarse affine minimizing classes holds exactly when $\ker J = \ker \bar{J}$. For every possible fine first jet, the condition is $\Delta(H, \operatorname{im} J) = 0$. If $\Delta \succeq 0$, vanishing loss on every orbit direction is sufficient. This includes a fixed Markov morphism equipped with the Fisher metrics of its input and output experiments in the same normalization; arbitrary independently reweighted metrics need not satisfy that premise.

Proof. Equivariance gives $dRJ = \bar{J}$. Thus the coarse residual using A^F is dRr_X. Its normal equation is $\bar{g}(dRr_X, \bar{J}\xi) = 0$. Fine orthogonality gives $g(r_X, J\xi) = 0$, so the coarse equation is equivalent to eq. (3.38). Once the representative minimizes both costs, their classes are $A^F + \ker J$ and $A^F + \ker \bar{J}$, proving the equality criterion. Every horizontal vector is realizable as a residual by choosing the first jet equal to that vector, which gives the all-jets statement. When $\Delta \succeq 0$, Cauchy–Schwarz for this semidefinite form makes $\Delta(v, J\xi) = 0$ whenever $\Delta(J\xi, J\xi) = 0$. □

Contraction alone is insufficient. Two independent unit-variance normal location laws with $J\xi = (\xi, \xi)$ and means $(x, -x)$ have Fisher matrix I and minimizing coefficient zero. Adding independent unit normal noise only to the first sample is a fixed equivariant channel. Its coarse Fisher matrix is $\operatorname{diag}(1/2, 1)$, whose cost $\frac{1}{2}(1+a)^2 + (-1+a)^2$ is minimized at $a = 1/3$. The minimizing connection has changed despite Fisher contraction. Its preservation is an additional requirement beyond the dynamical and response conditions in chapter 6.

Chapter 4

Coupled inference on agent laws

An interaction between probability laws can be specified directly, without first interpreting it as an observation likelihood. This chapter chooses such an interaction for both belief and model laws, derives its source rows, and supplies a covariant dissipative dynamics. A normalized model of records defines a second construction. Their relation depends on what is held fixed and on whether the interaction compares laws or sampled contents.

4.1 A potential for the two interacting sectors

Fix a context and a finite set of agents. Agent i has state and model spaces $\mathsf{K}_i, \mathsf{M}_i$, both standard Borel. In this chapter its joint law is restricted explicitly to $\zeta_i = q_i \otimes s_i$. Choose a proper model reference r_i and a normalized evaluator $G_i : \mathsf{M}_i \rightsquigarrow \mathsf{K}_i$. The reference joint is

$$R_i(dk, dm) = r_i(dm)G_i(m, dk), \qquad \mathcal{E}_0 = \sum_i D_{\mathrm{KL}}(q_i \otimes s_i \| R_i). \tag{4.1}$$

The evaluator couples the sectors even though their recognition law is a product. Where the terms are finite, disintegration gives

$$D_{\mathrm{KL}}(q_i \otimes s_i \| R_i) = D_{\mathrm{KL}}(s_i \| r_i) + \int D_{\mathrm{KL}}(q_i \| G_i(m, \cdot))s_i(dm). \tag{4.2}$$

This is a conditional relative-entropy identity: the model marginal contributes the first term and the state conditional contributes the second. In particular, the expectation of a log evaluator in this objective is different from the log of the predictive mixture $s_i G_i$.

Let $\mathbf{q} = (q_i)_i$ and $\mathbf{s} = (s_i)_i$. For each allowed directed edge $j \to i$, choose normalized channels $\mathcal{K}^q_{ij}[\mathbf{q}] : \mathsf{K}_j \rightsquigarrow \mathsf{K}_i$ and $\mathcal{K}^s_{ij}[\mathbf{s}] : \mathsf{M}_j \rightsquigarrow \mathsf{M}_i$. Their dependence on the laws is part of the specified interaction. Define

$$\begin{aligned} \nu^q_{ij} &= q_j \mathcal{K}^q_{ij}[\mathbf{q}], \quad D^q_{ij} = D_{\mathrm{KL}}(q_i \| \nu^q_{ij}), \\ \nu^s_{ij} &= s_j \mathcal{K}^s_{ij}[\mathbf{s}], \quad D^s_{ij} = D_{\mathrm{KL}}(s_i \| \nu^s_{ij}). \end{aligned} \tag{4.3}$$

The source sets $\mathcal{N}^q_i, \mathcal{N}^s_i$ are finite and nonempty, with positive normalized reference rows π^q_i, π^s_i. Belief routing uses β_i and model routing uses γ_i on their respective sets.

Definition 4.1 (Coupled law potential). For fixed $T_q, T_s > 0$, set

$$\begin{aligned}\mathcal{E} = \mathcal{E}_0 &+ \sum_i \left[\sum_{j \in \mathcal{N}_i^q} \beta_{ij} D_{ij}^q + T_q D_{\mathrm{KL}}(\beta_i \| \pi_i^q) \right] \\ &+ \sum_i \left[\sum_{j \in \mathcal{N}_i^s} \gamma_{ij} D_{ij}^s + T_s D_{\mathrm{KL}}(\gamma_i \| \pi_i^s) \right].\end{aligned} \tag{4.4}$$

The reference laws, evaluators, source priors, and temperatures are held fixed in the variations below. The admissible domain consists of laws for which all displayed divergences are finite.

This potential is a constitutive choice at the level of agent laws. No distribution over these law configurations is required to define it. General probability laws enter eq. (4.4); differentiability will require stronger hypotheses. An identity self-channel has $D_{ii}^q = 0$ throughout its domain, with the same convention in the model sector. Independent opposite attending edges remain independent data.

4.2 Source selection and reduction

Proposition 4.2 (Gibbs rows and the exact reduced potential). Hold the agent laws fixed and suppose every source cost is finite. Define

$$\begin{aligned} Z_i^q &= \sum_j \pi_{ij}^q e^{-D_{ij}^q/T_q}, \quad \beta_{ij}^* = \frac{\pi_{ij}^q e^{-D_{ij}^q/T_q}}{Z_i^q}, \\ Z_i^s &= \sum_j \pi_{ij}^s e^{-D_{ij}^s/T_s}, \quad \gamma_{ij}^* = \frac{\pi_{ij}^s e^{-D_{ij}^s/T_s}}{Z_i^s}. \end{aligned} \tag{4.5}$$

These are the unique minimizing rows. With

$$\mathcal{E}_{\mathrm{red}} = \mathcal{E}_0 - T_q \sum_i \log Z_i^q - T_s \sum_i \log Z_i^s, \tag{4.6}$$

the exact identity is

$$\mathcal{E} = \mathcal{E}_{\mathrm{red}} + T_q \sum_i D_{\mathrm{KL}}(\beta_i \| \beta_i^*) + T_s \sum_i D_{\mathrm{KL}}(\gamma_i \| \gamma_i^*). \tag{4.7}$$

Proof. For one row, $\log \beta_j^* = \log \pi_j - D_j/T - \log Z$. Substitute this expression into $T D_{\mathrm{KL}}(\beta \| \beta^*)$ to obtain the row objective plus $T \log Z$. Summing proves eq. (4.7). Relative entropy is nonnegative and vanishes exactly when its two probability laws agree, which proves uniqueness, including competitors on the simplex boundary. □

On a differentiable family, direct differentiation of the finite sums in eq. (4.5) gives

$$d\mathcal{E}_{\mathrm{red}} = d\mathcal{E}_0 + \sum_{ij} \beta_{ij}^* dD_{ij}^q + \sum_{ij} \gamma_{ij}^* dD_{ij}^s. \tag{4.8}$$

Indeed, $-T_q d\log Z_i^q = \sum_j \beta_{ij}^* dD_{ij}^q$, and the other sector is identical. There is no derivative of an optimized row to add. Every dependence of its cost on the laws still belongs in dD. If a source prior were varied, this formula would also contain $-T_q \sum_{ij} \beta_{ij}^* d\log \pi_{ij}^q$ and its model counterpart.

4.3 The derivative includes every interaction role

To display the variation, choose densities against fixed references on a regular dominated family. Write a for the density of a receiver law and b for that of its compared law. For mass-preserving variations and justified differentiation under the integrals,

$$dD_{\mathrm{KL}}(a\|b) = \int \log(a/b)da - \int (a/b)db. \tag{4.9}$$

Here the notation $\int f da$ denotes integration against the signed variation of the law, rather than differentiation of the sample coordinate. The formula follows by differentiating $a\log(a/b)$; the additional integral of da is zero.

For a belief interaction the compared-law variation is

$$d\nu_{ij}^q = (dq_j)\mathcal{K}_{ij}^q[\mathbf{q}] + q_j D\mathcal{K}_{ij}^q[\mathbf{q}](d\mathbf{q}). \tag{4.10}$$

The second term differentiates the prescribed channel itself. Thus an agent contributes when it is a receiver, when it is a source, and whenever its law changes a channel elsewhere. For fixed discrete kernels, the incoming contribution from $j \to i$ to a variation at source value v is

$$-\beta_{ij} \sum_u \frac{q_i(u)}{\nu_{ij}^q(u)} \mathcal{K}_{ij}^q(v,u). \tag{4.11}$$

The model sector has the same expression with q, β replaced by s, γ. If the channels are constructed from law-dependent moments or generators, eq. (4.10) also differentiates those moments and generators. Keeping their numerical values fixed computes a different vector field.

An identity self-channel illustrates why the variable identification matters. Here $a = b$ and $da = db$, so eq. (4.9) is zero throughout the state space. Freezing the source as an old snapshot while moving the receiver replaces this zero functional by a divergence from that snapshot. Its first derivative still vanishes at equality, but its curvature and its force after displacement differ. More generally, directed source selection does not make the gradient of eq. (4.4) one-way: a source appears in the energy of every row that attends to it.

4.4 Fisher dynamics with equal mobilities

The kinetic metric in this chapter uses its declared product recognition family. It is the product specialization of proposition 3.2, with separate Fisher blocks for the source rows. The contextual tensor in definition 3.3 instead evaluates changes along the base through a connection; it does not replace the vertical metric used for inference at a fixed context.

Theorem 4.3 (Local flow and exact dissipation on positive tables). Suppose the state and model alphabets are finite, all r_i and evaluator entries $G_i(m,k)$ are strictly positive, and the channel entries are twice continuously differentiable functions of the interior probability tables. Assume every compared-law entry is positive there. Let u^a range over all tables $q_i, s_i, \beta_i, \gamma_i$. Equip their product with the Fisher metric

$$g_u(v,w) = \sum_a \sum_z \frac{v_z^a w_z^a}{u_z^a}, \qquad \sum_z v_z^a = \sum_z w_z^a = 0. \tag{4.12}$$

For every interior initial condition there is a unique maximal interior solution of

$$\dot{u}_z^a = -u_z^a \left(h_z^a - \sum_w u_w^a h_w^a \right), \qquad h_z^a = \frac{\partial \mathcal{E}}{\partial u_z^a}. \tag{4.13}$$

Along this solution,

$$\frac{d\mathcal{E}}{dt} = -\sum_a \mathrm{Var}_{u^a}(h^a) \le 0. \tag{4.14}$$

Proof. The stated positivity and regularity make $\mathcal{E}$ twice continuously differentiable in interior simplex coordinates. The right side of eq. (4.13) is therefore locally Lipschitz, giving a unique local solution and its maximal continuation in the interior. Its entries sum to zero within each table. Centering h^a removes any ambiguity from extending $\mathcal{E}$ off the normalized tables. For every tangent vector v, the metric pairing of $u^a(h^a - \mathbb{E}_{u^a} h^a)$ with v^a equals $\sum_z h_z^a v_z^a$. This identifies the Fisher gradient. Finally, substitute eq. (4.13) into $\dot{\mathcal{E}} = \sum_{a,z} h_z^a \dot{u}_z^a$ to obtain the sum of negative variances. Interior continuation does not assert that the trajectory stays away from the boundary for all time. □

All blocks have unit mobility. A persistent slow model sector would have to follow from the interactions and their state, rather than from an inserted small multiplier in its equation. One may instead optimize the source rows at each state and apply the same construction to $\mathcal{E}_{\mathrm{red}}$ and the matter tables alone; eq. (4.8) gives its complete derivative.

For a regular finite-dimensional family with positive-definite Fisher tensor $g_{ab}(\theta)$, the corresponding equation is $\dot{\theta}^a = -(g^{-1})^{ab}\partial_b \mathcal{E}$ [1]. Whenever the derivatives exist, contraction gives $\dot{\mathcal{E}} = -(\partial\mathcal{E})^T g^{-1}(\partial\mathcal{E})$. Bimeasurable re-expression of samples, carrying all kernels and reference laws with it, preserves relative entropy and Fisher score inner products. On a family preserved by that re-expression, the gradient therefore transforms as a tangent vector. These statements concern fixed coordinate changes; coordinates that themselves evolve require their coordinate-velocity term.

4.5 What a fixed record model changes

Choose a proper prior $P_0(dy, dJ)$ on represented contents and source labels, and a normalized record kernel $\mathscr{R}(do \mid y, J)$. Then $\mathbb{P} = P_0\mathscr{R}$ is normalized by integrating the record first. At a declared record density $\ell_o(y,J)$ with $0 < Z_o = \int \ell_o dP_0 < \infty$, its posterior is $\Pi_o = \ell_o P_0 / Z_o$. Substitution proves

$$\mathcal{F}(Q;o) = D_{\mathrm{KL}}(Q\|P_0) - \mathbb{E}_Q \log \ell_o = D_{\mathrm{KL}}(Q\|\Pi_o) - \log Z_o, \tag{4.15}$$

when the split terms are defined. The right side supplies the extended definition otherwise. This is the variational free energy of a fixed model. A product family for Q is permitted and its block updates remain coupled through the likelihood factors. Each block derivative contains every factor incident on it.

With independent source priors and one likelihood factor reading each label, a factorized source row has cost $C_{ij} = \mathbb{E}_{Q_Y}[-\log \ell_i(o_i \mid Y, j)]$ and optimizer $\beta_{ij} \propto \pi_{ij} e^{-C_{ij}}$. This expected log likelihood differs from both the peer divergence and the likelihood integrated over a prior. For an isolated source record, exact source inference gives $\mathbb{P}(J = j \mid o) \propto \pi_j \mathbb{E}_{P_Y} \ell_j(o \mid Y)$. Equality of these different rows needs additional hypotheses.

Proposition 4.4 (Entropy and the likelihood Jensen gap). On standard Borel spaces, let $k(w \mid v)$ be a density of a normalized kernel into a receiver space with sigma-finite reference λ. Let the receiver law be $q = a\lambda$ and the source law be p. Put $b(w) = \int k(w \mid v)p(dv)$. Assume $0 < b(w) < \infty$ for q-almost every w, $k > 0$ almost everywhere under $q \otimes p$, and finiteness of $\mathbb{E}_q|\log a|$, $\mathbb{E}_q|\log b|$, and $\mathbb{E}_{q\otimes p}|\log k|$. Define $p^w(dv) = k(w \mid v)p(dv)/b(w)$ on this full-measure set. Then

$$\mathbb{E}_{q\otimes p}[-\log k(W \mid V)] = H_\lambda(q) + D_{\mathrm{KL}}(q\|pK) + \int D_{\mathrm{KL}}(p\|p^w)q(dw). \tag{4.16}$$

Proof. The density ratio gives $D_{\mathrm{KL}}(p\|p^w) = \log b(w) - \int \log k(w \mid v)p(dv)$. Integrating against q and using $-\int \log b dq = H_\lambda(q) + D_{\mathrm{KL}}(q\|pK)$ proves the identity. The last term vanishes exactly when $p^w = p$ for q-almost every w. □

The entropy here is relative to the declared reference. A record likelihood normalized in o need not be normalized in the receiver coordinate w, so proposition 4.4 requires that additional kernel interpretation. Even when it applies, the entropy and conditional divergence can affect the dynamics.

There is also an exact obstruction to universal equality. Freeze a channel K and receiver q. Suppose p_1K, p_2K have positive densities that differ on a set of positive q-measure, with both divergences finite. Strict convexity of $-\log$ gives

$$D_{\mathrm{KL}}(q\|(tp_1 + (1-t)p_2)K) < tD_{\mathrm{KL}}(q\|p_1K) + (1-t)D_{\mathrm{KL}}(q\|p_2K), \qquad 0 < t < 1. \tag{4.17}$$

This follows pointwise on the differing-density set and then by integration. By contrast, the expectation $\mathbb{E}_{q\otimes p}e(W, V)$ of any fixed integrable sample energy is affine in p. It cannot equal the divergence at both endpoints and their strict mixture. The obstruction concerns a fixed sample energy; it does not exclude a direct law potential, a frozen-template episode, or an explicitly different probability model whose variables are law points.

Chapter 5

Prediction and Selective Communication

A model law is useful through the records it predicts. Communication can preserve those predictions while discarding other distinctions, and a receiver can combine a message with information already present in its model. This chapter gives exact identities for both operations. The retained prediction is always named; agreement about one output is not automatically agreement about every future interaction.

5.1 Model presentations and their evaluators

Let $G_i : \mathsf{M}_i \rightsquigarrow \mathsf{K}_i$ be a normalized local model and let $H_i : \mathsf{K}_i \rightsquigarrow \mathsf{O}_i$ be a normalized record kernel. Their composition

$$K_i^{\mathrm{rec}}(m, A) = \int H_i(k, A) G_i(m, dk), \qquad p_i^{\mathrm{rec}} = s_i K_i^{\mathrm{rec}}, \tag{5.1}$$

gives the model's predictive record law. An evaluator may also take a preparation, accessible history, or chosen interaction as input; these are held fixed when two predictions are compared. More general normalized evaluators need not factor through G_i and H_i.

Suppose a model link $\Theta_{ij}^s : \mathsf{M}_j \to \mathsf{M}_i$ is a bimeasurable bijection. Exact compatibility of the two evaluators, after identifying their record alphabets, means

$$K_i^{\mathrm{rec}}(\Theta_{ij}^s m, A) = K_j^{\mathrm{rec}}(m, A). \tag{5.2}$$

Consequently, $s_i = (\Theta_{ij}^s)_{\#} s_j$ implies equal predictive record laws. The compatibility is additional information: two equal laws on a one-point model space can predict opposite deterministic bits through different evaluators.

Theorem 5.1 (Exact predictive information identity). Let s, t be probability laws on a standard Borel model space and let K be a common normalized record kernel. Assume $D_{\mathrm{KL}}(s\|t) < \infty$. Set $\mu(dm, do) = s(dm)K(m, do)$ and $\nu(dm, do) = t(dm)K(m, do)$, and denote their conditional model laws given o by s^o, t^o. Then

$$D_{\mathrm{KL}}(s\|t) = D_{\mathrm{KL}}(sK\|tK) + \int D_{\mathrm{KL}}(s^o\|t^o)(sK)(do). \tag{5.3}$$

Proof. The common kernel cancels in the joint Radon–Nikodym derivative, giving $D_{\mathrm{KL}}(\mu\|\nu) = D_{\mathrm{KL}}(s\|t)$. Disintegrating the same joint laws by their record marginal factors their density ratio into

the record ratio and the conditional model ratio. The logarithm is their sum. Integrating the first term gives $D_{\mathrm{KL}}(sK\|tK)$, and integrating the second first over m gives the displayed conditional term. Finiteness makes the decomposition unambiguous. □

The second term is the model distinction that the record leaves unresolved. Its vanishing is an exact sufficiency condition. A constant evaluator can make all predictions agree while retaining no model distinction at all. Useful agreement therefore requires a specified predictive task as well as a comparison of laws. For a common history kernel, the same identity holds with the full finite record history in place of o.

If the model link depends on the current laws, theorem 5.1 applies to the resulting fixed comparison at that configuration, provided eq. (5.2) holds there. Preservation of compatibility during an update is a separate condition on the dynamics and the evaluator.

5.2 What the receiver reconstructs

A sending map $T_j : E_j \to \mathsf{H}$ induces the message law $\nu_j = (T_j)_{\#}q_j$. Different laws can have the same message law. For linear maps $T_j : E_j \to \mathsf{H}$ and $O_i : \mathsf{H} \to E_i$,

$$\operatorname{rank}(O_i T_j) \le \operatorname{rank}(T_j). \tag{5.4}$$

Writing a message into a larger space cannot by itself recover distinctions discarded by T_j. A richer reconstruction uses the receiver's conditional model.

For a received value h and a declared likelihood $\ell_i(h \mid k)$, Bayesian assimilation is

$$q_i^+(dk) = \frac{\ell_i(h \mid k)\, q_i^-(dk)}{\int \ell_i(h \mid k')\, q_i^-(dk')}, \tag{5.5}$$

when the denominator is finite and positive. An unmeasured coordinate can change because it is conditionally related to the measured one in q_i^-. The observation selects among possibilities already related by the receiver's model.

Proposition 5.2 (Exact conditional reconstruction). Let $T : E \to \mathsf{H}$ be measurable on standard Borel spaces. Let q, p be laws with $D_{\mathrm{KL}}(q\|p) < \infty$, and set $\nu = T_{\#}q$. Choose a regular conditional law $p(dk \mid h)$ for p given $T = h$. Define

$$\widehat{q}(dk) = \int p(dk \mid h)\, \nu(dh). \tag{5.6}$$

Then $T_{\#}\widehat{q} = \nu$ and

$$D_{\mathrm{KL}}(q\|\widehat{q}) = \int D_{\mathrm{KL}}\big(q(\cdot \mid h)\|p(\cdot \mid h)\big)\, \nu(dh). \tag{5.7}$$

In particular, $\widehat{q} = q$ exactly when the two conditional laws agree for ν-almost every h.

Proof. Absolute continuity gives $\nu \ll T_{\#}p$, so the conditional version of p is defined at every h needed up to a ν-null set. That conditional is supported on $T^{-1}(h)$ almost surely. Hence its mixture has pushforward ν. The laws q and $\widehat{q}$ have the same h-marginal and respective conditional laws

$q(\cdot \mid h)$ and $p(\cdot \mid h)$. Factoring their density ratio by h proves eq. (5.7). Nonnegativity and the equality condition of relative entropy give the last claim. □

This reconstruction preserves the communicated marginal and supplies the discarded conditional from p. It makes precise the sense in which a receiver contributes structure that the message alone does not specify. The proposition assumes a common content presentation; a link or a separately declared correspondence is needed before applying it across different fibers.

Proposition 5.3 (Common recovery by likelihood descent). Let $C : \mathsf{X} \to \mathsf{Z}$ be a measurable map between standard Borel spaces. Let P_0 be a probability law and let $P_\alpha = L_\alpha P_0$ be a family of probability laws dominated by it. Choose the reference conditional $R_0(z, dx) = P_0(dx \mid C = z)$, and write $\bar{L}_\alpha(z) = \mathbb{E}_{P_0}[L_\alpha \mid C = z]$. Then

$$(C_\# P_\alpha) R_0(dx) = \bar{L}_\alpha(C(x)) P_0(dx). \tag{5.8}$$

Consequently this one kernel recovers every P_α exactly when each L_α has a $\sigma(C)$-measurable version. No finite-divergence assumption is needed.

Proof. For each measurable $A \subseteq \mathsf{Z}$, $P_\alpha(C^{-1}A) = \int_A \bar{L}_\alpha(z)(C_\# P_0)(dz)$, by the defining property of conditional expectation. For a measurable $B \subseteq \mathsf{X}$, disintegration therefore gives

$$\begin{aligned} ((C_\# P_\alpha) R_0)(B) &= \int \bar{L}_\alpha(z) R_0(z, B)(C_\# P_0)(dz) \\ &= \int_B \bar{L}_\alpha(C(x)) P_0(dx). \end{aligned} \tag{5.9}$$

Equality with P_α is equivalent to equality of their Radon–Nikodym densities, $L_\alpha = \bar{L}_\alpha \circ C$ almost surely. Conditional expectation fixes exactly the integrable $\sigma(C)$-measurable functions, proving both directions. □

A posterior-specific conditional always reconstructs that posterior from its own marginal. It need not reconstruct the reference or other posteriors. The common kernel in proposition 5.3 makes the stronger requirement explicit. Regular conditional laws and their almost-sure uniqueness are understood in the standard Borel sense [13].

Corollary 5.4 (Recovery without reading the exterior). Let P_0 now be a joint reference law of (X, E) on standard Borel spaces, and let $Z = C(X)$. Retain (Z, E) but reconstruct X using only $R_0(z, dx) = P_0(dx \mid Z = z)$, leaving E unchanged. This reconstructs P_0 exactly if and only if X and E are conditionally independent given Z. Under that condition it also reconstructs a dominated family $P_\alpha = L_\alpha P_0$ exactly if and only if each likelihood ratio descends to (Z, E).

Proof. The lifted reference has conditional law $R_0(z, dx)$ given (z, e). Because R_0 is supported on $C^{-1}(z)$ almost surely, its joint law with (Z, E) agrees with the true law exactly when $P_0(dx \mid Z, E) = P_0(dx \mid Z)$ almost surely. This is the stated conditional independence. In that case the exterior-preserving lift is the reference conditional for the map $(x, e) \mapsto (C(x), e)$. Apply proposition 5.3 to that map. □

5.3 A predictive coding criterion

Lossy source coding makes fidelity part of the problem definition [21]. Decoder side information is a realized variable correlated with the source, as in Wyner–Ziv coding [25]; knowing a prior without such a realization does not supply that variable. A few noiseless real numbers also do not specify a finite-bit channel. A communication rate requires a declared coding, precision, or noise model.

For a predictive criterion, let $P(X, Y, Z)$ be a declared preparation law. The source X may be represented content or a complete fine configuration; the choice is fixed in each application. Let Z be accessible receiver information and let Y be a response to be preserved. An encoder $T(du \mid X)$ gives the conditional independence

$$U - X - (Y, Z). \tag{5.10}$$

The decoder predicts Y from (U, Z). This is the setting of an information bottleneck, with the relevant variable explicitly named [22].

Theorem 5.5 (Predictive distortion and the optimal decoder). On standard Borel spaces, assume the displayed information quantities are finite and eq. (5.10) holds. For any predictive decoder $D(dy \mid u, z)$ with finite expected relative entropy,

$$\begin{aligned} &\mathbb{E} D_{\mathrm{KL}}\big(P(Y \mid X, Z) \| D(Y \mid U, Z)\big) \\ &\quad = I(Y; X \mid U, Z) + \mathbb{E} D_{\mathrm{KL}}\big(P(Y \mid U, Z) \| D(Y \mid U, Z)\big). \end{aligned} \tag{5.11}$$

The optimal decoder is $P(dy \mid u, z)$. Its distortion is

$$\mathcal{D}(T) = I(Y; X \mid U, Z) = I(Y; X \mid Z) - I(Y; U \mid Z). \tag{5.12}$$

Proof. Conditional independence gives $P(Y \mid X, U, Z) = P(Y \mid X, Z)$. In the logarithmic ratio defining the left side of eq. (5.11), insert the intermediate conditional law $P(Y \mid U, Z)$. The first resulting expectation is the definition of $I(Y; X \mid U, Z)$. In the second, averaging over X gives the conditional law $P(Y \mid U, Z)$, yielding the remaining divergence. The second term vanishes at the stated decoder and is nonnegative for every decoder. Finally, expand $I(Y; X, U \mid Z)$ in both orders and use $I(Y; U \mid X, Z) = 0$. □

The functional $I(X; U \mid Z) + \lambda \mathcal{D}(T)$, for $\lambda \geq 0$, is one possible criterion for selecting a representation. Its first term is an information cost, not automatically the length of an individual message. An operational coding theorem requires its own source and block-coding assumptions. Transmitting a sample from $(T_j)_{\#} q_j$ and transmitting that whole probability law are also different protocols.

For parent formation, choose X to include the fine configuration and choose Y to be a future boundary response over a specified horizon and interaction. Zero distortion means that U and Z are sufficient for this target under this preparation. A subsequent interaction may read information that this target ignored. Exact inheritance therefore requires the dynamical condition developed in chapter 6, not only predictive sufficiency for one chosen response.

Chapter 6

Inherited Parents and Recursive Formation

A collection becomes a parent when its retained state supports inherited interaction and evolution. A probability law obtained by compression is a first part of this construction. It does not determine how that law will change, what the parent can communicate, or which information its children must retain. We give exact conditions for these further steps and a finite example in which inheritance works through two levels. The example supplies its reductions; formation of the reductions remains a separate question.

6.1 Two objects that a parent must retain

Definition 6.1 (Sample compression and configuration retention). Let $x \in \mathsf{S}$ be a complete agent configuration and let Q_x be its declared law on a standard Borel space Y of represented contents. A normalized sample channel $C : \mathsf{Y} \rightsquigarrow \mathsf{Z}$ gives the retained law

$$Q_x^c(A) = \int C(y, A) Q_x(dy). \tag{6.1}$$

A configuration retention is a measurable map $R : \mathsf{S} \to \mathsf{S}^c$. Its value includes the parent laws and the additional data needed for its declared interactions and update.

The distinction applies to both sectors. A parent carrying q_I and s_I also needs the evaluator that makes s_I predictive. Boundary channels, source rows, source marks, and memory can be further components of $R(x)$. Equal marginal laws do not imply equality of these components. When a sample channel reads random links or source variables, those variables belong to the joint input law of the channel. Substitution of a mean link or a selected source is a different retention.

A law of represented contents and a probability law over complete configurations are different objects. Coupled deterministic evolution of law-valued agents requires no probability distribution over x. If such a distribution is used below, its preparation or transition mechanism is stated. All constructions admit general probability laws. A Gaussian parent is one possible family, subject to its own closure conditions.

6.2 Exact deterministic inheritance

Proposition 6.2 (An update descends precisely when it respects retained states). Let S, S^c be measurable spaces, let $R : \mathsf{S} \to \mathsf{S}^c$ be a measurable surjection with a measurable right inverse ι,

and let $F : \mathsf{S} \to \mathsf{S}$ be measurable. There is a measurable parent update F^c satisfying

$$R \circ F = F^c \circ R, \tag{6.2}$$

if and only if

$$R(x) = R(x') \quad \Longrightarrow \quad R(F(x)) = R(F(x')). \tag{6.3}$$

The parent update is unique. Finite spaces automatically admit the required right inverse.

Proof. Equation (6.2) implies the implication in (6.3). Conversely, define $F^c = R \circ F \circ \iota$. It is measurable. Since $R(\iota(R(x))) = R(x)$, the implication gives $F^c(R(x)) = R(F(x))$. Every parent state has a preimage, so this equation also fixes F^c uniquely. □

Repeated substitution gives $R \circ F^n = (F^c)^n \circ R$ for every nonnegative integer n. This is an exact statement about the chosen update, including its order and retained parameters. Replacing that update by descent of a newly defined parent objective requires another argument.

For a differentiable version, suppose R is continuously differentiable and the fine and parent vector fields B, B^c have unique local solutions. Their flows satisfy $R \circ \Phi_t = \Phi^c_t \circ R$ on their common existence interval if

$$DR_x B(x) = B^c(R(x)). \tag{6.4}$$

Indeed, the chain rule makes $R(\Phi_t(x))$ a parent solution with initial value $R(x)$, and uniqueness identifies it. Conversely, differentiation of the flow identity at zero gives (6.4). For a moving map R_t, the left side becomes $\partial_t R_t(x) + DR_t(x)B_t(x)$. Membership changes therefore need an explicit handoff map; a differential identity on one fixed partition does not specify them.

Proposition 6.3 (A sufficient metric condition for gradient inheritance). Let $R : (\mathsf{S}, g) \to (\mathsf{S}^c, g_c)$ be a smooth surjective submersion between finite-dimensional Riemannian manifolds. Suppose the smooth potentials satisfy $E = \chi \circ E_c \circ R$, with $\chi' > 0$. Assume there is a smooth positive function d such that

$$g_c(DR_x u, DR_x v) = d(x)^2 g(u, v), \qquad u, v \in (\ker DR_x)^\perp. \tag{6.5}$$

Then their negative gradient fields obey

$$DR_x[-\operatorname{grad}_g E(x)] = a(x)[-\operatorname{grad}_{g_c} E_c](R(x)), \qquad a(x) = \chi'(E_c(R(x)))d(x)^2. \tag{6.6}$$

In particular, a Riemannian submersion and $E = E_c \circ R$ give same-time inheritance.

Proof. For $v \in \ker DR_x$, differentiation of the potential identity gives $dE_x(v) = 0$. Thus $w = \operatorname{grad}_g E(x)$ lies in $(\ker DR_x)^\perp$. The restriction of DR_x to this horizontal space is an isomorphism onto $T_{R(x)}\mathsf{S}^c$. Given any coarse tangent z, take its horizontal lift v. Then

$$\begin{aligned} g_c(DR_x w, z) &= d(x)^2 g(w, v) = d(x)^2 dE_x(v) \\ &= d(x)^2 \chi'(E_c(R(x)))d(E_c)_{R(x)}(z). \end{aligned} \tag{6.7}$$

Nondegeneracy of g_c identifies $DR_x w$ with $a(x) \operatorname{grad}_{g_c} E_c(R(x))$. Negating proves the identity. When $a = 1$, eq. (6.4) applies. □

For $a > 0$, a fine trajectory projects onto the same oriented coarse gradient orbit after the time change $d\tau/dt = a(x(t))$. An autonomous field in the original time additionally requires the product $a(x)\,\mathrm{grad}_{g_c} E_c(R(x))$ to be constant on retention fibers. Factorization $a = \bar{a} \circ R$ suffices; dependence on discarded state need not satisfy this requirement. The proposition is one sufficient route to inheritance. Fisher contraction and preservation of the collective connection in chapter 3 are different conditions.

Fiber minimization alone does not supply its potential hypothesis. For $R(u, v) = v$ on Euclidean $\mathbb{R}^2$, let

$$E(u, v) = \tfrac{1}{2}(u - v)^2 + \tfrac{1}{2}v^2, \qquad E_c(v) = \inf_u E(u, v) = \tfrac{1}{2}v^2. \tag{6.8}$$

Equality $E = E_c \circ R$ holds only on $u = v$. More generally, $dE(\partial_u) = u - v$ does not vanish on any open set, so there is no identity $E = \chi(E_c \circ R)$ on an open set for any χ.

Definition 6.4 (Loss of dynamical roles under product recognition). Take $Q_x = \mathrm{Ber}(p_1) \otimes \mathrm{Ber}(p_2)$, with $x = (p_1, p_2) \in [0, 1]^2$, and the coupled deterministic law

$$\dot{p}_1 = -p_1 + \lambda(p_2 - p_1), \qquad \dot{p}_2 = -2p_2 + \lambda(p_1 - p_2), \qquad \lambda \geq 0. \tag{6.9}$$

The linear system has a unique global solution and its vector field points inward on every face of the square. Retain only the law of $K_1 + K_2$. The configurations $(0.8, 0.2)$ and $(0.2, 0.8)$ both give probabilities $(0.16, 0.68, 0.16)$ for the values $0, 1, 2$. However,

$$\frac{d}{dt}\mathbb{E}(K_1 + K_2) = -(p_1 + 2p_2), \tag{6.10}$$

equals -1.2 and -1.8 respectively. Their retained laws therefore have different derivatives. No autonomous update of that retained law can reproduce both trajectories.

This obstruction occurs while recognition remains a product. The missing datum is which probability belongs to which dynamical role. Correlated recognition is neither a prerequisite for this counterexample nor its automatic remedy. Coupled mean-field dynamics are admissible; the proposed retention must still pass its own inheritance test.

6.3 Exact stochastic inheritance

Proposition 6.5 (Finite strong lumpability). Let T be a transition matrix on a finite configuration space S and let $R : \mathsf{S} \to \mathsf{S}^c$ be surjective. A single parent transition matrix reproduces the retained process for every initial law if and only if, for every $z' \in \mathsf{S}^c$,

$$\sum_{y : R(y) = z'} T(x, y) = \sum_{y : R(y) = z'} T(x', y) \quad \text{whenever } R(x) = R(x'). \tag{6.11}$$

In that case the common sum defines $T^c(R(x), z')$ uniquely.

Proof. The common sums are nonnegative and sum to one, so they define a transition matrix. Conditional on any retained history ending at z, the hidden configuration lies in $R^{-1}(z)$. Every state in that fiber gives the same next retained distribution $T^c(z, \cdot)$; averaging over its conditional law leaves this distribution unchanged. Induction gives the parent path law from the pushed initial law.

Conversely, start the fine process at each point mass δ_x. A common parent kernel must give the same one-step distribution whenever these initial states have the same image. This is (6.11). □

This is the finite strong-lumpability criterion [14]. On standard Borel spaces the same argument applies when $T(x, R^{-1}A)$ factors through R as a measurable probability kernel. The qualification is part of the construction. Closure for every initial law is stronger than closure for one chosen preparation.

An equilibrium conditional average of fine transitions defines a normalized candidate parent kernel. It does not prove (6.11): the hidden conditional law during a transient can differ from the equilibrium conditional. Likewise, agreement of invariant distributions does not establish agreement of histories.

6.4 Boundary responses and composition

A parent must inherit what it does to the rest of the population. Fix a declared boundary protocol a, including the preparation and receiving conditions. Let $K_a : \mathsf{S} \rightsquigarrow \mathsf{O}_a$ be its normalized response kernel and let $D_a : \mathsf{O}_a \rightsquigarrow \mathsf{O}_a^c$ retain the response of interest. An inherited kernel K_a^c satisfies

$$(K_a D_a)(x, A) = K_a^c(R(x), A). \tag{6.12}$$

Here $(KD)(x, A) = \int D(o, A)K(x, do)$. Under the measurable right-inverse hypothesis of proposition 6.2, such a kernel exists exactly when $K_a D_a$ is constant on each fiber of R. Define it using the right inverse to prove sufficiency; substitution proves necessity. When several outputs are used jointly, this condition applies to their joint kernel. Preserving each output marginal alone need not preserve their dependence.

Proposition 6.6 (Exact elimination through a conditional separator). Let Z, U, V have a declared joint reference law $\rho(dz, du, dv) = \pi(dz)\rho_{UV}(du, dv \mid z)$ on standard Borel spaces. Denote its conditional marginals by ρ_U, ρ_V. For bounded nonnegative likelihood factors $L_U(u, z), L_V(v, z)$, define

$$F_U(z) = \int L_U(u, z)\rho_U(du \mid z), \qquad F_V(z) = \int L_V(v, z)\rho_V(dv \mid z). \tag{6.13}$$

The identity

$$\int L_U(u, z)L_V(v, z)\rho_{UV}(du, dv \mid z) = F_U(z)F_V(z) \tag{6.14}$$

holds π-almost everywhere for every such pair precisely when U and V are conditionally independent given Z. Under that condition, a nonnegative factor $L_Z(z)$ with finite positive total evidence gives the unnormalized retained measure

$$m_Z(dz) = L_Z(z)F_U(z)F_V(z)\pi(dz). \tag{6.15}$$

Proof. Conditional independence factors ρ_{UV} as $\rho_U \otimes \rho_V$ almost surely. Tonelli's theorem then gives eqs. (6.14) and (6.15). Conversely, take indicator factors from countable generating families closed under finite intersections for the two standard Borel spaces. Outside one π-null set, the conditional joint and product laws agree on every generating rectangle. Uniqueness of probability measures extends this equality to the product sigma algebra, proving conditional independence. □

The shared factor L_Z appears once. If the hidden blocks are not conditionally independent, their joint conditional factor must be retained. This elimination statement concerns the declared reference and likelihood; it does not replace the law-dependent interaction potential by a fixed joint model.

For a child channel $\mathcal{K}_{ij} : \mathsf{Y}_j \rightsquigarrow \mathsf{Y}_i$ and retentions $C_j : \mathsf{Y}_j \rightsquigarrow \mathsf{Z}_J$ and $C_i : \mathsf{Y}_i \rightsquigarrow \mathsf{Z}_I$, the corresponding square is $\mathcal{K}_{ij}C_i = C_j\mathcal{K}^c_{IJ}$, where $\mathcal{K}^c_{IJ} : \mathsf{Z}_J \rightsquigarrow \mathsf{Z}_I$. The rightmost expression means first retain the sender and then apply the parent channel. The retained sender must determine every distinction visible to the retained receiver. In a linear realization this reduces to $C_iM_{ij} = M^c_{IJ}C_j$; the kernel-inclusion criterion and its proof are given in the supplement.

Proposition 6.7 (Inheritance composes). Suppose $R_{10} : \mathsf{S}_0 \to \mathsf{S}_1$ and $R_{21} : \mathsf{S}_1 \to \mathsf{S}_2$ intertwine deterministic updates at successive levels. Then $R_{20} = R_{21} \circ R_{10}$ intertwines the first and third updates. The analogous assertion holds for finite lumped Markov transitions. Let $K^\ell_a : \mathsf{S}_\ell \rightsquigarrow \mathsf{O}_\ell$ be boundary kernels, with response retentions $D_{10} : \mathsf{O}_0 \rightsquigarrow \mathsf{O}_1$ and $D_{21} : \mathsf{O}_1 \rightsquigarrow \mathsf{O}_2$. If

$$K^0_aD_{10} = R_{10}K^1_a, \qquad K^1_aD_{21} = R_{21}K^2_a, \tag{6.16}$$

where maps on the right are their deterministic kernels, then $K^0_aD_{10}D_{21} = R_{10}R_{21}K^2_a$.

Proof. For deterministic updates, $R_{21}R_{10}F^0 = R_{21}F^1R_{10} = F^2R_{21}R_{10}$, where juxtaposition in this identity denotes function composition. For Markov transitions and their deterministic retention kernels, $T^0R_{10}R_{21} = R_{10}T^1R_{21} = R_{10}R_{21}T^2$. For boundary kernels, associate the successive integrals and substitute (6.16). All integrands for events are nonnegative, so interchanging the integration order is valid. □

When changes of gauge presentation act on the retained spaces, the retentions and boundary kernels must transform compatibly with those actions. Re-expressing a commuting square then gives the same square in the new coordinates. Neither a parent frame chosen for this purpose nor the square itself determines a base connection. If the parent is also required to retain the collective Fisher connection, the additional condition in proposition 3.14 must be checked.

6.5 A finite two-level construction

Let a complete configuration be $x = (a, b, c, r) \in \{0, 1\}^4$. The coordinates are labels of internal configurations, with r an internal binary register and $\rho(x) = r$ its readout. To attach actual agent laws, choose

$$q_a = \mathrm{Ber}\left(\frac{1+2a}{4}\right), \qquad s_a = \mathrm{Ber}\left(\frac{2+a}{4}\right), \qquad \zeta_a = q_a \otimes s_a. \tag{6.17}$$

Thus both sectors vary, with tied parameters in this supplied family; b, c are additional internal marks. A model presentation $m \in \{0, 1\}$ has evaluator

$$K(m, \{1\}) = \frac{1}{4} + \frac{m}{2}. \tag{6.18}$$

The corresponding predictive law is $\nu_a = s_aK = \mathrm{Ber}(1/2 + a/8)$, which distinguishes the two values of a.

At each step draw independent bits $\xi \sim \mathrm{Ber}(1/4)$, $\eta \sim \mathrm{Ber}(1/3)$, and $\upsilon \sim \mathrm{Ber}(1/5)$, independently of preceding draws, and set

$$a' = a \oplus \xi, \qquad b' = b \oplus a \oplus \eta, \qquad c' = c \oplus b \oplus \upsilon. \tag{6.19}$$

Independently of these bit draws, conditional on the current configuration, give r' law ν_a. This supplies a transition of an internal register; the evaluator alone did not force this choice. The symbol $\oplus$ denotes addition modulo two. The dynamics are coupled: a affects the next b, and b affects the next c.

Retain $R_{10}(a, b, c, r) = (a, b, r)$ and then $R_{21}(a, b, r) = (a, r)$. For $f_p(0) = 1 - p$ and $f_p(1) = p$, the first parent kernel is

$$\begin{aligned} T^1((a, b, r), (a', b', r')) = f_{1/4}(a' \oplus a)\nu_a(\{r'\}) \\ \cdot f_{1/3}(b' \oplus b \oplus a). \end{aligned} \tag{6.20}$$

Summing the fine transition over c' gives this expression, independently of c. Summing it over b' gives $T^2((a, r), (a', r')) = f_{1/4}(a' \oplus a)\nu_a(\{r'\})$, independently of b. Both reductions satisfy proposition 6.5. Their configuration spaces have sizes sixteen, eight, and four, and both parents retain the laws in (6.17), their evaluator, and the register. Its readout descends by the identity on record values. The first parent additionally retains b.

Both parents reproduce the entire finite record history as well as the retained configuration history. Indeed, write the fine path probability as its initial mass times its transition factors. Summing over hidden coordinates gives the parent path law by proposition 6.5; the record at each step is already a coordinate of that path. The next record has probability $1/2 + a/8$ of being one, so the inherited response is nonconstant. No independence of the initial coordinates is needed.

This construction has nontrivial reductions and coupled dynamics, but its retentions were specified in advance. It proves exact two-level inheritance, without proving that the dynamics choose either reduction.

6.6 Averaged laws under smooth peer dynamics

The arithmetic mixture of child laws is a normalized candidate parent law. It need not determine either its own derivative or its boundary responses. The following example establishes both failures within the smooth peer-KL family of chapter 4, with identical agents, a coupled evaluator, and equal Fisher mobilities. It concerns this specified retention; it is not an obstruction to every possible parent state.

Definition 6.8 (A coupled binary model with law-dependent channels). Use counting measure on $k, m \in \{-1, 1\}$ and the strictly positive laws

$$q_x(k) = \frac{1 + kx}{2}, \qquad s_y(m) = \frac{1 + my}{2}, \qquad x, y \in (-1, 1). \tag{6.21}$$

Choose $J = 1/2$, reference $r(m) = 1/2$, and normalized evaluator $G_J(k \mid m) = e^{Jkm}/(2\cosh J)$. For the declared product recognition family, the local variational term is exactly

$$\begin{aligned} E_0(x,y) &= \sum_{k,m} q_x(k) s_y(m) \log \frac{q_x(k) s_y(m)}{r(m) G_J(k \mid m)} \\ &= \psi(x) + \psi(y) - Jxy + \log\cosh J, \\ \psi(x) &= \frac{1+x}{2}\log(1+x) + \frac{1-x}{2}\log(1-x). \end{aligned} \tag{6.22}$$

All expectations are finite. The mixed derivative is $-J$, so changing either sector changes the force on the other.

In either sector, let a sender with parameter x use the normalized kernel

$$K_x(\ell \mid k) = \frac{1+\rho(x)k\ell}{2}, \qquad \rho(x) = \frac{1+x^2}{4}, \qquad q_x K_x = q_{f(x)}, \quad f(x) = \frac{x+x^3}{4}. \tag{6.23}$$

Every entry exceeds $1/4$. The channel transmits the sampled sender bit, and its transmission coefficient depends nontrivially on the sender law. For passive frame covariance, present each alphabet as the atoms $\{\pm u_i\}$ of a one-dimensional fiber, with $u_i \neq 0$, and define $\mathcal{K}_{ij}(u_j k, \{u_i \ell\}) = K_{x_j}(\ell \mid k)$. Under $g_i \in \mathrm{GL}(1,\mathbb{R})$, push the laws forward and replace u_i by $g_i u_i$. The reference parameters remain fixed, and

$$\mathcal{K}'_{ij}(g_j v, g_i A) = \mathcal{K}_{ij}(v, A). \tag{6.24}$$

Independent sector frames give the same construction for s; pushing G_J through its model and belief frames preserves its normalization.

For two identical agents, set

$$\begin{aligned} d(x,z) &= D_{\mathrm{KL}}(q_x \| q_z), \\ P(x_1,x_2) &= d(x_1, f(x_2)) + d(x_2, f(x_1)), \\ E &= \sum_{i=1}^{2} E_0(x_i, y_i) + P(x_1,x_2) + P(y_1,y_2). \end{aligned} \tag{6.25}$$

Each source row contains only the other agent, so its normalized weight is one and its row-relative-entropy term vanishes. This is a specialization of the declared peer potential. The local term is the displayed KL expectation; no further identification of the entire law-dependent potential with a fixed-joint-law ELBO is assumed. The product Fisher metric and its negative gradient flow are

$$g = \sum_i \left(\frac{dx_i^2}{1-x_i^2} + \frac{dy_i^2}{1-y_i^2} \right), \qquad \dot{x}_i = -(1-x_i^2)\partial_{x_i} E, \quad \dot{y}_i = -(1-y_i^2)\partial_{y_i} E. \tag{6.26}$$

Both mobilities are one. The vector field is smooth on $(-1,1)^4$.

Proposition 6.9 (The mixture retention fails the dynamical inheritance test). For the flow in (6.26), the retention

$$R(x_1, y_1, x_2, y_2) = \left(\frac{q_{x_1} + q_{x_2}}{2}, \frac{s_{y_1} + s_{y_2}}{2} \right) = (q_{\bar{x}}, s_{\bar{y}}) \tag{6.27}$$

does not support an autonomous parent vector field on the whole interior. The same failure holds if the averaged joint law $\frac{1}{2}\sum_i q_{x_i} \otimes s_{y_i}$ is also retained.

Proof. For $0 < |a| < 1$ and $0 < |c| < 1$, take $A = (0, c, 0, c)$ and $B = (a, c, -a, c)$ in the coordinate order of (6.27). Both retain (q_0, s_c) and the same averaged joint law $q_0 \otimes s_c$. These are equal full binary probability laws, not merely equal estimates of their means. The identities $f(-x) = -f(x)$ and $d(-x, -z) = d(x, z)$ imply

$$P(x_1, x_2) = P(x_2, x_1) = P(-x_1, -x_2), \qquad \partial_1 P(a, -a) = -\partial_2 P(a, -a), \quad \nabla P(0, 0) = 0. \tag{6.28}$$

At B the two Fisher prefactors are equal. Consequently the peer forces and the two atanh terms cancel in $\dot{\bar{x}}$, giving

$$\dot{\bar{x}}(A) = Jc, \qquad \dot{\bar{x}}(B) = Jc(1 - a^2). \tag{6.29}$$

Both model-sector derivatives can also be written exactly. Define

$$h(c) = 2 \operatorname{atanh} c - \operatorname{atanh} f(c) + f'(c) \frac{f(c) - c}{1 - f(c)^2}. \tag{6.30}$$

Indeed, direct differentiation gives $\partial_x d(x, z) = \operatorname{atanh} x - \operatorname{atanh} z$ and $\partial_z d(x, z) = (z - x)/(1 - z^2)$. Thus

$$\begin{aligned} (\dot{y}_1, \dot{y}_2)(A) &= -(1 - c^2)(h(c), h(c)), \\ (\dot{y}_1, \dot{y}_2)(B) &= -(1 - c^2)(h(c) - Ja, h(c) + Ja), \\ \dot{\bar{y}}(A) = \dot{\bar{y}}(B) &= -(1 - c^2)h(c). \end{aligned} \tag{6.31}$$

The retained belief derivatives nevertheless differ by $-Jca^2 \neq 0$. For $a = 1/2, c = 1/4$, they are $1/8$ and $3/32$. Hence $DR_x B(x)$ is not constant on a retention fiber, contradicting the necessary condition (6.4). The identical retained joint laws establish the additional assertion. □

The parameter a may be arbitrarily small. A neighborhood of the synchronized configurations therefore does not make this retention exact. Unlike definition 6.4, the example uses identical dynamical roles.

Proposition 6.10 (The same retention loses a boundary response). Choose a child uniformly, draw its belief bit, and transmit that bit through (6.23) to an external receiver. This declared protocol has a normalized binary response law with mean $b(x_1, x_2) = (f(x_1) + f(x_2))/2$. For any m, a with $|m| + |a| < 1$, compare belief parameters (m, m) and $(m + a, m - a)$, keeping both model parameters equal to c. The retained marginal and averaged joint laws agree, but

$$b(m + a, m - a) - b(m, m) = \frac{3}{4} m a^2. \tag{6.32}$$

For $ma \neq 0$, their full binary response laws differ.

Proof. Expand $f(x) = (x + x^3)/4$ and use $((m + a)^3 + (m - a)^3)/2 = m^3 + 3ma^2$. At $m = a = 1/4$, the two response means are $20/256$ and $17/256$. Thus the response kernel is not constant on a retention fiber, violating (6.12). □

Proposition 6.11 (Interior synchronization cannot first occur at finite time). For (6.26), the diagonal $\Delta = \{x_1 = x_2,\ y_1 = y_2\}$ is invariant. A solution starting outside Δ cannot enter it at a finite time while remaining in the interior.

Proof. Agent exchange preserves the potential and metric, hence the vector field. On its fixed set Δ, the corresponding component derivatives agree, which proves invariance. The smooth field has unique local solutions in both time directions. If an interior solution meets Δ, uniqueness from that meeting point identifies it locally, backward as well as forward, with the solution in Δ. Along the original existence interval, the set of times at which it lies in Δ is therefore open; it is also closed by continuity. Connectedness of that interval makes it the entire interval, contradicting the initial condition. □

This result permits asymptotic synchronization, state-dependent threshold selection, and exact quotients that do not require synchronization. It does not prove convergence. An alternative retained state or a restricted invariant family must be tested against both inheritance conditions; the calculations above do not exclude such constructions.

6.7 Memory and formation

When a proposed retention fails, one exact option is to retain more of the configuration. For a declared stochastic process, another is to retain the conditional law of the hidden configuration given the accessible retained history. That conditional law determines future predictions. In a finite model it is a probability vector with an exact recursive update, derived in the supplement. Its attainable values need not form a finite set. Introducing such a filter presupposes a process and a preparation law; it is not compulsory for deterministic coupled agents.

Theorem 6.12 (Information lost by a retained history). Let $\mathbb{P}$ be a probability law on $\mathsf{Z}_0 \times \cdots \times \mathsf{Z}_{T-1}$, where each record space is standard Borel and T is finite. Set $H_t = (Z_0, \dots, Z_{t-1})$, with H_0 a point. For measurable maps into standard Borel statistic spaces, let $M_t = f_t(H_t)$. Write $P_t^M(\cdot \mid m)$ for a regular conditional law of Z_t given $M_t = m$. Let $\widetilde{T}_t$ be any normalized kernel from the statistic space to Z_t, and let $\widetilde{\mathbb{P}}$ be the causal path law whose next-record kernel is $\widetilde{T}_t(f_t(h), \cdot)$ at history h. Then, in $[0, \infty]$,

$$\begin{aligned} D_{\mathrm{KL}}(\mathbb{P} \| \widetilde{\mathbb{P}}) = &\sum_{t=0}^{T-1} I_{\mathbb{P}}(Z_t; H_t \mid M_t) \\ &+ \sum_{t=0}^{T-1} \mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}\big(P_t^M(\cdot \mid M_t) \| \widetilde{T}_t(M_t, \cdot)\big). \end{aligned} \tag{6.33}$$

The first sum is the minimum path divergence among these predictors, attained by choosing $\widetilde{T}_t = P_t^M$. Its t-th term vanishes exactly when the true next-record law has a version depending on H_t only through M_t.

Proof. Let $P_t(\cdot \mid h)$ be the true conditional law given $H_t = h$. The relative-entropy chain rule expresses the path divergence as

$$\sum_{t=0}^{T-1} \mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}\big(P_t(\cdot \mid H_t) \| \widetilde{T}_t(M_t, \cdot)\big). \tag{6.34}$$

To decompose its t-th summand, compare the true joint law of (H_t, Z_t) with the law obtained by keeping its H_t marginal and drawing Z_t from $\widetilde{T}_t(M_t, \cdot)$. Disintegrate these two laws by (M_t, Z_t). The divergence of their marginals is $\mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}(P_t^M(\cdot \mid M_t) \| \widetilde{T}_t(M_t, \cdot))$. In the second joint law the conditional history given (M_t, Z_t) is the true conditional history given M_t alone. The conditional divergence term is consequently $I_{\mathbb{P}}(Z_t; H_t \mid M_t)$.

The chain rule on standard Borel spaces is an equality of nonnegative extended quantities. A singular marginal or conditional contributes infinite divergence; no subtraction of infinite terms is used. Applying it to each summand of eq. (6.34) gives eq. (6.33). Choosing the true statistic-conditional rows makes every term in the second sum zero. Conditional mutual information vanishes exactly when the corresponding conditional joint law is the product of its conditional marginals. Since M_t is a function of H_t, this is equivalent to $P_t(\cdot \mid H_t) = P_t^M(\cdot \mid M_t)$ almost surely. □

The finite-window case is a Markov approximation of a record process [8, Sec. 7.3]; the theorem permits any measurable history statistic. Its two terms separate forgetting information from predicting incorrectly with the information kept. A statistic alone need not have a compact recursive realization. To update from the retained state and new record only, one additionally needs $M_{t+1} = U_t(M_t, Z_t)$, or a sufficient enlargement of M_t. The theorem concerns one preparation and protocol. Sufficiency for every admitted intervention requires the corresponding conditional-law identity for each such protocol.

A handoff initializes the parent from the actual fine state or its declared conditional law, including the source marks and memory read by future interactions. When evidence is shared across levels, its likelihood belongs to the joint model once. Reusing a child's record through a complete same-episode parent posterior requires the appropriate conditional or cavity accounting. A subsequent predictive prior is a different operation.

6.8 Statistical partitions and their limits

An independence partition is a property of a declared joint law. A tuple of individual agent laws does not itself specify a cross-agent joint law; taking their product would impose independence in the input. The following constructions apply when such a joint law has been supplied.

Proposition 6.13 (The finest independent partition). Let V be a finite nonempty index set and let $\mathbb{P}$ be a probability law on $\prod_{i \in V} \mathsf{Y}_i$, with standard Borel coordinate spaces. Call a partition π independent when

$$\mathbb{P} = \bigotimes_{B \in \pi} \mathbb{P}_B, \tag{6.35}$$

where $\mathbb{P}_B$ is its marginal on block B. There is a unique finest independent partition $\pi_*(\mathbb{P})$. Relabeling the coordinates relabels this partition, and separate bimeasurable changes of their presentations preserve it.

Proof. The one-block partition is independent. Suppose π and σ are independent. Marginalizing the σ factorization to a block $B \in \pi$ gives

$$\mathbb{P}_B = \bigotimes_{\substack{D\in\sigma \\ B\cap D\neq\varnothing}} \mathbb{P}_{B\cap D}. \tag{6.36}$$

Substituting these marginals into the π factorization proves independence of the common refinement consisting of the nonempty intersections $B \cap D$. There are finitely many partitions of V. Refining all independent ones therefore gives an independent partition finer than every one of them, which is necessarily unique. Relabelings and coordinatewise bimeasurable maps commute with taking marginals and product laws. Applying the inverse transformation gives the reverse implication, so the set of independent partitions and its unique finest member transform as stated. □

This statistical target is the finest-independent clustering problem [19]. Its existence does not require Gaussian laws or densities, but does not identify statistical independence with agency. In particular, it makes no statement about autonomous dynamics or future boundary responses.

Corollary 6.14 (Conditional independence partitions). Let the same coordinate tuple have a joint law with an already retained standard Borel variable C. Define

$$\mathcal{I}_C(\pi) = \int D_{\mathrm{KL}}\left(\mathbb{P}(dy_V \mid c) \middle\| \bigotimes_{B\in\pi} \mathbb{P}_B(dy_B \mid c)\right) \mathbb{P}_C(dc). \tag{6.37}$$

There is a unique finest partition for which $\mathcal{I}_C(\pi) = 0$.

Proof. Zero divergence means equality of the conditional joint and product laws for $\mathbb{P}_C$-almost every c. There are finitely many partitions. Outside the union of their relevant null sets, the common-refinement argument in proposition 6.13 applies pointwise to every conditionally independent partition. Their common refinement is therefore conditionally independent and uniquely finest. □

Choosing $C = Y_V$ makes every partition conditionally independent while retaining the whole tuple. The corollary does not choose the context that a useful parent should retain.

Definition 6.15 (Weak coupling can destroy every independent split). For any even $N \geq 4$, let $y_i \in \{-1, 1\}$ and define the strictly positive finite law

$$\mathbb{P}_\epsilon(y) = Z_\epsilon^{-1} \exp\left(J \sum_{b=1}^{N/2} y_{2b-1}y_{2b} + \epsilon \sum_{b=1}^{N/2-1} y_{2b}y_{2b+1}\right), \qquad J > 0, \quad \epsilon \geq 0, \tag{6.38}$$

where Z_ϵ is the sum of the numerator over the finite alphabet. At $\epsilon = 0$, distinct pairs are independent and each pair is dependent: its odds ratio is $e^{4J} \neq 1$. Thus the finest partition consists of those pairs.

For $\epsilon > 0$ the interaction path is connected. Any nontrivial partition cuts an edge. Fix all coordinates except the endpoints of one cut edge, whose coupling is $J_e \in \{J, \epsilon\}$. Positivity makes every such conditional law well defined. Its odds ratio is $e^{4J_e} \neq 1$, since all terms involving just one endpoint cancel in that ratio. Factorization across the proposed partition would make these endpoints conditionally independent after the other coordinates were fixed, forcing ratio one. No nontrivial independent partition exists. Hence $\pi_*(\mathbb{P}_\epsilon) = \{V\}$ for every $\epsilon > 0$, although continuity of each of the finitely many probabilities gives $\mathbb{P}_\epsilon \to \mathbb{P}_0$ in total variation. Exact independence is therefore not a continuous soft-membership criterion.

Proposition 6.16 (Symmetry constrains a deterministic hard selector). Let $|V| \geq 3$, let all coordinate spaces equal Y, and suppose a deterministic partition selector Π reads only a law on Y^V. Assume relabeling naturality: $\Pi(\sigma_\# \mathbb{P}) = \sigma\Pi(\mathbb{P})$ for every permutation σ of V. If $\mathbb{P}$ is invariant under all permutations, then $\Pi(\mathbb{P})$ is either the singleton partition or the one-block partition.

Proof. The selected partition must be invariant under every permutation. If it had a block B with $2 \leq |B| < |V|$, choose distinct $i, j \in B$ and $k \notin B$. The transposition exchanging j and k sends B to a distinct block that still contains i. Two distinct blocks of a partition cannot intersect. Therefore no such B exists, leaving exactly the two stated possibilities. □

A distribution over partitions, soft memberships, asymmetric realized histories, or additional geometric inputs are outside this proposition. An information-based hierarchy can also be retained before choosing a cut [5]. None of these possibilities supplies the required dynamics without a further construction. The symmetry result does not exclude symmetry breaking from asymmetric initial conditions.

6.9 The formation obligation

Definition 6.17 (Inherited recursive formation). An inherited recursive-formation construction specifies population dynamics that selects nontrivial blocks from an admitted class of initially unpartitioned configurations, with a declared persistence criterion and a handoff determined by the state or accessible history. Each selected block must carry inherited belief and model laws, evaluators, boundary responses, and dynamics. The resulting parents must then select a block containing at least two first-stage parents under those inherited interactions, without resetting their source data or supplying a newly tuned attraction at the next level.

Persistence can mean an invariant block relation in deterministic dynamics or a specified path event in stochastic dynamics. The earlier finite construction settles inheritance for a supplied hierarchy. It does not select that hierarchy or establish persistence. An extension with explicit fusion events is constructed in chapter 7. Under the original smooth potential, chapter 12 supplies a conditional four-agent realization: a retained-state threshold selects two parents and then their union, with exact dynamics, extensive responses, and collective geometry on a prepared common-precision family. Its additive quotients are available initially. Obtaining such sufficient interfaces from less structured preparations, or deriving a preferred membership criterion, remains open.

Chapter 7

Finite Fusion with Exact Statistical Inheritance

We now construct a population whose dynamics selects persistent blocks and then fuses the resulting parents. Pairwise fusion penalties occur in convex clustering [11], and suitable nonsmooth consensus protocols produce finite-time agreement [6]. The construction here also specifies the represented laws, model evaluator, gauge transport, parent metric, and boundary kernels. Its peer potential is a new nonsmooth square-root-KL interaction. It is distinct from the smooth KL potential of the preceding dynamics chapter.

7.1 A law family closed under geometric averaging

Work over one context, with state and model spaces $\mathsf{K} = \mathsf{M} = (0, \infty)$. Write $\mathrm{LN}(\mu, \sigma^2)$ for the law of e^Z when $Z \sim N(\mu, \sigma^2)$. An agent of integer multiplicity $n \geq 1$ carries the product law

$$q_n = \mathrm{LN}(\mu^q, 1/n), \qquad s_n = \mathrm{LN}(\mu^s, 1/n), \qquad \zeta_n = q_n \otimes s_n. \tag{7.1}$$

Multiplicity is retained state information. The variances are fixed within each type; only the two log means evolve. Laws are first written in a common reference coordinate. A fixed positive frame $U_i^q = U_i^s =: U_i \in \mathrm{GL}^+(1)$ represents both sectors locally by multiplication. Frames do not evolve in this example.

Choose constants $a > c > 0$ and define the normalized reference and evaluator

$$r_n = \mathrm{LN}\left(0, \frac{a}{n(a^2 - c^2)}\right), \qquad G_n(m, dk) = \mathrm{LN}\left(\frac{c}{a} \log m, \frac{1}{an}\right)(dk). \tag{7.2}$$

The two occurrences of a lognormal expression in this equation denote ordinary probability laws and kernels. In logarithmic coordinates the joint reference $r_n(dm)G_n(m, dk)$ has precision

$$nA, \qquad A = \begin{pmatrix} a & -c \\ -c & a \end{pmatrix}. \tag{7.3}$$

Its eigenvalues $n(a - c)$ and $n(a + c)$ are positive. The predictive law is

$$s_n G_n = \mathrm{LN}\left(\frac{c}{a}\mu^s, \frac{(c/a)^2 + 1/a}{n}\right). \tag{7.4}$$

Thus the evaluator depends on the represented model value. Physical references and evaluators are obtained by pushing both coordinates through the common local frame.

Proposition 7.1 (Local energy and metric). For every $n \geq 1$ and every pair of real means, let

$$C = a - 1 - \frac{1}{2}\log(a^2 - c^2), \qquad \mathcal{U}(\mu^q, \mu^s) = \frac{a}{2}\big((\mu^q)^2 + (\mu^s)^2\big) - c\mu^q\mu^s. \tag{7.5}$$

Then

$$E_{0,n} = D_{\mathrm{KL}}(q_n \otimes s_n \| r_n G_n) = C + n\mathcal{U}, \qquad g_n = n\big((d\mu^q)^2 + (d\mu^s)^2\big). \tag{7.6}$$

Proof. The logarithm is a measurable bijection, so it preserves KL divergence. The recognition law in log coordinates has covariance I_2/n, while the reference has covariance A^{-1}/n. The Gaussian KL formula gives

$$E_{0,n} = \frac{1}{2}\left[\operatorname{tr} A - 2 - \log\det A + n(\mu^q, \mu^s)A(\mu^q, \mu^s)^{\mathsf{T}}\right], \tag{7.7}$$

which is the stated expression. The score for either mean is $n(Z - \mu)$, whose squared expectation is n. The product law has zero cross score expectation, giving the displayed Fisher metric. □

For independent synchronized children of multiplicities n_j, set $n_I = \sum_{j\in I} n_j$. Their retained reference samples are

$$k_I = \prod_{j\in I} k_j^{n_j/n_I}, \qquad m_I = \prod_{j\in I} m_j^{n_j/n_I}. \tag{7.8}$$

Weighted log averages have variance $\sum_j (n_j/n_I)^2/n_j = 1/n_I$. The same calculation for the joint reference gives $r_{n_I}G_{n_I}$. Conditional log observation noise has variance $\sum_j (n_j/n_I)^2/(an_j) = 1/(an_I)$, so the evaluator itself is inherited. Physical samples are first returned through U_j^{-1} and the retained sample is expressed in a chosen parent frame U_I.

7.2 Messages and state-dependent scalar links

Define a normalized message kernel by

$$B_n(k, dw) = \mathrm{LN}(\log k, 1 - 1/n)(dw), \tag{7.9}$$

interpreting $B_1(k, dw) = \delta_k(dw)$. It restores unit log variance: $q_nB_n = \mathrm{LN}(\mu^q, 1)$ and $s_nB_n = \mathrm{LN}(\mu^s, 1)$. These messages give all multiplicities a common comparison family.

Model prediction also commutes with this interface. Put $h = (c/a)^2 + 1/a$ and define $B_n^h(k, dw) = \mathrm{LN}(\log k, h(1 - 1/n))(dw)$, again using the identity at $n = 1$. Both routes in the kernel identity

$$G_nB_n^h = B_nG_1 \tag{7.10}$$

have conditional log mean $(c/a)\log m$ and variance $h - (c/a)^2/n$. Integrating against s_n gives

$$(s_nB_n)G_1 = (s_nG_n)B_n^h = \mathrm{LN}\left(\frac{c}{a}\mu^s, h\right). \tag{7.11}$$

Thus a parent can emit the original predictive message by either normalized route.

For each sector $\nu \in \{q, s\}$, choose a fixed $0 < \chi_\nu < 1$ and set

$$L_{ij}^\nu = \chi_\nu(\mu_i^\nu - \mu_j^\nu), \qquad \Theta_{ij}^\nu = U_i \exp(L_{ij}^\nu)U_j^{-1}. \tag{7.12}$$

Under a local change of frame $U_i \mapsto h_i U_i$, the physical transport changes to $h_i \Theta^\nu_{ij} h_j^{-1}$. The reference means and L^ν_{ij} remain unchanged. This supplies nonidentity, law-dependent positive scalings when means differ. The chosen scalar links have trivial cycle product; no curvature claim is needed.

Transport of the reference message from j shifts its log mean to $\mu^\nu_j + \chi_\nu(\mu^\nu_i - \mu^\nu_j)$. Consequently the directed comparison cost

$$d^\nu_{ij} = \frac{\sqrt{2D_{\mathrm{KL}}\big(\mathrm{LN}(\mu^\nu_i, 1) \| (\exp L^\nu_{ij})_{\#}\, \mathrm{LN}(\mu^\nu_j, 1)\big)}}{1 - \chi_\nu} = |\mu^\nu_i - \mu^\nu_j| \tag{7.13}$$

is gauge invariant. The equality includes the dependence of the link on both endpoint laws. For unequal means, differentiating its log-mean residual gives

$$\partial_{\mu^\nu_i}\big[\mu^\nu_i - \mu^\nu_j - L^\nu_{ij}\big] = 1 - \chi_\nu. \tag{7.14}$$

This factor cancels the denominator in (7.13). Holding the link fixed while differentiating would give a different dynamics.

7.3 The continuous subgradient dynamics

For multiplicities n_i with total $N = \sum_i n_i$, choose $\lambda > 0$ and the potential

$$\mathcal{F}(\boldsymbol{\mu}) = \sum_i n_i\big[C + \mathcal{U}(\mu^q_i, \mu^s_i)\big] + \lambda \sum_{i<j} n_i n_j \big(|\mu^q_i - \mu^q_j| + |\mu^s_i - \mu^s_j|\big). \tag{7.15}$$

All contacts are fixed and present. If source rows are recorded, take $\beta_{ij} = \gamma_{ij} = n_j/N$, including self entries. Summing child entries gives exactly the same parent row. Self comparison costs vanish, while their row mass records the corresponding contact multiplicity. There is no Gibbs row adaptation. The local term has the stated KL origin; no fixed sample-model ELBO identity is asserted for the entire potential.

With $M = \mathrm{diag}(n_1, n_1, \ldots)$, define the flow on $\mathbb{R}^{2r}$, for r current agents, by

$$M\dot{\boldsymbol{\mu}} \in -\partial\mathcal{F}(\boldsymbol{\mu}). \tag{7.16}$$

Solutions are absolutely continuous in the external time $t \geq 0$ and satisfy this inclusion almost everywhere. There are no boundaries or stochastic driving terms. Both law sectors have the same basic mobility; the factor $1/n_i$ comes from their inherited Fisher metric. In coordinates,

$$\dot{\mu}^q_i = -a\mu^q_i + c\mu^s_i - \lambda \sum_{j \neq i} n_j \sigma^q_{ij}, \tag{7.17}$$

$$\dot{\mu}^s_i = -a\mu^s_i + c\mu^q_i - \lambda \sum_{j \neq i} n_j \sigma^s_{ij}, \tag{7.18}$$

where $\sigma^\nu_{ij} = -\sigma^\nu_{ji}$ belongs to $\mathrm{Sign}(\mu^\nu_i - \mu^\nu_j)$ and $\mathrm{Sign}(0) = [-1, 1]$.

Lemma 7.2 (Uniqueness and persistence). Two solutions of (7.16) with the same masses obey

$$\|\boldsymbol{\mu}(t) - \widetilde{\boldsymbol{\mu}}(t)\|_M \le e^{-(a-c)t}\|\boldsymbol{\mu}(0) - \widetilde{\boldsymbol{\mu}}(0)\|_M. \tag{7.19}$$

On any interval where a solution exists, synchronized unit-mass children remain synchronized in both sectors.

Proof. The quadratic part is $(a-c)$-strongly convex in the M metric and the absolute-value part is convex. For subgradients $\xi, \widetilde{\xi}$ at two configurations, monotonicity gives

$$\langle \xi - \widetilde{\xi}, \boldsymbol{\mu} - \widetilde{\boldsymbol{\mu}} \rangle \ge (a-c)\|\boldsymbol{\mu} - \widetilde{\boldsymbol{\mu}}\|_M^2. \tag{7.20}$$

Substitution from (7.16) and integration proves (7.19). Swapping two equal unit-mass children preserves the initial configuration, potential, and metric. Uniqueness therefore preserves their equality. This argument concerns the full two-sector system, not a restricted scalar equation. □

7.4 An invariant class with explicit global solutions

Theorem 7.3 (Finite sticking on a common-offset family). Let all initial masses be one and suppose

$$\mu_i^q(0) = u_i^0 + v_0, \qquad \mu_i^s(0) = u_i^0 - v_0, \tag{7.21}$$

where the u_i^0 are distinct real numbers and $v_0 \neq 0$. There is a unique global solution of the full inclusion for these initial data. Both sectors have the form

$$\mu_i^q = u_i + v, \qquad \mu_i^s = u_i - v, \qquad v(t) = v_0 e^{-(a+c)t}. \tag{7.22}$$

Blocks formed by equal u_i persist and follow the mass-weighted version of the same equation.

Proof. Set $\kappa = a - c > 0$. Equal sector orderings reduce (7.18) to

$$\dot{u}_i \in -\kappa u_i - \lambda \sum_{j \neq i} n_j \operatorname{Sign}(u_i - u_j), \qquad \dot{v} = -(a+c)v. \tag{7.23}$$

Introduce the increasing auxiliary time and rescaled positions

$$\tau(t) = \frac{e^{\kappa t} - 1}{\kappa}, \qquad u_i(t) = e^{-\kappa t} z_i(\tau(t)). \tag{7.24}$$

Since $\dot{\tau} = e^{\kappa t} > 0$, direct differentiation gives

$$z_i' \in -\lambda \sum_{j \neq i} n_j \operatorname{Sign}(z_i - z_j). \tag{7.25}$$

An ordered block has constant velocity $\lambda(M_{\text{right}} - M_{\text{left}})$ between collisions. Adjacent blocks approach with relative speed $\lambda(n_I + n_J) > 0$. At their collision, join their masses, retain the common position, and use the new velocity $\lambda(M_{\text{right}} - M_{\text{left}})$. This equals the mass-weighted average of their velocities just before collision. Choosing zero internal subgradients verifies the original inclusion for every constituent child after collision. Each event reduces the number of blocks, so at most

$N-1$ events occur, and the resulting piecewise-linear curve exists for all $\tau \geq 0$. Reversing (7.24) constructs a global full-system solution. Lemma 7.2 makes it unique. □

The class (7.21) has a common difference between its two sector means. This is an initial-data restriction, not an imposed constraint force. It is a proper submanifold of the full parameter space. The two laws of each agent remain distinct at every finite time because $v(t) \neq 0$. The auxiliary τ is a monotone reparameterization of this solution, not a geometric or operational clock.

7.5 Two formation stages without a supplied partition

Theorem 7.4 (Four agents form two parents and then one). In theorem 7.3, take four agents with the unordered initial set

$$\{u_i^0\} = \{-3d, -2d, 2d, 3d\}, \qquad d > 0. \tag{7.26}$$

Two size-two blocks form at t_1 and remain distinct until $t_2 > t_1$, where they fuse under their inherited equations, with

$$t_1 = \frac{1}{\kappa} \log\left(1 + \frac{\kappa d}{2\lambda}\right), \qquad t_2 = \frac{1}{\kappa} \log\left(1 + \frac{5\kappa d}{4\lambda}\right). \tag{7.27}$$

Proof. In sorted order, the initial z velocities are $(3\lambda, \lambda, -\lambda, -3\lambda)$. The two outer gaps close at $\tau_1 = d/(2\lambda)$, leaving parent positions $-3d/2$ and $3d/2$. Their inherited masses are two and their velocities are 2λ and -2λ. Their gap closes after another $3d/(4\lambda)$, giving $\tau_2 = 5d/(4\lambda)$. Inverting (7.24) gives (7.27). □

Exact reflection symmetry is unnecessary. For four sorted initial positions, let their adjacent gaps be $g_1, g_2, g_3 > 0$, and put $H = \max(g_1, g_3)$. If

$$D = g_2 + \frac{g_1 + g_3}{2} - 2H > 0, \tag{7.28}$$

both outer pairs form by $\tau_* = H/(2\lambda)$. To check that no middle collision intervenes, first note that $D > 0$ implies $g_2 > H$, so a middle collision cannot occur first. Suppose $g_1 \leq g_3$. At the first outer collision the middle gap is $g_2 - g_1$; afterward it decreases at speed 3λ until the other outer pair collides at $g_3/(2\lambda)$. Its value there is $D > 0$. The case $g_3 \leq g_1$ is the reflected calculation. The two parent gap then decreases at speed 4λ, and the final time is

$$\tau_{\text{final}} = \tau_* + \frac{D}{4\lambda} = \frac{2g_2 + g_1 + g_3}{8\lambda}. \tag{7.29}$$

Condition (7.28) is strict and defines an open set relative to the ordered common-offset family. Taking all label permutations gives an admitted class with no assigned labels for blocks. The initial laws are unequal across agents, and the interaction rule contains no partition. Membership is selected by attained equality.

7.6 The inherited handoff and its scope

At a collision, restrict a block I to its common means. Summing its local energies and pulling back its metric gives

$$\sum_{i\in I} n_i(C+\mathcal{U}) = n_I(C+\mathcal{U}) = E_{0,n_I} + (n_I-1)C, \qquad g_I = n_I\big((d\mu_I^q)^2 + (d\mu_I^s)^2\big). \tag{7.30}$$

Every external pair term sums to $n_I n_J|\mu_I^\nu - \mu_J^\nu|$. Zero internal subgradients therefore lift each solution of the inherited equation to a fine solution. Uniqueness identifies that lift with the original trajectory. This also proves persistence for inherited masses by expanding a mass into synchronized unit children. With R retaining block means, masses, and frames, the corresponding parameter flows satisfy the following identity. For a fixed formed partition, keep both component index sets fixed even when further components coincide; a later merger is a separate retention composed with R. Then

$$R \circ \Phi_t = \Phi_t^c \circ R \tag{7.31}$$

on every synchronized stratum reached by the construction, for all future times. Weighted geometric averaging is associative, so later collisions give the same retained laws as direct averaging. The state is continuous at a collision; the velocity changes through the inherited mass sum.

Along the constructed continuous trajectory, ordinary differentiation on each interval between collisions gives

$$\frac{d\mathcal{F}}{dt} = -\sum_i n_i\big((\dot\mu_i^q)^2 + (\dot\mu_i^s)^2\big) \quad \text{almost everywhere.} \tag{7.32}$$

The objective is continuous at collisions when the constants above are retained. Comparing with the stationary configuration having every mean zero in (7.19) proves convergence to that configuration for the admitted initial data. No discrete numerical update has been specified, so this calculation makes no discrete-step assertion.

Sample retention is genuinely noninjective: different positive sample vectors have the same geometric mean. For n synchronized unit children, their log samples conditional on the retained log mean $\bar Z$ have law

$$(Z_1,\dots,Z_n) \mid \bar Z \sim N\left(\bar Z\mathbf{1}, I_n - \frac{\mathbf{1}\mathbf{1}^{\mathsf T}}{n}\right). \tag{7.33}$$

The residual is independent of the mean: their joint Gaussian cross covariance is zero. Exponentiation gives a normalized joint reconstruction kernel. Its one-child marginal is (7.9). Independent residuals for all children would be incorrect because the off-diagonal conditional covariances are $-1/n$. Generative joint samples similarly use the covariance of $(\log m, \log k)$ under $s_1 G_1$; their residual kernel is determined by that covariance, not by the recognition product law. The local KL residual $(n-1)C$ records the corresponding state-independent information loss under the reference comparison.

Parameter retention has a narrower meaning. On an exactly synchronized stratum, child means are redundant copies; retaining their common value does not identify different nonsynchronized configurations. Away from that stratum, a block average need not determine its external sign forces. Equation (7.31) makes no claim about such hidden parameter fibers.

The admitted boundary protocol draws fresh messages through B_n, fresh predictive messages through (7.11), or complete fresh represented samples through the joint reconstruction. To type an accessible stored readout, augment the complete population configuration by $o \in \mathsf{O} = (0, \infty)$ and set $\rho(\boldsymbol{\mu}, \boldsymbol{n}, o) = o$. At declared epochs, the transition Γ advances the deterministic parameter flow and overwrites o using the selected normalized scalar output kernel. Between epochs, o is fixed, and it never enters the mean equation. Retention copies this accessible o exactly; it does not claim access to cached private residual samples. For vector readouts the same definition uses the corresponding finite product space and joint kernel. Equal parameter trajectories and conditional output kernels imply equal joint accessible readout laws for every finite sequence of such queries, by multiplying successive conditional kernels. Reusing persistent latent samples or introducing record feedback requires another state specification.

Open problem 7.5 (Formation under smooth peer KL dynamics). The finite mergers above depend on a constitutive nonsmooth interaction. For a locally Lipschitz vector field with an invariant synchronization manifold, an interior trajectory cannot first reach that manifold in finite time: local backward uniqueness would identify its preceding segment with the manifold trajectory. Establishing recursive formation for the smooth KL dynamics therefore requires a different formation criterion or an additional mechanism, together with its own inherited closure calculation.

Chapter 8

Sufficient States for Smooth KL Evolution

The smooth potential of chapter 4 can admit exact reductions while its agents remain different. Such a reduction requires control of the conditional information that enters the retained force. We first identify that information, then construct a closed state made of general probability laws. The construction keeps the original KL potential and unit Fisher mobilities. Its conditional reconstruction channels are additional constitutive data within that potential; they are not a consequence of an arbitrary law-dependent transport. They provide a sufficient-state architecture, but do not yet explain selection of agent groups.

8.1 The force that survives marginalization

Let Q be a positive law on a standard Borel space, let C be a measurable statistic, and write $\mu = C_{\#}Q$. Suppose a differentiable functional has variational derivative h_Q and Fisher velocity

$$\dot{Q} = -Q(h_Q - \mathbb{E}_Q h_Q). \tag{8.1}$$

All identities below are immediate for finite positive tables. On general spaces we assume a dominated differentiable path, square-integrable scores, and integrability that permits the displayed conditional expectations and differentiations. Statements about flow inheritance are on an interval where the fine and proposed retained equations have unique regular solutions. They do not assert unrestricted existence on every space of probability measures.

Proposition 8.1 (Conditional mean force). The retained velocity is

$$\dot{\mu} = -\mu\left(\mathbb{E}_Q[h_Q \mid C] - \mathbb{E}_Q h_Q\right). \tag{8.2}$$

Consequently, an autonomous velocity for μ exists precisely when the centered conditional mean on the right is determined by μ on the admitted class.

Proof. For every bounded measurable f, integrate $f(C)$ against eq. (8.1) and condition on C. This gives the weak form of eq. (8.2). Two laws with the same marginal have the same marginal velocity if and only if their centered conditional forces agree almost everywhere under that marginal. With the stated uniqueness hypothesis, equality of these velocities gives equality of retained flows. □

For $Q(dz, dv) = \mu(dz)\kappa_z(dv)$ and a fixed reference $P(dz, dv) = \nu(dz)\pi_z(dv)$, the local KL force illustrates the missing information:

$$\mathbb{E}_Q[\log(dQ/dP) \mid z] = \log \frac{d\mu}{d\nu}(z) + H(z), \qquad H(z) = D_{\mathrm{KL}}(\kappa_z \| \pi_z). \tag{8.3}$$

Thus a scalar total conditional loss does not usually determine the retained force. Its dependence on z matters. The Fisher metric makes the same separation: if a score is $f(z, v) = a(z) + b(z, v)$, with $\mathbb{E}_\mu a = 0$ and $\mathbb{E}_{\kappa_z} b = 0$, then

$$g_Q(fQ, fQ) = \int a(z)^2 \mu(dz) + \int \left[\int b(z, v)^2 \kappa_z(dv) \right] \mu(dz). \tag{8.4}$$

The cross term vanishes by conditional centering. This calculation is consistent with the statistical behavior of Fisher information under sufficient statistics [2]. Dynamical sufficiency asks a different question: it preserves the specified evolution and responses, and need not preserve inference about every discarded parameter.

8.2 Conditional reconstruction within the original potential

For each agent take $\mathsf{K}_i = \mathsf{Z}_i \times \mathsf{V}_i$ and $\mathsf{M}_i = \mathsf{W}_i \times \mathsf{U}_i$, all standard Borel. Use positive public densities against fixed reference probabilities, so conditional versions share common full-measure public sets. Disintegrate

$$q_i(dz, dv) = \mu_i(dz)\kappa_{i,z}(dv), \qquad s_i(dw, du) = \sigma_i(dw)\lambda_{i,w}(du). \tag{8.5}$$

Neither conditional need be independent of its retained coordinate. Choose fixed private reference probabilities A_i^q, A_i^s and a fixed normalized public reference

$$R_i^c(dz, dw) = r_i^c(dw)G_i^c(w, dz), \qquad R_i = R_i^c \otimes A_i^q \otimes A_i^s, \tag{8.6}$$

with the product reordered into state and model coordinates. The evaluator G_i^c can couple μ_i and σ_i. Independence of the private reference factors is an assumption of this construction, not a restriction of the general theory.

Let $B_{ij}^q[\boldsymbol{\mu}] : \mathsf{Z}_j \rightsquigarrow \mathsf{Z}_i$ be normalized channels, with the regularity needed for the original variation. Define the full channels by

$$\mathcal{K}_{ij}^q[\mathbf{q}]((z_j, v_j), d(z_i, v_i)) = B_{ij}^q[\boldsymbol{\mu}](z_j, dz_i)\kappa_{i,z_i}(dv_i). \tag{8.7}$$

The model channel uses $B_{ij}^s[\boldsymbol{\sigma}]$ and λ_{i,w_i} in the same way. Source sets may exclude self edges; if self comparisons are included, use the primitive identity convention separately. Public channels may be informative and nonconstant functions of the retained laws. For example, on the public space $\{-1, 1\}$, the channel $B_\mu(z, \{z'\}) = (1 + \rho(\mu)zz')/2$, with $\rho(\mu) = (1 + (\mathbb{E}_\mu Z)^2)/4$, is strictly positive, transmits source information, and varies with the law. Public/private maps and reference factors are declared interface data; carrying them through a bimeasurable change of presentation preserves these constructions.

Proposition 8.2 (Exact cancellation of private peer information). Where the divergences are defined,

$$\begin{aligned} q_j \mathcal{K}^q_{ij} &= (\mu_j B^q_{ij})\kappa_i, \\ D_{\mathrm{KL}}(q_i \| q_j \mathcal{K}^q_{ij}) &= D_{\mathrm{KL}}(\mu_i \| \mu_j B^q_{ij}). \end{aligned} \tag{8.8}$$

This is an identity on the full admitted class, including correlated q_i. Its derivative includes the sender, receiver, and channel derivatives.

Proof. Integration over the source conditional gives the first identity. The compared laws have the same receiver conditional $\kappa_{i,z}$, so their density ratio depends only on z. Integrating its logarithm proves the second identity. Differentiating the resulting functional identity on any regular path proves the derivative statement. Holding κ_i fixed only in the denominator would violate that identity and would define another force. □

Let $\mathcal{E}_c$ denote exactly eq. (4.4) applied to $\mu_i, \sigma_i, R^c_i, B^q_{ij}, B^s_{ij}$, with the same rows, source priors, and temperatures. Define

$$\begin{aligned} \ell^q_{i,z} &= \log(d\kappa_{i,z}/dA^q_i), \quad H^q_i(z) = \int \ell^q_{i,z} d\kappa_{i,z}, \\ \ell^s_{i,w} &= \log(d\lambda_{i,w}/dA^s_i), \quad H^s_i(w) = \int \ell^s_{i,w} d\lambda_{i,w}. \end{aligned} \tag{8.9}$$

In the following formulas, assume these log densities are bounded on the time interval in question. This convenient sufficient condition allows arbitrary dominated law shapes and justifies the exponential moments and derivatives used below. We do not impose a Gaussian or finite-dimensional exponential family.

Theorem 8.3 (Exact marginal and conditional dynamics). For the architecture above the original energy is

$$\mathcal{E} = \mathcal{E}_c + \sum_i \int H^q_i(z)\mu_i(dz) + \sum_i \int H^s_i(w)\sigma_i(dw). \tag{8.10}$$

Write $h^q_i = \delta\mathcal{E}_c/\delta\mu_i$ and $h^s_i = \delta\mathcal{E}_c/\delta\sigma_i$, including every interaction role. The full unit-mobility Fisher dynamics gives

$$\begin{aligned} \dot{\mu}_i &= -\mu_i \left[h^q_i + H^q_i - \mathbb{E}_{\mu_i}(h^q_i + H^q_i)\right], \\ \partial_t \ell^q_{i,z} &= H^q_i(z) - \ell^q_{i,z}. \end{aligned} \tag{8.11}$$

The equations for σ_i, ℓ^s_i are obtained by exchanging sectors. Source rows evolve by their original Fisher equations with the public costs, or may be exactly optimized as in proposition 4.2.

Proof. Apply the relative-entropy chain rule to the local reference in eq. (8.6) and use eq. (8.8). This proves eq. (8.10). Up to a constant independent of (z, v), the full derivative with respect to q_i is $h^q_i(z) + \ell^q_{i,z}(v)$. This can be checked directly from the log-density ratio in the local KL; every peer contribution is a functional of μ alone. Conditional averaging gives the first equation of eq. (8.11). Subtract its logarithmic derivative from the full logarithmic derivative of $q_i = \mu_i\kappa_i$. The public score and global centering cancel, leaving the second equation. The private terms contain no source rows, so their variations do not change the row equations. □

The dependence of H_i^q on z is the only private contribution to the retained belief velocity in this architecture. It cannot generally be discarded. Its evolution will require more than its present value.

8.3 An exact law-valued completion

Definition 8.4 (Conditional log-density spectrum). For a private reference A and conditional density κ_z, define

$$\xi_z = (\ell_z)_{\#}A, \qquad \ell_z = \log(d\kappa_z/dA). \tag{8.12}$$

This probability law records how much reference mass carries each log-density value. It forgets which private point carries that value. It obeys

$$\int e^r \xi_z(dr) = 1, \qquad H(z) = \int r e^r \xi_z(dr). \tag{8.13}$$

Theorem 8.5 (Closed conditional-spectrum evolution). For theorem 8.3, the state consisting of all μ_i, σ_i, conditional spectra ξ_i^q, ξ_i^s, and source rows is autonomous. Each spectrum satisfies, in weak form,

$$\frac{d}{dt}\int f(r)\xi_z(dr) = \int f'(r)(H(z) - r)\xi_z(dr), \tag{8.14}$$

for continuously differentiable test functions on the bounded range in use. For a time interval t starting at zero, its exact solution is

$$\begin{aligned} a &= e^{-t}, & Z_z(a) &= \int e^{ar}\xi_z^0(dr), \\ \xi_z^t &= (r \mapsto ar - \log Z_z(a))_{\#}\xi_z^0. \end{aligned} \tag{8.15}$$

The entropy profile obeys

$$\dot{H}(z) = -\operatorname{Var}_{e^r\xi_z}(r). \tag{8.16}$$

Proof. The reference A is fixed. Differentiate $\int f(\ell_z(v))A(dv)$ and substitute the conditional equation in eq. (8.11); this proves eq. (8.14). Direct differentiation shows that

$$\frac{d\kappa_z(t)}{dA}(v) = \frac{\exp(a\ell_z^0(v))}{Z_z(a)} \tag{8.17}$$

solves the normalized conditional equation. Pushing A through its logarithm gives eq. (8.15). Setting $f(r) = re^r$ in the weak equation gives $\dot{H} = H(H+1) - \int (r^2+r)e^r\xi_z(dr)$, which is eq. (8.16). The probability $e^r\xi_z(dr)$ is normalized by eq. (8.13). All private contributions to eq. (8.11) are now functions of the retained spectra. The remaining scores and row equations depend only on retained public data, proving closure. □

This is a configuration retention. Its spectra can be infinite-dimensional, and there is no claim that they give a fixed finite number of parameters. If represented as a sample statistic, the map $v \mapsto \ell_z(v)$ depends on the current law; its motion is exactly the term differentiated in eq. (8.14). It must not be frozen while the law changes. The autonomous equations are inherited equations.

They have not been identified with a new unrestricted Fisher gradient on probability laws over the spectrum coordinate.

Corollary 8.6 (Different internal conditionals with the original public flow). Suppose that, for each agent and each sector separately, the initial spectrum is independent of the public coordinate. Then this property persists, and the public laws and source rows follow exactly the original coarse KL/Fisher system generated by $\mathcal{E}_c$.

Proof. The same initial spectrum gives the same solution of eq. (8.15) at every public coordinate. Consequently $H_i^q(z)$ is constant in z, and $H_i^s(w)$ is constant in w. Each constant disappears under its own marginal centering in eq. (8.11). □

Equal spectra do not require equal conditionals. With a private reference uniform on two points, the conditionals $(3/4, 1/4)$ and $(1/4, 3/4)$ share

$$\xi = \tfrac{1}{2}\delta_{\log(3/2)} + \tfrac{1}{2}\delta_{\log(1/2)}. \tag{8.18}$$

A public bit may select either conditional, producing correlation between public and private variables. Another law with the same public marginal may use the first conditional at both public values and be a product. Both have the same public trajectory in this architecture, although their internal laws differ. Different agents may have different public laws and different spectra. No equality between agents is imposed.

This class is invariant, but it is not reached for the first time at finite interior time. For any $a > 0$, the spectral update in eq. (8.15) is invertible on its image. Indeed, if ξ^t is given, the initial log value is recovered by the map

$$y \longmapsto \frac{y}{a} - \log \int e^{r/a} \xi^t(dr). \tag{8.19}$$

Substituting eq. (8.15) and using $\int e^r \xi^0(dr) = 1$ proves the assertion. Equal spectra at a finite time therefore had to be equal initially. The same inversion applied to eq. (8.17) proves that distinct conditionals cannot first become equal at finite time. Neither fact forbids an autonomous quotient that keeps the unequal spectra.

8.4 Boundary records and feedback

Public-only response kernels descend immediately: the channels B_{ij}^q, B_{ij}^s and evaluator G_i^c are determined by the retained public laws. A private-sensitive response needs more information. In eq. (8.18), let the probabilities of record 1 at the two private points be $(3/4, 1/4)$. The two conditionals yield record probabilities $5/8$ and $3/8$. After record 1, their private posteriors are $(9/10, 1/10)$ and $(1/2, 1/2)$. Thus an unmarked spectrum can fail both the boundary test and the next-update test.

There is an exact completion for a specified family of records. Fix a measurable mark $T : \mathsf{V} \to \mathsf{D}$, with D standard Borel, through which every allowed private readout factors. Retain

$$\eta_z = (\ell_z, T)_{\#} A. \tag{8.20}$$

The correlation between the log-density value and the mark is retained. For a normalized record kernel with strictly positive density $L(o \mid z, d)$, assume the following integrals are finite and nonzero and each updated law stays in the admitted regular domain. A bounded log likelihood is a sufficient local condition for the conditional log-density bounds. Put

$$\begin{aligned} N_z(o) &= \int e^r L(o \mid z, d)\eta_z(dr, dd), \\ N(o) &= \int N_z(o)\mu(dz). \end{aligned} \tag{8.21}$$

Proposition 8.7 (Exact marked feedback). The record density is $N(o)$. Exact conditioning of this sector gives the closed updates

$$\begin{aligned} \mu^+(dz) &= \frac{N_z(o)}{N(o)}\mu(dz), \\ \eta_z^+ &= \big((r, d) \mapsto (r + \log L(o \mid z, d) - \log N_z(o), d)\big)_{\#}\eta_z. \end{aligned} \tag{8.22}$$

Between records, eqs. (8.14) and (8.15) transport the first coordinate of η_z and leave its mark fixed. Hence the marked state determines any finite sequence of these records, feedback updates, and original smooth-flow intervals.

Proof. Since $d\kappa_z = e^{\ell_z} dA$, integration of the likelihood gives eq. (8.21). Bayes' rule gives the displayed marginal update and $d\kappa_z^+/dA = e^{\ell_z} L(o \mid z, T)/N_z(o)$. Taking the logarithm and pushing A gives the marked update. The fine reference remains fixed. The conditional equation after the update is therefore the same equation with this new conditional law. Inductively, the inherited flow gives the next retained state, the inherited record kernel gives the same next-record law, and the inherited update gives the same posterior retained state. Actions selected from retained histories preserve this induction. Any reference or likelihood parameters changed by records must also be retained and updated by a rule that depends only on this state. □

The update is sectorwise. A product of separate belief and model likelihoods preserves the recognition restriction $q_i \otimes s_i$. A likelihood jointly coupling the two sectors need not preserve that product; exact joint conditioning would require a joint recognition state, or a separately declared projection. Marked feedback can also destroy equality of unmarked spectra across public coordinates. The augmented marked state still closes, but the simpler conclusion of corollary 8.6 then need not hold.

For an explicit retained sample law, use

$$Q^c(dz, dr, dd) = \mu(dz)e^r\eta_z(dr, dd). \tag{8.23}$$

It is normalized and reproduces every declared marked record. It is the pushforward under the current, law-dependent statistic $(z, v) \mapsto (z, \ell_z(v), T(v))$. Different assignments of density values to private points within a mark can give the same retained state. If the permitted records distinguish every private point, the mark may have to retain that point, eliminating this compression. The boundary protocol therefore fixes what can be forgotten. Predictive-state constructions likewise

make sufficiency relative to future observations [20]; the formulas here specify the state and update for this particular KL architecture.

8.5 What is still required for formation

Positive source temperature does not give exact isolation by itself. For optimized rows, every allowed finite-cost edge has positive weight. For a dynamic row, set

$$a_{jk} = \log \frac{\beta_j/\pi_j}{\beta_k/\pi_k}. \tag{8.24}$$

The Fisher row equation from eq. (4.4) gives

$$\dot{a}_{jk} = -(D_j - D_k) - T_q a_{jk}. \tag{8.25}$$

Multiplying by $e^{T_q t}$ and integrating shows that finite cost differences keep every such ratio finite on a regular finite interval. Initially positive weights therefore stay positive there. Uniformly bounded cost differences also bound these ratios for all time and prevent asymptotic disappearance of an allowed weight. A formed collective may nevertheless keep nonzero boundary interactions; isolation is not required for inheritance.

The construction above compresses internal conditional information. It does not combine several distinct public agent laws into one smaller collective state. Retaining their complete labeled tuple and calling it a parent would leave that aggregation problem untouched. Nor does recognizing a persistent source-preference group establish a compatible collective retention. The distinction between a coarse observable and an autonomous macroscopic evolution is also expressed by flow intertwining in Barnett and Seth [4].

Open problem 8.8 (Endogenous aggregation with conditional information retained). Within the original smooth potential, construct an initially unpartitioned population whose evolution selects persistent groups, together with a noninjective collective retention that forgets some of their distinct public degrees of freedom while determining all inherited forces, channels, and declared feedback. The conditional-spectrum state may contribute part of that retention. Its existence for each agent does not establish the collective construction, and no subsequent formation stage follows from it alone.

Chapter 9

State-Controlled Erasure and Persistent Groups

There is a conditional first-stage construction inside the original smooth KL potential. A prescribed channel can lose its dependence on some internal laws when a dynamically evolving gate coordinate enters a flat part of that channel. The remaining agents can be distinct, their preferred groups can change, and the retained dynamics can then become exactly autonomous. The mechanism is an engineered constitutive choice. It supplies neither a general explanation of sufficient interfaces nor a second formation stage. The general laws of the preceding chapter remain the framework; the existence example below uses an explicitly invariant family of positive finite laws.

9.1 A homogeneous population with internal gate coordinates

Write $b_x(k) = (1 + kx)/2$ on $\{-1, 1\}$, set $d(x, z) = D_{\mathrm{KL}}(b_x \| b_z)$, and put $\psi(x) = d(x, 0)$. Four agents carry

$$q_i = b_{x_i} \otimes b_{z_i} \otimes b_{h_i}, \qquad s_i = b_{y_i} \otimes b_{w_i} \otimes b_{\widehat{h}_i}. \tag{9.1}$$

The first coordinate is public, the second private, and the third controls a channel. Their unit Fisher metric has diagonal entries $1/(1 - u^2)$ for each mean u. The product preparation within each sector is an assumption. It is invariant under the full table dynamics specified below, rather than a projection imposed after each step.

Take $J = \log 3$ and $h_* = -1/2$. Use the proper reference $r = b_0 \otimes b_0 \otimes b_{h_*}$ and normalized evaluator

$$G_J(k \mid m) = \frac{e^{Jkm}}{2 \cosh J} \tag{9.2}$$

on the public coordinates, with independent private output b_0 and gate output b_{h_*}. The local term is exactly

$$\begin{aligned} E_{0,i} = {} & \psi(x_i) + \psi(y_i) - J x_i y_i + \log \cosh J \\ & + \psi(z_i) + \psi(w_i) + d(h_i, h_*) + d(\widehat{h}_i, h_*). \end{aligned} \tag{9.3}$$

All agents have the same reference and evaluator. Every nonself agent is an allowed source, with uniform prior and fixed temperature $T > 0$. We use the exact Gibbs-reduced version of the original potential. The bits are reference coordinates: a change of presentation carries the reference laws, channels, and readout maps with it, so the KL and Fisher covariance statements apply.

Define $\chi(r) = 0$ for $r \le 0$, $\chi(r) = 1$ for $r \ge 1$, and, for $0 < r < 1$,

$$\chi(r) = \frac{e^{-1/r}}{e^{-1/r} + e^{-1/(1-r)}}. \tag{9.4}$$

This increasing function is smooth, all its derivatives match at the endpoints, and $M_\chi = \max |\chi'| < \infty$. Fix $h_0 = 1/4$ and $\delta > 0$, and set

$$H_q = \frac{1}{4}\sum_i h_i, \quad H_s = \frac{1}{4}\sum_i \widehat{h}_i, \quad \alpha(H) = \chi((H - h_0)/\delta). \tag{9.5}$$

The gate is a nonlocal function of internal laws. Its threshold and flat region are constitutive data. There is no external time schedule.

Choose gains $0 < \varepsilon, \varrho < 1$. On a belief edge $j \to i$, let the public output kernel be $(1-\varepsilon)b_{x_i} + \varepsilon\delta_{k_{\mathrm{pub}}}$, and independently draw private output from $b_{\alpha(H_q)\varrho z_j}$ and gate output from b_{h_*}. Its compared law is

$$\nu_{ij}^q = b_{(1-\varepsilon)x_i + \varepsilon x_j} \otimes b_{\alpha(H_q)\varrho z_j} \otimes b_{h_*}. \tag{9.6}$$

This normalized channel is strictly positive on the interior and depends smoothly on the current marginal means. It transmits public samples and uses the source's private law. The model channel exchanges (x, z, h) with $(y, w, \widehat{h})$. Define

$$D_\varepsilon(x, v) = d(x, (1-\varepsilon)x + \varepsilon v). \tag{9.7}$$

The original peer cost becomes

$$D_{ij}^q = D_\varepsilon(x_i, x_j) + d(z_i, \alpha(H_q)\varrho z_j) + d(h_i, h_*). \tag{9.8}$$

No fusion cost has been added.

The definitions extend to arbitrary positive tables on each three-bit space by taking the indicated marginal means. At a product table, the derivative of the receiver entropy is additive in its coordinates. Each compared law is a product, and every channel derivative is a function of a single coordinate because the channel parameters depend on marginal means. The local evaluator also gives additive scores within each sector. The full Fisher score is consequently a sum of coordinate scores on eq. (9.1). Differentiating a product shows that the full table flow preserves this family and induces the stated Bernoulli Fisher metric. This argument includes incoming and channel derivatives.

9.2 Finite entry into an exactly erasing region

Lemma 9.1 (Permanent gate closure under the original flow). For sufficiently small $\delta > 0$ and then sufficiently small $\varrho > 0$ relative to δ, there is an open class within eq. (9.1), with

$$|z_i|, |w_i| < p = 3/4, \qquad h_i, \widehat{h}_i \in (h_0 + \delta/2, h_0 + 3\delta/4), \tag{9.9}$$

for which both gates reach zero in finite time and remain zero.

Proof. The exact envelope derivative gives

$$\dot{h}_i = -(1-h_i^2)\left[2(\operatorname{atanh} h_i - \operatorname{atanh} h_*) + \frac{\alpha'(H_q)}{4}\sum_{a,j}\beta^*_{aj}\partial_\alpha d(z_a, \alpha\varrho z_j)\right]. \tag{9.10}$$

The factor two comes from the local term and the receiver's source-independent gate cost. For $|z_a|, |z_j| \le p$ and $\varrho \le 1/2$,

$$\partial_\alpha d(z_a, \alpha\varrho z_j) = \varrho z_j \frac{\alpha\varrho z_j - z_a}{1-\alpha^2\varrho^2 z_j^2}. \tag{9.11}$$

Its absolute value is bounded by $B\varrho$, where one may take $B = 3p^2/[2(1-p^2/4)]$. The four row sums equal one, so the correction to the gate force is bounded by $B\varrho M_\chi/\delta$.

On $K = [1/8, 1/2]$, define the positive constants

$$\begin{aligned} m &= \min_{h\in K} 2(1-h^2)(\operatorname{atanh} h - \operatorname{atanh} h_*), \\ M &= \max_{h\in K} 2(1-h^2)(\operatorname{atanh} h - \operatorname{atanh} h_*), \qquad V = M + m/2. \end{aligned} \tag{9.12}$$

Choose $B\varrho M_\chi/\delta \le m/2$. While its coordinates stay in K, each gate coordinate has $\dot{h}_i \le -m/2$ and $|\dot{h}_i| \le V$. Also choose δ so that the initial interval lies in K and

$$\frac{3V\delta}{2m} < h_0 - 1/8. \tag{9.13}$$

The sector mean reaches h_0 by time $3\delta/(2m)$, before any coordinate can leave K. On $H_q = h_0$, both α and α' vanish. Thereafter

$$\dot{h}_i = -2(1-h_i^2)(\operatorname{atanh} h_i - \operatorname{atanh} h_*). \tag{9.14}$$

All these coordinates start above h_* and decrease toward it. Their mean therefore cannot leave the closed-gate region. The model-sector proof is identical.

It remains to justify the private bounds used in the argument. At $\varrho = 0$ the private field is $\dot{z} = -2(1-z^2)\operatorname{atanh} z$, strictly inward at $z = \pm p$. Private derivatives of eq. (9.8) converge uniformly to this field as $\varrho \to 0$, uniformly over normalized source rows and $0 \le \alpha \le 1$. The incoming terms are of order ϱ; differentiating a private mean does not differentiate α. Compactness therefore preserves the strict inward signs for sufficiently small ϱ. The same holds for w. This closes the interval argument. □

Let τ be the later of the two gate-closure times. The construction uses a smooth but nonanalytic channel: its exactly flat region makes finite entry into exact erasure possible. Decreasing a small nonzero analytic gate would not establish this conclusion.

9.3 A change of preferred groups without synchronization

Theorem 9.2 (Persistent heterogeneous pairs). For sufficiently small $\varepsilon > 0$, admit two public states in $(l, u)^2$ and two in $(-u, -l)^2$, with $l = 0.49$, $u = 0.51$, including every assignment of these

signs to the four labels. These boxes are forward invariant. After τ, each agent's unique preferred source in both sectors is the other agent in its box. There are initial private states for which the preferred pairs are instead crossed between these boxes before closure. The public states within each final pair can remain distinct at every finite time.

Proof. For the local public flow, the inequalities

$$Jl - \operatorname{atanh} l > 0, \qquad \operatorname{atanh} u - Ju > 0 \tag{9.15}$$

give strict inward signs on every face of the two boxes. They follow by differentiating $Jv - \operatorname{atanh} v$ near its zero $v = 1/2$. Set $c = (1-\varepsilon)x + \varepsilon v$. Direct differentiation gives

$$\begin{aligned} \partial_1 D_\varepsilon(x,v) &= \operatorname{atanh} x - \operatorname{atanh} c + (1-\varepsilon)\frac{c-x}{1-c^2}, \\ \partial_2 D_\varepsilon(x,v) &= \varepsilon \frac{c-x}{1-c^2}. \end{aligned} \tag{9.16}$$

Both are uniformly $O(\varepsilon^2)$ on $[-u,u]^2$; the first-order terms cancel in the first line. The full public force has one outgoing row and three incoming sources, each weighted by a number in $[0,1]$. A sufficiently small ε preserves every strict inward margin, independently of the private state and the gate.

After gate closure, the private and gate costs in a row do not depend on its source, so Gibbs ranking uses D_ε alone. Taylor's theorem in the first argument of d gives

$$\frac{1}{2}(x-c)^2 \le d(x,c) \le \frac{(x-c)^2}{2(1-u^2)}. \tag{9.17}$$

A same-box source has separation at most $u - l$, whereas an opposite-box source has separation at least $2l$. Since $(u-l)^2/(1-u^2) < (2l)^2$, the same-box partner is the unique preferred source throughout the invariant region. Cross-pair weights remain positive.

For an explicit initial reorganization, choose private means $\pm\zeta$, with $0 < \zeta < p$, in both sectors, assigning each same-private-sign pair one positive and one negative public member. Initially $\alpha \ge 1/2$. The private advantage of the same-sign partner over an opposite-sign source is

$$d(\zeta, -\alpha\varrho\zeta) - d(\zeta, \alpha\varrho\zeta) = 2\zeta \operatorname{atanh}(\alpha\varrho\zeta) \ge \varrho\zeta^2. \tag{9.18}$$

The public cost range is at most $2u^2\varepsilon^2/(1-u^2)$. Choosing this smaller than $\varrho\zeta^2$ makes the initial preferred pairs the crossed pairs. Strict inequalities persist under small perturbations of the initial private values. Thus the final persistent grouping is not an unchanged initial source assignment.

Choose initial public states within a box separated by a fixed positive distance. Their speeds are uniformly bounded, whereas $\tau \le 3\delta/(2m)$. Taking δ sufficiently small keeps them distinct through τ. Afterward the public field is autonomous, smooth, and permutation equivariant. Its equality diagonal for a pair is invariant. Backward uniqueness precludes first entry into that diagonal at a finite time. This proves finite-time distinctness; it does not give a uniform lower bound on their separation as $t \to \infty$. □

The parameter choices are compatible: first choose a sufficiently small δ, then ϱ sufficiently small relative to δ, and finally ε sufficiently small that ε^2 is small relative to ϱ and the fixed public inward margins. All means stay in compact subsets of $(-1, 1)$, giving global continuation in this admitted family.

9.4 The exact inherited state

For $t \geq \tau$ the reduced potential separates exactly:

$$\begin{aligned}\mathcal{E}_{\mathrm{red}} &= \mathcal{E}_{\mathrm{pub}}(\mathbf{x}, \mathbf{y}) \\ &\quad + 2\sum_i [\psi(z_i) + \psi(w_i) + d(h_i, h_*) + d(\widehat{h}_i, h_*)].\end{aligned} \tag{9.19}$$

Here $\mathcal{E}_{\mathrm{pub}}$ is the original coupled local KL and Gibbs-reduced peer potential with costs D_ε. The product Fisher metric separates as well. Discarding the private and gate laws therefore gives an exact autonomous public system without fitting another potential or mobility.

This independence is generally absent before gate closure. To see that this is an actual change of the vector field, expand one reduced row with $\alpha > 0$ fixed. If $A_{ij} = (x_j - x_i)^2/[2(1 - x_i^2)]$, its first mixed public-private term is

$$\frac{\alpha\varrho\varepsilon^2}{T} z_i \operatorname{Cov}_{j\neq i}(A_{ij}, z_j). \tag{9.20}$$

This follows by expanding $D_\varepsilon = \varepsilon^2 A + O(\varepsilon^3)$, $d(z_i, \alpha\varrho z_j) = \psi(z_i) - \alpha\varrho z_i z_j + O(\varrho^2)$, and the logarithm of the row normalization. The covariance is under the uniform source prior. It is not identically zero even within the crossed-preference preparation. Fix $v = (-b, -a, a, b)$ with $l < a < b < u$, keep $z = \zeta(1, -1, 1, -1)$ fixed, and vary only public $x_i = rv_i$ near $r = 1$. The summed coefficient is

$$\frac{2\zeta^2 r^2}{9}\left[\frac{a^2 - 3b^2 + 6ab}{1 - r^2b^2} + \frac{b^2 - 3a^2 + 6ab}{1 - r^2a^2}\right]. \tag{9.21}$$

Both numerator factors are positive on the stated intervals, so this expression has positive derivative in $r > 0$. Two fixed admissible values of ζ therefore give different public directional gradients at leading order $\varrho\varepsilon^2$. The differentiated remainder is $O(\varrho\varepsilon^3 + \varrho^2\varepsilon^2)$ on these compact sets. For sufficiently small positive gains, equal public data thus have different public derivatives while the gate is open. This expansion establishes nonidentity of the preclosure force; eq. (9.19) is an exact postclosure identity.

For each selected pair I, retain its size and the empirical measure

$$\eta_I = \frac{1}{|I|}\sum_{i\in I} \delta_{(x_i, y_i)}, \qquad M = \sum_I |I|\eta_I. \tag{9.22}$$

This is a deterministic configuration statistic, not an assumed probability distribution over uncertain population configurations. Let $V[v; M]$ be the public Fisher vector field at the public state $v = (x, y)$, computed from $\mathcal{E}_{\mathrm{pub}}$. It is determined by v and the counting measure M: source sums use $M - \delta_v$, and the derivative includes both receiver and incoming source terms from eq. (9.16).

The normalized Gibbs denominator is computed from the same counting measure. Permutation equivariance makes this definition independent of labels, including multiplicities.

Proposition 9.3 (Exact parent evolution and public responses). The inherited parent equations are

$$\frac{d}{dt}\int f(v)\eta_I(dv) = \int \nabla f(v)\cdot V[v;M]\eta_I(dv) \tag{9.23}$$

for every continuously differentiable test function. On the realizable finite-atom states they determine the retained evolution uniquely. They also determine every declared fresh public response whose source selection depends only on public states, including joint responses with a shared sampled child.

Proof. Differentiate the finite sum in eq. (9.22) using the exact public equations. This gives eq. (9.23). Equal retained parent measures admit enumerations that differ only by within-pair permutations. Uniqueness and permutation equivariance of the public system give identical evolved measures, proving the retained uniqueness assertion. For a fresh response, select the child or children using the declared normalized public kernel, then integrate their public sample laws against the retained empirical measures. Both the selection probabilities and sample laws are determined by the retained states. For shared-source outputs, use one shared draw rather than multiplying separate marginals; the same calculation preserves their joint law. □

The parent can carry belief and model marginals $q_I = \int b_x \eta_I(d(x,y))$ and $s_I = \int b_y \eta_I(d(x,y))$, with normalized public evaluator G_J. These two marginals alone are insufficient: η_I retains the paired public roles and receiver-specific routing information. The inherited equations are eq. (9.23), not a newly postulated Fisher flow for the two mixture marginals.

The physical compression here discards the private and gate laws: the admitted family has twenty-four continuous means, and the retained empirical state keeps eight public means, up to within-pair permutations. It keeps every public child state. Cached private readouts, feedback that changes the erased channels, and protocols that later inspect discarded private coordinates are outside the response class. The reference, gate, initial sign-box preparation, and boundary restrictions are all supplied assumptions. The dynamics select changing source preferences and a persistent grouping within them, but do not discover the erasing architecture. The two final sign boxes remain separated, so this example supplies no recursive second-stage formation and no further compression of the distinct public states.

Chapter 10

Public compression and inherited responses

A nonlinear projection of the original smooth KL flow can discard continuous public differences without an erasure channel or finite-time equality. This autonomous parent nevertheless fails to predict ordinary child sampling. A separate identifiability result explains why an unrestricted identity-channel Gibbs interface retains every exposed child law.

10.1 An exact collective phase for general laws

Let A and B be probability measures on standard Borel spaces K and M. Suppose there are measurable functions $\varphi : \mathsf{K} \to \{-1, 1\}$ and $\psi : \mathsf{M} \to \{-1, 1\}$ with zero reference means. For example, either space may be a balanced two-point factor times an arbitrary probability space. The agent laws below need not factor over these coordinates. Fix $|J| < 1/4$ and the normalized reference

$$R(dk, dm) = \frac{e^{J\varphi(k)\psi(m)}}{\cosh J} A(dk)B(dm). \tag{10.1}$$

Its model marginal is B, and its evaluator is $G(m, dk) = e^{J\varphi(k)\psi(m)}A(dk)/\cosh J$. Thus

$$D_{\mathrm{KL}}(q_i \otimes s_i \| R) = D_{\mathrm{KL}}(q_i \| A) + D_{\mathrm{KL}}(s_i \| B) - J\mathbb{E}_{q_i}\varphi\mathbb{E}_{s_i}\psi + \log\cosh J. \tag{10.2}$$

For $J \neq 0$ the belief and model laws interact through the evaluator.

Take $N \geq 2$ identical agents, identity peer channels, all other agents as sources, uniform nonself source priors, and fixed positive temperatures. Use the exact Gibbs-reduced potential of eq. (4.6), with unit Fisher mobility. No source or channel derivative is frozen. Introduce the Banach spaces

$$\begin{aligned} \mathsf{E}_q &= \{u \in L^\infty(A) : \mathbb{E}_A u = 0\}, \quad q_u = \frac{e^u}{\mathbb{E}_A e^u} A, \\ \mathsf{E}_s &= \{v \in L^\infty(B) : \mathbb{E}_B v = 0\}, \quad s_v = \frac{e^v}{\mathbb{E}_B e^v} B, \end{aligned} \tag{10.3}$$

and $\mathsf{E} = \mathsf{E}_q \times \mathsf{E}_s$, with the maximum norm. These are full bounded log-density charts, rather than a finite-parameter family. Set $z_i = (u_i, v_i)$, $\bar{z} = N^{-1}\sum_i z_i$, and $\delta_i = z_i - \bar{z}$. Write Φ_t for the resulting fine flow in E^N.

On the diagonal $z_1 = \cdots = z_N = z$, every peer force vanishes. The diagonal vector field is

$$f(u, v) = \big(-u + J\varphi\mathbb{E}_{s_v}\psi, -v + J\psi\mathbb{E}_{q_u}\varphi\big). \tag{10.4}$$

Denote its flow by ϕ_t. This equation is defined for all $z \in \mathsf{E}$, for positive and negative finite times. In particular, it can be used below without extending a locally defined diagonal dynamics.

Theorem 10.1 (A compressive phase projection under the original KL flow). For the preceding data there is a forward-invariant neighborhood U of the common reference configuration in E^N on which

$$\Pi(x) = \lim_{t\to\infty} \phi_{-t}\big(\overline{\Phi_t x}\big) \tag{10.5}$$

exists in E and is continuously differentiable. It satisfies

$$\Pi(z,\dots,z) = z, \qquad \Pi(\Phi_t x) = \phi_t(\Pi x), \qquad D\Pi_{(z,\dots,z)}(h_1,\dots,h_N) = \frac{1}{N}\sum_i h_i. \tag{10.6}$$

After shrinking U if necessary, Π is a submersion with nontrivial fibers parametrized locally by $N-1$ public log-density differences. The parent laws q_{u^c}, s_{v^c}, where $(u^c, v^c) = \Pi(x)$, follow eq. (10.4) exactly. If two fine child states (q_i, s_i) are initially different, they remain different at every finite time in this neighborhood.

Proof. Exponential normalization, bounded density ratios, the finite Gibbs sums, and their derivatives are smooth maps on bounded log-density balls. Consequently the full reduced Fisher vector field V is smooth there. Centering a logarithmic Fisher equation against A or B removes its scalar normalizer and gives an ordinary differential equation in the Banach chart.

At the origin every peer KL cost has zero first variation, so the Gibbs rows equal $1/(N-1)$ and their first variations vanish. The receiver and sender derivatives each contribute one copy of the complete-graph difference operator. Therefore, with $\kappa_N = 2N/(N-1)$,

$$\begin{aligned} (DV(0)h)_i &= \mathcal{A}h_i - \kappa_N(h_i - \bar{h}), \\ \mathcal{A}(u,v) &= \big(-u + J\varphi\mathbb{E}_B[\psi v], -v + J\psi\mathbb{E}_A[\varphi u]\big). \end{aligned} \tag{10.7}$$

The off-diagonal operator in $\mathcal{A}$ has norm at most $|J|$. Hence its common and difference semigroups have bounds $e^{-(1-|J|)t}$ and $e^{-(1+\kappa_N-|J|)t}$ in the corresponding maximum norms.

Here are the local estimates needed for the limit, including its regularity. Choose $\mu > 0$ and b strictly below these two decay rates, respectively, with

$$b > L := 1 + 2|J|. \tag{10.8}$$

This is possible because $\kappa_N > 2$. Taylor expansion and the variation-of-constants formula, on a sufficiently small ball in the norm of $(\bar{z}, \delta)$, give a forward-invariant neighborhood and

$$\|\Phi_t x\| + \|D\Phi_t(x)\| \le Ce^{-\mu t}, \qquad \|\delta(t)\| + \|D_x\delta(t)\| \le Ce^{-bt}. \tag{10.9}$$

The constants are uniform on a smaller neighborhood. To detail the second estimate, the diagonal is invariant, so the difference equation is linear in δ with coefficient $\mathcal{A} - \kappa_N I + O(\|x\|)$. Its differentiated equation has the same stable homogeneous part and an additional forcing bounded by $C\|\delta(t)\|\|D\Phi_t(x)\|$. Variation of constants bounds that forcing by e^{-bt} times a convergent integral of $e^{-\mu t}$. Small losses in the two decay exponents absorb the coefficient perturbations. The

first estimate follows in the same way from the strictly stable full linearization, after choosing the neighborhood small enough that its nonlinear derivative perturbation is less than the chosen decay margins.

Permutation equivariance implies that the average derivative of V in a zero-sum direction vanishes at every diagonal point. Thus the exact average equation has the form

$$\dot{\bar{z}} = f(\bar{z}) + g(\bar{z}, \delta), \qquad g = O(\|\delta\|^2), \quad D_{\bar{z}} g = O(\|\delta\|^2), \quad D_\delta g = O(\|\delta\|). \tag{10.10}$$

It follows from eq. (10.9) that both $g(\Phi_t x)$ and its derivative with respect to x are bounded by Ce^{-2bt}.

The normalized feature expectations in eq. (10.4) have uniformly bounded first and second derivatives. In particular, f is globally Lipschitz with constant at most L, has at most linear growth, and has a bounded second derivative. Its complete flow obeys

$$\|D\phi_{-t}\| \le e^{Lt}, \qquad \|D^2\phi_{-t}\| \le Ce^{2Lt}, \qquad t \ge 0. \tag{10.11}$$

Differentiation in time gives the convergent integral

$$\Pi(x) = \bar{z}(0) + \int_0^\infty D\phi_{-t}\big|_{\bar{z}(t)} g(\bar{z}(t), \delta(t))dt. \tag{10.12}$$

The integrand is bounded by $Ce^{-(2b-L)t}$. Its derivative in x is the sum of a term with $D\phi_{-t}D_x g$ and one with $D^2\phi_{-t}[D_x\bar{z}, g]$. They are bounded by integrable multiples of $e^{-(2b-L)t}$ and $e^{-(2b-2L)t}$. Uniform convergence proves continuous differentiability.

On the diagonal the residual and its first derivative vanish, proving the retraction and derivative assertions. Shifting the limit in eq. (10.5) by a finite time proves the intertwining identity. The averaging derivative has the diagonal embedding as a bounded right inverse. The Banach implicit function theorem therefore gives the claimed local fibers, including continuously varying heterogeneous preparations. Finally, if two complete child states first became equal at a finite time, exchanging their labels would produce the same configuration at that time. Permutation equivariance and local uniqueness backward to the initial time would imply that they were initially equal, a contradiction. □

The parent is a virtual collective state, generally different from either the arithmetic or geometric pool. The restricted potential $ND_{\mathrm{KL}}(q^c \otimes s^c \| R)$ and metric Ng_{Fisher} give eq. (10.4). Fine states remain unchanged by the construction. They approach the common reference, so the theorem supplies neither persistent nonzero separation nor selected membership.

The construction is an exact dynamical lumping [12] with stable-foliation structure [7]. The proof uses Banach charts directly; it does not extend a finite-dimensional theorem by analogy. The limit supplies no finite-cost evaluation algorithm.

10.2 An ordinary response that the phase does not preserve

The failure of boundary sufficiency can be seen inside the preceding example. Take two binary agents, $q_i = s_i = \mathrm{Ber}_{\pm 1}(x_i)$, $0 < J < 1/4$, and set $x_1 = m + a$, $x_2 = m - a$. This is an invariant

subfamily. The exact single-agent diagonal equation is

$$f_b(x) = -(1-x^2)(\operatorname{atanh} x - Jx). \tag{10.13}$$

Both source rows have one entry. Including the incoming peer derivative gives

$$\dot{x}_i = -(1-x_i^2)[2\operatorname{atanh} x_i - \operatorname{atanh} x_j - Jx_i] - (x_i - x_j). \tag{10.14}$$

Expansion at the reference state yields

$$\begin{aligned} \dot{m} &= f_b(m) + (6-3J)ma^2 + O((|m|+|a|)^5), \\ \dot{a} &= -(5-J)a + O((|m|+|a|)^3). \end{aligned} \tag{10.15}$$

In this two-agent finite-dimensional subfamily the same integral argument gives a C^3 phase: choose b close to $5-J$, so $2b > 4L$, and differentiate the integral three times. The differentiated residuals retain two exponentially decaying difference factors, while the inverse-flow derivatives grow at most as e^{4Lt}.

Let p be the mean parameter of the phase law. Retraction, child exchange, and simultaneous bit reversal imply $p = m + cma^2 + o((|m|+|a|)^3)$. Substitution into $Dp(V) = f_b(p)$ determines the coefficient:

$$p = m + \frac{3(2-J)}{2(5-J)}ma^2 + o((|m|+|a|)^3). \tag{10.16}$$

Indeed, the coefficient equation is $(6-3J) - 2(5-J)c = 0$. Along $m = a > 0$ sufficiently small, $p \neq m$. The heterogeneous state and the diagonal state (p,p) have exactly the same phase parent, whereas choosing a child uniformly and sampling its public bit gives means m and p, respectively. This response is not constant on the phase fibers. Replacing it by sampling the phase law would specify a new interface.

10.3 What unrestricted Gibbs probes retain

Consider a block of exposed belief laws $q_j = f_j\lambda$ against a probability reference λ, with f_j bounded above and away from zero. Fix positive source weights w_j and a temperature $T > 0$. An identity-channel receiver with law $r\lambda$ sees the block normalizer

$$Z(r) = \sum_j w_j \exp[-D_{\mathrm{KL}}(r\lambda \| q_j)/T]. \tag{10.17}$$

The probability of selecting this block against a known outside source determines $Z(r)$, so the normalizer is an observable routing response without revealing a child label.

Proposition 10.2 (Unrestricted Gibbs response identifies the exposed child laws). If two finite source collections have equal eq. (10.17) for every positive receiver density in an L^∞ neighborhood of 1, then their weighted empirical measures of densities are equal:

$$\sum_j w_j \delta_{f_j} = \sum_k w'_k \delta_{f'_k}. \tag{10.18}$$

Repeated identical laws may be combined by adding their weights; distinct exposed laws cannot be forgotten by a retention preserving this entire response class.

Proof. Multiplication by the known receiver-entropy factor gives

$$e^{\int r\log r d\lambda/T} Z(r) = \sum_j w_j e^{\int r\log f_j d\lambda/T}. \tag{10.19}$$

List the distinct normalized densities in the union of the two collections. No two of their logarithms differ by a constant. There is therefore a bounded mean-zero function h for which the numbers $v_j = \int h\log f_j d\lambda$ are all different: each equality excludes one proper hyperplane in the finite-dimensional span of centered log-density differences, and finitely many such hyperplanes cannot cover that span. Put $r = 1 + th$ for sufficiently small $|t|$. Subtracting the two expressions in eq. (10.19) gives $\sum_j c_j e^{tv_j/T} = 0$. Differentiation through orders zero to one less than the number of distinct laws gives an invertible Vandermonde system. Every c_j vanishes. Its positive factor $e^{\int \log f_j d\lambda/T}$ leaves exactly equality of the total weights on each density. □

In particular, a single ordinary law with one positive weight reproduces all these responses only if every exposed child law already equals it. The proposition concerns fixed identity comparison channels and unrestricted receiver probes. It is not an impossibility result for arbitrary channels, restricted interfaces, or isolated dynamical factors such as theorem 10.1.

For a restricted affine receiver family $r_\theta = r_0 + \sum_a \theta_a h_a$, the response instead identifies

$$\mu = \sum_j w_j e^{b_j/T}\delta_{v_j}, \qquad b_j = \int r_0\log f_j d\lambda, \qquad (v_j)_a = \int h_a \log f_j d\lambda. \tag{10.20}$$

Equal slopes can be combined by adding the tilted weights. This possible boundary compression still requires a dynamical check. The incoming Fisher force from one Gibbs receiver is $\dot f_j = \alpha_j(r_\theta - f_j)$, and consequently

$$\dot b_j = \alpha_j\left(\int \frac{r_0 r_\theta}{f_j} d\lambda - 1\right), \qquad (\dot v_j)_a = \alpha_j \int \frac{h_a r_\theta}{f_j} d\lambda. \tag{10.21}$$

These reciprocal moments are additional information unless the admitted family supplies a closure identity.

10.4 The remaining recursive requirement

For unique flows, block retention R_I and external state e require controlled projectability:

$$DR_I(x_I)V_I(x_I;e) = \bar V_I(R_I x_I; R_{\mathrm{ext}} e). \tag{10.22}$$

Boundary kernels, including memory of a reused child, must factor through the retained data. Joining within a closed admissible family must also descend:

$$R_{I\cup J}\circ \mathcal{J}_{I,J} = \bar{\mathcal{J}}_{I,J}\circ (R_I, R_J). \tag{10.23}$$

Chapter 11

Boundary Closure and Recursive Composition

There is an exact hierarchy of public compressions under the original smooth potential with a coupled evaluator and identity peer channels. Each level preserves fresh uniform-descendant responses, their subsequent evolution, and a declared class of record conditioning. The construction supplies an invariant family and its hierarchy. It establishes recursive inheritance on that family, while leaving the selection of the family and hierarchy open.

11.1 A general-law family on a supplied tree

Let A and B be probability references on standard Borel spaces with coordinates r and w. Choose a positive joint density $\rho(r, w)$ against $A \otimes B$, normalized to one, with bounded logarithm. Its disintegration supplies a proper model reference and normalized evaluator as in eq. (4.1). It may couple its arguments; for example, balanced signs $\phi(r), \psi(w)$ give $\rho = \exp(J\phi\psi)/\cosh J$.

Fix a binary tree of depth $d \geq 1$ with $N = 2^d$ leaves. Each leaf is a primitive agent. Append one independent Haar circle for every internal tree node to the belief sample space, and a separate set to the model sample space. The references extend by these independent circle laws. At internal node v, fix a measurable phase function $\alpha_v(r)$ and define

$$\sigma_v(r, \theta_v) = \operatorname{sign} \cos(\theta_v - \alpha_v(r)). \tag{11.1}$$

The sign on its null zero set is immaterial. Conditional on the base, the signs are independent and fair. The circles are original public sample coordinates. Different phase functions change the laws in those same coordinates; they are held fixed along a continuous trajectory.

Label the left and right edges by $\epsilon = +1, -1$. A leaf i defines a signed path

$$u_j(i, \theta) = \epsilon_j(i)\sigma_{v_{j-1}(i)}(r, \theta_{v_{j-1}(i)}), \qquad j = 1, \ldots, d. \tag{11.2}$$

Let U_d be the uniform law on $\{-1, 1\}^d$. Choose any positive bounded profile $f(r, u)$, with bounded logarithm and $\int f dA dU_d = 1$, and set

$$q_i(r, \theta) = f(r, u(i, \theta)), \qquad f_0(r) = \mathbb{E}_{U_d} f(r, u). \tag{11.3}$$

These are densities against A times all the Haar factors. Each leaf has the same base marginal $f_0 A$, but its full public law can differ from every other leaf law. Define model densities from an independent profile $g(w, z)$ and model circles in the same way. The base spaces and the profile

functions are unrestricted within the stated regularity domain; no Gaussian or finite-parameter family is imposed.

All physical peer channels are identities, every other agent is an allowed source, and the original source priors are uniform. If two leaves share exactly k initial edges, there are $n_k = 2^{d-k-1}$ possible sources in that class for each receiver. Give each such source weight b_k/n_k, where b is a probability vector on $k = 0, \dots, d-1$. The belief and model vectors b^q, b^s evolve separately. Uniform original rows are included by taking

$$\pi_k = \frac{n_k}{N-1}, \qquad b^q_k = b^s_k = \pi_k. \tag{11.4}$$

The tree therefore enters the prepared relation among the agent laws and row classes. The fine potential and its uniform priors contain no additional preference for a chosen pair.

11.2 The exact inherited profile equations

For each k, define an operator P_k on bit profiles: keep the first k bits, reverse bit $k+1$, and average all later bits. It is a Markov operator, self-adjoint for U_d. It describes an average over a class of original source labels; it is not a replacement physical communication channel. Define

$$D_k(f) = \int f(r,u)[\log f(r,u) - (P_k \log f)(r,u)]dA(r)dU_d(u). \tag{11.5}$$

Theorem 11.1 (Exact coupled hierarchy dynamics). On its regular positive domain, the family eq. (11.3) is invariant under the original full Fisher flow, including source-row evolution. Its inherited profile energy and metric are the original restrictions divided by their common factor N. The complete profile equations retain the incoming source derivatives.

Proof. For any two leaves in class k, the first k path bits agree, bit $k+1$ is opposite, and later bits lie on disjoint branches. Integrating their independent signs gives exactly $D_{\mathrm{KL}}(q_i \| q_j) = D_k(f)$. Every leaf path has uniform bit law. Each local product law therefore has the same base marginals f_0, g_0 in its evaluator term. Direct substitution gives $\mathcal{E} = NF$, with

$$\begin{aligned} F = &\int f \log f dA dU_d + \int g \log g dB dU_d - \int f_0(r) g_0(w) \log \rho(r,w) dA dB \\ &+ \sum_{k=0}^{d-1} b^q_k D_k(f) + T_q D_{\mathrm{KL}}(b^q \| \pi) + \sum_{k=0}^{d-1} b^s_k D_k(g) + T_s D_{\mathrm{KL}}(b^s \| \pi). \end{aligned} \tag{11.6}$$

The metric restricts to N times

$$g^c = \int \frac{(\delta f)^2}{f} dA dU_d + \int \frac{(\delta g)^2}{g} dB dU_d + \sum_{a=q,s} \sum_k \frac{(\delta b^a_k)^2}{b^a_k}. \tag{11.7}$$

Thus no new mobility is introduced.

Self-adjointness gives the exact derivative

$$\delta D_k(f) = \int \delta f \left[\log f + 1 - P_k \log f - \frac{P_k f}{f} \right] dA dU_d. \tag{11.8}$$

The last ratio is the incoming source contribution. Define

$$V^q(r) = -\int g_0(w)\log\rho(r,w)dB(w), \qquad V^s(w) = -\int f_0(r)\log\rho(r,w)dA(r). \tag{11.9}$$

Up to an irrelevant additive constant, the belief force and evolution are

$$\begin{aligned} h^q &= 2\log f + V^q - \sum_k b_k^q\left[P_k\log f + \frac{P_k f}{f}\right], \\ c_q &= \int f h^q dA dU_d, \qquad \dot f = -f(h^q - c_q). \end{aligned} \tag{11.10}$$

Exchange the sectors for $\dot g$. With $D_k^q = D_k(f)$ and $D_k^s = D_k(g)$, the rows obey

$$\dot b_k^a = -b_k^a\left[D_k^a + T_a\log\frac{b_k^a}{\pi_k} - \sum_l b_l^a\left(D_l^a + T_a\log\frac{b_l^a}{\pi_l}\right)\right]. \tag{11.11}$$

To establish full-flow invariance, fix a receiver and a public sample point. The leaves in its class-k source subtree run through every suffix bit string once, regardless of the physical signs at that point. Their averages of $\log q_j$ and q_j are therefore $P_k\log f$ and $P_k f$. Incoming weights have the same class values. The original fine force at that leaf is precisely eq. (11.10) evaluated at its path bits. All leaf normalizations agree because every path-bit vector is uniform under the reference. Source costs also agree within their classes, preserving the row form. Hence the full field stays in the profile family, and the restricted gradient is its actual evolution. Bounded logarithms, bounded $\log\rho$, and positive rows give local Lipschitz equations in the bounded-profile charts, with unique solutions on their regular existence intervals. □

The same retentions apply to the original Gibbs-reduced flow. Replace each row by $b_k^{a,*} = \pi_k\exp(-D_k^a/T_a)/\sum_l \pi_l\exp(-D_l^a/T_a)$ and use the matter metric. The envelope identity in eq. (4.8) gives the displayed profile force with these weights, including its incoming term. All symmetry and response arguments below remain unchanged.

The inherited cost is still a relative entropy. Let

$$J_k(dr,du,dv) = A(dr)f(r,u)U_d(du)P_k(u,dv), \tag{11.12}$$

and let J_k^{rev} exchange u,v. Reversibility gives $D_k(f) = D_{\mathrm{KL}}(J_k\|J_k^{\mathrm{rev}})$. It is generally different from the divergence to the averaged source law:

$$D_k(f) - D_{\mathrm{KL}}(fA\otimes U_d\|(P_k f)A\otimes U_d) = \int f[\log(P_k f) - P_k\log f], dA dU_d \geq 0. \tag{11.13}$$

Replacing the exact inherited cost by the latter divergence would discard this term and change the dynamics.

11.3 Recursive retention and the response process

At each base point, arrange the values of $f(r, u)$ in their binary profile tree. Retain this tree with unordered children at every internal node. Its leaves still carry density values, and its nested grouping is retained. Flattening it into a multiset would lose which values share a prefix and could change D_k.

For a physical subtree rooted at depth l, apply this retention only to its descendant values, keeping the dependence on the base and physical ancestor coordinates. This defines its parent descriptor. Include the corresponding model descriptor, the shared row-class data, primitive counts, and the remaining tree connections. At a lowest pair, the unordered values x, y can equivalently be written as a mean $m = (x + y)/2$ and a squared contrast $v = ((x - y)/(x + y))^2$. A first parent therefore retains more than its mixture law, but does not retain the two full child laws on their original circle coordinates.

Theorem 11.2 (Recursive boundary inheritance on the supplied hierarchy). These parent descriptors inherit the joint controlled dynamics at every level of the same fine system. Joining two sibling descriptors gives their unordered pair. The join commutes with evolution and with fresh uniform-descendant response kernels. Common record likelihoods depending on the base preserve these statements under arbitrary finite adaptive sequences, whenever the updates remain in the regular domain.

Proof. At a fixed physical point, changing an internal sign exchanges the two branches below that node. Thus the map from physical leaf labels to their bit strings is an automorphism of the rooted binary tree. Forgetting its realization removes the phase at the joined node and leaves exactly the unordered profile subtree.

Tree automorphisms preserve common-prefix length, U_d, and every P_k. They consequently preserve eqs. (11.6) and (11.7) and commute with eqs. (11.10) and (11.11). This remains true for measurable base-dependent branch exchanges: the equations have no derivatives of the base coordinate, the local evaluator reads only f_0, g_0, and the global integrals are unchanged. The quotient flow is therefore independent of the chosen ordered representatives. The finite tree quotient admits measurable representatives, for example by recursively ordering the finite tuples of its values. This also establishes measurability of the descended updates.

For partial retentions, the same argument applies to the product of automorphism groups of the retained subtrees. The complement supplies the retained sibling and ancestor data needed for their interactions; the joint parent network closes. Thus this is controlled closure for complements in the stated prepared family, rather than autonomy under every external law. Combining sibling trees and then forgetting their ordering is the same operation as retaining their union directly. Repeated compatible joins give the same descriptor, and joins of disjoint subtrees commute. These are representation changes inside one fixed N-agent interaction graph. They do not add or delete fine edges, and do not authorize an arbitrary reassociation of the leaves into another hierarchy.

For a response at a node of depth l, choose a descendant freshly and uniformly, draw its full original public sample, and conceal its label. At every physical sample point the descendant labels run

through every suffix bit string. The response density is exactly

$$f_l(r, u_1, \dots, u_l) = \mathbb{E}_{u_{l+1},\dots,u_d} f(r, u_1, \dots, u_d). \tag{11.14}$$

It is the mean of the retained subtree values, with all unused Haar factors left in the original output space. At the root it is $f_0(r)$. Choosing a sibling parent uniformly and then a descendant uniformly gives exactly the same full sample law. The model sector follows the same rule with a separate fresh label.

Fix a normalized positive record kernel $L_e^q(do \mid r)$ with bounded positive density on the admitted update domain. It may be applied to the original sample as a readout. Every child has base marginal $f_0 A$, so a common record has the same component normalization

$$Z_o = \int f_0(r) L_e^q(o \mid r) dA(r). \tag{11.15}$$

Conditioning every mixture component preserves uniform labels and gives

$$f^+(r, u) = \frac{L_e^q(o \mid r) f(r, u)}{Z_o}. \tag{11.16}$$

This multiplies every value in a profile tree at the given base point by the same factor. It therefore commutes with all retentions and joins. The model update is identical in form, and rows remain unchanged at the record update. Equality of retained states gives equality of the next response law and the retained update. Induction, using the inherited continuous flow between records, proves equality of every finite history law under policies choosing the next admitted experiment from the observed history. At a representation join, the already observed history is carried forward as controller state; no past record is erased. □

The fresh-sampling condition matters. For a pair $q_\pm = m(1 \pm a\sigma)$, two samples from one reused child have joint density

$$m(k)m(k')[1 + a(k)a(k')\sigma(k)\sigma(k')], \tag{11.17}$$

which can reveal the discarded phase. A shared child label for a joint belief/model response can likewise expose a relative contrast. Unrestricted external identity-channel probes can distinguish these laws, as in chapter 10. These protocols are outside the stated inherited interface.

11.4 Public information lost at every depth

Choose, only as an initial witness,

$$f(r, u) = f_0(r) \prod_{j=1}^{d} [1 + a_j(r) u_j], \qquad 0 < a_j(r) < 1, \tag{11.18}$$

with regular bounds and $\int f_0 dA = 1$. The evolution subsequently uses the unrestricted positive profile f, and is not assumed to preserve this product. Before joining at depth l, the two child-parent response densities differ by their factor $1 \pm a_{l+1}\sigma_v$. Rotating α_v changes these full public laws, while

their joined unordered profile tree stays fixed. Strict differences persist on a nonzero regular time interval by continuity. The same construction works at every depth and in both sectors.

These losses are continuous in total variation. For a small phase shift δ, the sign disagreement has Haar mass $|\delta|/\pi$. If two branch response densities at a node are x, y, then, including the base and ancestor integrations, the total variation distance induced in one child-parent response is

$$\frac{|\delta|}{2\pi}\int \mathbb{E}_{U_l}|x(r,U_l)-y(r,U_l)|dA(r). \tag{11.19}$$

Each phase can be an arbitrary measurable function of the base. The quotient therefore discards spatial patterns in the original public laws, beyond a permutation of primitive agents. It retains the amplitude profile tree, whose size can grow with the number of descendants; no fixed-size parent representation is asserted.

The sample-coordinate density is piecewise constant in these circles, while the law-valued Fisher evolution remains smooth in time on the regular profile domain. Squared Hellinger distance along a nontrivial phase rotation is of order $|\delta|$, so these paths have no finite Fisher tangent and are not continuous in the bounded-log-density norm. The result does not give a smooth dimension reduction on an open law chart. It gives exact recursive public compression on a prepared invariant class, with general base laws, a coupled evaluator, identity interactions, and no prescribed erasure gate. The tree and the class are supplied, and no endogenous formation time has been derived.

The reference coordinates r, w in this construction belong to represented sample spaces. They are not identified with the contextual base $\mathcal{C}$. Applying chapter 3 would additionally require an admitted smooth context-parameterized family and compatible connections. The measurable retention proved here does not supply the collective-connection condition in proposition 3.14.

11.5 Why ordinary response completion can remove all compression

The preceding restrictions cannot generally be removed by adding a few response moments to the phase parent. Observation derivatives can already recover a whole fine configuration, as in the local observability method of Hermann and Krener [10]. The following calculation uses only the inverse function theorem.

Proposition 11.3 (Fresh responses locally identify a binary pair). For two original identity-peer agents with the coupled binary evaluator $G_J(k \mid m) = e^{Jkm}/(2\cosh J)$, $|J| < 1/4$, there is an open heterogeneous region where any retention preserving both fresh belief/model mixture responses and their future evolution is locally injective.

Proof. Write $q_i = b_{x_i}$, $s_i = b_{y_i}$, where $b_x(k) = (1+kx)/2$. The exact original equations, including incoming derivatives, are

$$\dot{x}_i = -(1-x_i^2)[2\,\mathrm{atanh}\,x_i - \mathrm{atanh}\,x_j - Jy_i] - (x_i - x_j), \tag{11.20}$$

and the exchanged equation for y. Set $x_{1,2} = m \pm a$, $y_{1,2} = n \pm b$. Fresh mixture responses are b_m, b_n. Their velocities have the convergent local expansions

$$\begin{aligned} F_x &= -(1-m^2)(\operatorname{atanh} m - Jn) + (6m - Jn)a^2 - 2Jmab + O(5), \\ F_y &= -(1-n^2)(\operatorname{atanh} n - Jm) + (6n - Jm)b^2 - 2Jnab + O(5). \end{aligned} \tag{11.21}$$

Here $O(5)$ denotes terms of total degree at least five. To check the coefficient, the uncoupled part of F_x is

$$-\tfrac{1}{2}\sum_{\pm}[1-(m\pm a)^2]\operatorname{atanh}(m\pm a) + 2ma[\operatorname{atanh}(m+a) - \operatorname{atanh}(m-a)], \tag{11.22}$$

whose expansion is $-m + 2m^3/3 + 6ma^2 + O(5)$.

For $\mathcal{H} = (m, n, F_x, F_y)$, its Jacobian determinant at $m = n = \epsilon c_1$, $a = b = \epsilon c_2$, $c_1 c_2 \neq 0$, is

$$\det D\mathcal{H} = 4\epsilon^4 c_1^2 c_2^2 (6-3J)(6-J) + O(\epsilon^6). \tag{11.23}$$

Indeed, the lower-right block is $2\epsilon^2 c_1 c_2$ times the matrix with diagonal $6-2J$ and off-diagonal $-J$, plus $O(\epsilon^4)$. Its determinant is positive for sufficiently small nonzero ϵ, and remains nonzero nearby. If two states in that neighborhood have the same inherited parent, their response curves agree. Equality at time zero and of their first derivatives gives equality of $\mathcal{H}$, hence equality of the states by local invertibility. No differentiability assumption on the retention is needed. □

This is a local obstruction in one original model. It neither removes the global child-exchange ambiguity nor excludes the selective protocols and invariant classes constructed above.

Chapter 12

Conditional Recursive Formation

The original smooth potential admits a conditional example of the formation obligation in definition 6.17. Four initially singleton agents select two parents and then their union through a fixed rule applied to their evolving laws. Every join discards a continuous distinction between public model laws. The inherited description preserves the original-time dynamics, a declared joint response interface, and a curved collective connection with a rank-two contextual tensor.

The construction specifies its statistical architecture. Children have matrix Wishart laws with common precision in each sector, product recognition, and comparison channels depending on the population moment. Individual shapes are constant over the contextual base. Responses use fresh independent samples and return extensive sums. Membership is selected by a declared threshold, rather than by a threshold derived uniquely from the interaction potential. The preparation class is open within this invariant family. Each additive quotient is available initially; the dynamics and rule select when it becomes the operative description. No law is reset at that event.

12.1 Matrix laws and the declared architecture

Let both represented spaces be the cone S_2^{++} of real symmetric positive-definite two-by-two matrices. A random matrix X or Y is represented content, distinct from a contextual coordinate. Write P, Q for precision matrices, to distinguish them from the principal bundle P. For $\nu > 1/2$ and $\mathsf{P} \succ 0$, define

$$W_2(\nu, \mathsf{P})(dX) = \frac{\det(\mathsf{P})^{\nu}}{\Gamma_2(\nu)} \det(X)^{\nu - 3/2} e^{-\operatorname{tr}(\mathsf{P}X)} dX, \qquad \Gamma_2(\nu) = \sqrt{\pi}\Gamma(\nu)\Gamma(\nu - \tfrac{1}{2}). \tag{12.1}$$

Here $dX = dx_{11}dx_{12}dx_{22}$. The continuous shape convention is the central Wishart convention discussed by Letac and Massam [16]; it does not require a Gaussian representation or an integer number of degrees of freedom.

Lemma 12.1 (Normalization and addition). The density in eq. (12.1) is normalized. Its mean is $\nu\mathsf{P}^{-1}$. Independent variables with laws $W_2(\nu, \mathsf{P})$ and $W_2(\mu, \mathsf{P})$ have sum law $W_2(\nu + \mu, \mathsf{P})$.

Proof. At $\mathsf{P} = I_2$, set $x_{11} = u > 0$, $x_{12} = \sqrt{u}z$, and $x_{22} = v + z^2$, with $v > 0$ and $z \in \mathbb{R}$. The determinant is uv, the trace is $u + v + z^2$, and the Jacobian is $\sqrt{u}$. The integral is $\Gamma(\nu)\Gamma(\nu - \frac{1}{2})\sqrt{\pi}$. Substitution $X = \mathsf{P}^{-1/2} Z \mathsf{P}^{-1/2}$ gives the general normalizer: congruence by a positive matrix B has Jacobian $\det(B)^3$ on this cone. Differentiating the normalizing integral with respect to P gives the stated mean.

The convolution density at $S \succ 0$ contains the integral

$$e^{-\operatorname{tr}(\mathsf{P}S)} \int_{0<X<S} \det(X)^{\nu-3/2} \det(S-X)^{\mu-3/2} dX. \tag{12.2}$$

The substitution $X = S^{1/2}US^{1/2}$ makes it a constant times $e^{-\operatorname{tr}(\mathsf{P}S)} \det(S)^{\nu+\mu-3/2}$. The constant is positive and finite by Tonelli's theorem and normalization of the convolution. Normalizing the resulting density determines it and proves the addition formula. □

Initially take $N = 4$ agents with

$$q_i = W_2(\alpha_i, \mathsf{P}), \qquad s_i = W_2(\eta_i, \mathsf{Q}), \qquad \zeta_i = q_i \otimes s_i, \tag{12.3}$$

where every shape exceeds $1/2$. The model shape η_i is unrelated to a source-row weight. Fix $\alpha_0, \eta_0 > 1/2$ and $\kappa > 0$. The local data are the proper reference $r_i = W_2(\eta_0, I_2)$ and the normalized constant evaluator $G_i(m, dX) = W_2(\alpha_0, I_2)(dX)$. For any admitted finite-moment configuration set

$$M_q = \sum_i \mathbb{E}_{q_i} X, \qquad \Lambda_q = I_2 + \frac{\kappa}{N} M_q, \qquad \mathcal{K}^q_{ij}[\mathbf{q}](k, dX) = W_2(\alpha_0, \Lambda_q)(dX). \tag{12.4}$$

The model sector uses $M_s = \sum_i \mathbb{E}_{s_i} Y$, $\Lambda_s = I_2 + \kappa M_s/N$, and target $W_2(\eta_0, \Lambda_s)$. Every source, including self, is allowed with prior $\pi_{ij} = 1/N$ and positive original temperatures. These self channels are nonidentity channels. Each channel ignores its particular source sample but reads the current laws through the total moment. Independent draws across receiver children specify the joint channel experiment used when their outputs are added.

All source costs in a receiver row agree, so the exact Gibbs row equals its prior. The Gibbs-reduced original potential is $\mathcal{E}_q + \mathcal{E}_s$, where

$$\mathcal{E}_q = \sum_i \left[D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, I_2)) + D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, \Lambda_q)) \right]. \tag{12.5}$$

The two sectors have separate population coupling. The chosen local evaluator adds no cross-sector coupling. Outside the Wishart family the displayed functional requires finite relative entropies and moments; the differential also requires integrable log densities and log determinants, and differentiation under the integrals along the admitted paths.

12.2 The complete smooth Fisher flow

Proposition 12.2 (Invariant family and original-time evolution). With the unit Fisher metric and all channel derivatives included, eq. (12.3) is invariant under eq. (12.5). In the belief sector,

$$C_q = \Lambda_q + \frac{\kappa}{N}(M_q - N\alpha_0\Lambda_q^{-1}) = I_2 + \frac{2\kappa}{N} M_q - \kappa\alpha_0\Lambda_q^{-1}, \tag{12.6}$$

$$\dot{\alpha}_i = -2(\alpha_i - \alpha_0), \qquad \dot{\mathsf{P}} = -2\mathsf{P} + I_2 + C_q. \tag{12.7}$$

The model equations exchange $\alpha, \alpha_0, \mathsf{P}, M_q, C_q$ with $\eta, \eta_0, \mathsf{Q}, M_s, C_s$. These solutions are global and regular on the stated family. Distinct initial shapes remain distinct at every finite time.

Proof. Put $c_0 = \alpha_0 - 3/2$. Expanding the normalized densities gives

$$\mathcal{E}_q = 2\sum_i \int q_i \log q_i dX - 2c_0 \sum_i \mathbb{E}_{q_i} \log\det X + \operatorname{tr} M_q + \operatorname{tr}(\Lambda_q M_q) \\ - N\alpha_0 \log\det \Lambda_q + 2N \log\Gamma_2(\alpha_0). \tag{12.8}$$

Since $d\Lambda_q = (\kappa/N) dM_q$, the differential of the last moment-dependent pair is $\operatorname{tr}(C_q dM_q)$. Thus the full variational force, up to a sample-independent scalar, is

$$h_i(X) = 2\log q_i(X) - 2c_0 \log\det X + \operatorname{tr}[(I_2 + C_q)X]. \tag{12.9}$$

The direct sender variation is zero because a mass-zero variation traversing a constant-in-input kernel gives zero. The channel variation is not zero: it is precisely the second term in the first expression for C_q. The unrestricted density equation is $\dot{q}_i = -q_i(h_i - \mathbb{E}_{q_i} h_i)$. Comparing the coefficients of $\log\det X$ and X in this equation with the derivative of eq. (12.1) gives eq. (12.7); normalization fixes the remaining scalar. This proves tangency of the full density field.

For regular Wishart data with possibly different precisions the same calculation gives $\dot{\mathsf{P}}_i = -2\mathsf{P}_i + I_2 + C_q$. Consequently

$$\mathsf{P}_i(t) - \mathsf{P}_j(t) = e^{-2t}(\mathsf{P}_i(0) - \mathsf{P}_j(0)). \tag{12.10}$$

Initially common precision remains common. On that family put $A = \sum_i \alpha_i$ and $\bar{\alpha} = A/N$. Then $M_q = A\mathsf{P}^{-1}$ and $\bar{\alpha}(t) = \alpha_0 + e^{-2t}(\bar{\alpha}(0) - \alpha_0)$. The precision field commutes with P, so its initial eigenvectors remain fixed and each positive eigenvalue solves

$$\dot{p} = f(p, \bar{\alpha}(t)) = -2p + 2 + \frac{2\kappa\bar{\alpha}(t)}{p} - \frac{\kappa\alpha_0 p}{p + \kappa\bar{\alpha}(t)}. \tag{12.11}$$

Let $m_- = \min(\bar{\alpha}(0), \alpha_0)$ and $m_+ = \max(\bar{\alpha}(0), \alpha_0)$. For $\varepsilon = \min(1/2, m_-/\alpha_0)$, $f > 0$ when $0 < p \le \varepsilon$. For $L = 2 + \kappa m_+$, $f \le -2$ when $p \ge L$. Every eigenvalue therefore stays between $\min(p(0), \varepsilon) > 0$ and $\max(p(0), L) < \infty$. Shapes stay between their initial values and their reference shape, all strictly above $1/2$. The smooth finite-dimensional field consequently has a unique global solution on this invariant family.

Slightly enlarged compact parameter neighborhoods dominate polynomial moments and logarithmic determinant derivatives in eq. (12.1). They justify finite KL, square-integrable scores, and the differentiations above. Integrating the log-density equation with factor e^{2t} shows that any positive dominated regular solution with these initial laws has only the same log-determinant and linear matrix statistics; its parameters satisfy the same ODE. Finally, shape differences equal their initial differences times e^{-2t}. □

The interaction is sensitive to other agents. For a symmetric moment variation H,

$$DC_q[H] = \frac{2\kappa}{N} H + \frac{\kappa^2\alpha_0}{N} \Lambda_q^{-1} H \Lambda_q^{-1}. \tag{12.12}$$

Its trace pairing with nonzero H is strictly positive. Varying another agent's shape produces $H = \mathsf{P}^{-1} d\alpha_j$ and changes the centered force on q_i. This coupling comes from the declared population moment; the source rows remain at their priors and do not select preferred neighbors.

12.3 Parent laws and exact joining

Theorem 12.3 (Additive parents with the original Fisher dynamics). For any disjoint current partition, retain

$$n_I = |I|, \qquad A_I = \sum_{i \in I} \alpha_i, \qquad B_I = \sum_{i \in I} \eta_i, \qquad q_I = W_2(A_I, \mathsf{P}), \quad s_I = W_2(B_I, \mathsf{Q}). \tag{12.13}$$

The parent evaluator, reference, and comparison channels are the corresponding sum-pushforwards of independent child outputs. Their shapes are $n_I \alpha_0, n_I \eta_0$, and their source priors are $\pi^c_{IJ} = n_J / N$. With the primitive $N = \sum_I n_I$ kept fixed, their ordinary unit-Fisher dynamics is

$$\dot{A}_I = -2(A_I - n_I \alpha_0), \qquad \dot{B}_I = -2(B_I - n_I \eta_0), \tag{12.14}$$

together with the same precision equations eq. (12.7). Joining disjoint blocks adds (n, A, B) and keeps the common precisions. It commutes with laws, channels, and evolution, and is associative.

Proof. Use the fixed sample map $(X_i, Y_i)_{i \in I} \mapsto (\sum_i X_i, \sum_i Y_i)$. Lemma 12.1 gives eq. (12.13). The same map on the independent reference or channel outputs gives $W_2(n_I \alpha_0, I_2)$ or $W_2(n_I \alpha_0, \Lambda_q)$, and the model exchange. The parent evaluator is constant in its model input, just as the fine evaluator is. Source normalization gives $\pi^c_{IJ} = n_J / N$ by summing primitive sender weights. Its row is again the exact Gibbs prior.

Total moments satisfy $M_q = \sum_I \mathbb{E}_{q_I} X_I$ and $M_s = \sum_I \mathbb{E}_{s_I} Y_I$. The parent original-KL energy in the belief sector is

$$\mathcal{E}^c_q = \sum_I \left[D_{\mathrm{KL}}(q_I \| W_2(n_I \alpha_0, I_2)) + D_{\mathrm{KL}}(q_I \| W_2(n_I \alpha_0, \Lambda_q)) \right]. \tag{12.15}$$

Its Λ_q terms are still $\operatorname{tr}(\Lambda_q M_q) - N \alpha_0 \log\det \Lambda_q$ because $\sum_I n_I = N$. Its complete moment derivative is therefore the same C_q. The density-force calculation in eq. (12.9), with reference shape $n_I \alpha_0$, gives eq. (12.14) and the original precision equation. This is the projected fine vector field at the same time, with no mobility correction.

Adding two parents preserves total moments, counts, and forcing. Shape equations add, and sample addition gives the joined law and normalized channel. Associativity follows from addition of the three retained scalars and of the sample matrices. □

The final single parent still has its inherited nonidentity internal-source channel. Replacing it by an identity self-comparison would delete the second KL term and change its flow. A parent source label is the pushforward of one primitive source-label draw; a batch exposing separate primitive labels would require their joint mark law. Those labels are not exposed by the interface below.

Exact evolution does not assert equality of fine and parent energies. For example, at $\mathsf{P} = I_2$ replace two shapes α_0, α_0 by $\alpha_0 + \epsilon, \alpha_0 - \epsilon$. The total moment and parent state are unchanged, but the

fine energy increases by $2D_{\mathrm{KL}}(W_2(\alpha_0+\epsilon,I_2)\|W_2(\alpha_0,I_2))+2D_{\mathrm{KL}}(W_2(\alpha_0-\epsilon,I_2)\|W_2(\alpha_0,I_2))>0$. The vector-field calculation, rather than energy factorization, proves inheritance.

12.4 Selected membership and persistent distinctions

Fix a threshold $\delta>0$ and begin with singletons. For a current block, read its mean belief-shape deviation

$$b_I=A_I/n_I-\alpha_0,\qquad \dot{b}_I=-2b_I,\qquad b_{I\cup J}=\frac{n_Ib_I+n_Jb_J}{n_I+n_J}. \tag{12.16}$$

Join blocks when their distance $|b_I-b_J|$ first reaches δ from above. Simultaneous threshold edges join their connected components. Recompute distances after each event. This rule reads only retained laws and counts, is invariant under relabeling, and contains no supplied membership tree. The threshold specifies a persistence criterion; it is additional constitutive data.

Theorem 12.4 (Two selected formation stages). Let $a_1<a_2<a_3<a_4$ be the sorted initial belief shapes, with consecutive gaps d_1,d_2,d_3, and suppose

$$\delta<\min(d_1,d_3),\qquad d_2>\max(d_1,d_3). \tag{12.17}$$

All four shapes exceed $1/2$, while model shapes vary independently above $1/2$. Under eq. (12.16), the two lower and the two upper shapes form parents at

$$t_L=\tfrac{1}{2}\log(d_1/\delta),\qquad t_R=\tfrac{1}{2}\log(d_3/\delta). \tag{12.18}$$

Their union forms at

$$t_2=\tfrac{1}{2}\log\left(\frac{d_2+(d_1+d_3)/2}{\delta}\right)>\max(t_L,t_R). \tag{12.19}$$

There are no intervening cross joins or cascades. Both first parents coexist for a positive time interval. Each joined child-block distance remains below threshold afterward, and all fine belief laws remain distinct at every finite time. Each join forgets a continuous model-shape distinction even if all membership labels and event times are retained.

Proof. Every initial belief-shape difference is multiplied by e^{-2t}. Suppose $d_1<d_3$. The lower pair reaches threshold first. Its distance to the closest remaining singleton is then $e^{-2t}(d_2+d_1/2)$, whose coefficient exceeds d_3. The upper pair therefore reaches threshold next, before any cross join. All cross distances are strictly above threshold at both events. The opposite order follows by exchanging upper and lower pairs. If $d_1=d_3$, exactly the two disjoint outer edges reach threshold together, and the component rule forms the same pairs. After these events the distance of the two parent scores is $e^{-2t}[d_2+(d_1+d_3)/2]$, giving eq. (12.19).

Continue any joined child-block score through the original fine dynamics. Its distance from its sibling is δ at their join and decreases as $e^{-2(t-t_{\mathrm{join}})}$ thereafter. Thus persistence follows from the equations, rather than only from storing a membership flag. Primitive shape differences never vanish at finite time. The two first-parent laws also remain distinct, because their mean belief shapes remain different.

For loss at a first join, fix all belief shapes and perturb two initial model shapes in that pair by $(\eta_i, \eta_j) \mapsto (\eta_i + \epsilon, \eta_j - \epsilon)$ for sufficiently small ϵ. The parent model sum, both population moments, all subsequent precision evolution, and every membership event and time are unchanged. Individual model laws change continuously. Distinct fixed belief shapes distinguish the labels, so this change is not a permutation. At the second join, transfer model shape between the two first-parent sums while keeping their union sum fixed. This gives a further continuous lost distinction and still leaves the belief-only selection schedule unchanged. □

A birth time and a parent belief sum can reveal a two-child belief gap. The proof therefore uses independent model-shape allocation for its loss witness. Already observed sample records remain with the controller; they are not erased or asserted to have the same law across different initial preparations. Loss concerns the current parent state and the future interface specified next.

The preparation class is nonempty and relatively open. For example take $\alpha_0 = \eta_0 = \kappa = 1$, $\delta = 1/10$, $(a_1, a_2, a_3, a_4) = (1, 6/5, 9/5, 21/10)$, and model shapes in a neighborhood of $(4/5, 11/10, 7/5, 17/10)$. The gaps are $(1/5, 3/5, 3/10)$, and the events occur at $\frac{1}{2}\log 2$, $\frac{1}{2}\log 3$, and $\frac{1}{2}\log(17/2)$. A sufficiently small neighborhood and every relabeling obey eq. (12.17). Openness is within the common-precision family, not among all positive density tuples. The smooth statistical retention applies on each fixed membership branch; no global smoothness across switching boundaries is required.

12.5 Exact joint response histories

The response interface is noninvasive. At each query, draw fresh samples independently across children and sectors from $\bigotimes_i (q_i \otimes s_i)$. For the current disjoint partition Π, form the entire vector

$$Z_\Pi = ((X_I, Y_I))_{I \in \Pi}, \qquad X_I = \sum_{i \in I} X_i, \quad Y_I = \sum_{i \in I} Y_i. \tag{12.20}$$

An action may choose any normalized readout kernel on this vector. It can report overlapping unions of whole current blocks using the shared batch, so no independence between such overlapping reports is assumed. Query times and readouts may depend measurably on carried record history and retained state. Recording an output does not reset a belief or model law.

Theorem 12.5 (Adaptive extensive-response inheritance). For every admitted preparation and every finite sequence of finite query times selected by the preceding protocol, the fine and parent descriptions have the same joint law of records, actions, query times, and retained states. This includes the membership events of theorem 12.4.

Proof. Independence across disjoint blocks and lemma 12.1 give the conditional batch kernel

$$K_\Pi = \bigotimes_{I \in \Pi} [W_2(A_I, \mathsf{P}) \otimes W_2(B_I, \mathsf{Q})]. \tag{12.21}$$

Initialize the retained process from the same fine configuration. Condition on the retained state and carried history at a query. The same policy chooses the next waiting time and readout in both descriptions. Theorem 12.3 gives equal retained states after waiting, and eq. (12.16) gives the same intervening events and addition handoffs. Freshness makes the next conditional batch law

exactly eq. (12.21); composition with the same normalized readout gives the same next-record kernel. Induction proves equality of every finite joint history. Independent external randomization can be conditioned on in the same proof. □

The theorem compares two descriptions of each given preparation. A final aggregate is not claimed to reconstruct previously discarded records or the distribution of earlier records from a different preparation. Queries cutting through a current block need additional intersection data. For example, same-batch sums over primitive subsets C, D have

$$\operatorname{Cov}(\operatorname{tr}(UX_C), \operatorname{tr}(VX_D)) = A_{C\cap D} \operatorname{tr}(\mathsf{P}^{-1}U\mathsf{P}^{-1}V), \tag{12.22}$$

which can expose a discarded shape sum. Repeated hidden-child sampling, child-specific Bayesian updates, and interventions changing the reference potential are outside the declared interface.

12.6 Inherited collective geometry

Use the trivial principal bundle $\mathcal{C}\times SO(2)$ and the common left sample action $X \mapsto gXg^T, Y \mapsto gYg^T$. It sends precisions to $g\mathsf{P}g^T, g\mathsf{Q}g^T$. The identity-precision references are invariant under this action. Use the collective product-law Fisher metric with unit weights, the common comparison frame, and the convention $D^A p = dp + J_p A$ of chapter 3. Let $\mathcal{C} = (a_-, a_+) \times (-\chi_0, \chi_0)$, with $0 < a_- < a_+$, and initialize the contextual section by

$$\mathsf{P}(0; a, \chi) = R_\chi \operatorname{diag}(e^a, e^{-a}) R_\chi^T, \qquad \mathsf{Q}(0) = \operatorname{diag}(e^b, e^{-b}), \qquad b > 0. \tag{12.23}$$

Here R_χ is plane rotation by χ. All individual shape parameters are constant over $\mathcal{C}$. Consequently the selection schedule is also constant over this contextual chart.

Theorem 12.6 (Exact connection and contextual-tensor inheritance). For every block, before and after either join, sum retention preserves its collective minimizing connection class and minimized contextual tensor. For eq. (12.23), that tensor has rank two and the connection has nonzero curvature at every finite time.

Proof. Differentiating eq. (12.1) at fixed shape gives the precision score and Fisher metric

$$\ell_U(X) = \alpha \operatorname{tr}(\mathsf{P}^{-1}U) - \operatorname{tr}(UX), \qquad g^F(U, V) = \alpha \operatorname{tr}(\mathsf{P}^{-1}U\mathsf{P}^{-1}V). \tag{12.24}$$

The second identity follows by differentiating the normalizing integral twice. For common precision variations (U, V), the summed fine score is

$$A_I \operatorname{tr}(\mathsf{P}^{-1}U) - \operatorname{tr}(UX_I) + B_I \operatorname{tr}(\mathsf{Q}^{-1}V) - \operatorname{tr}(VY_I). \tag{12.25}$$

It equals the parent score and is measurable from the retained sums. Both contextual and common rotation-orbit directions have this form. Their Fisher pairings and all mixed pairings are therefore preserved. The infinitesimal stabilizers agree as well: in each description they are the rotations commuting with both precisions. Thus the normal equations for the minimizing connection have the same solutions modulo the same isotropy, and the residual tensors agree. This also verifies the

mixed-loss criterion in proposition 3.14. The sample addition channel is fixed, so it contributes no moving-channel derivative.

For explicit curvature, let $p_\pm(t,a)$ be the two eigenvalues of $\mathsf{P}(t;a,\chi)$, and let $q_\pm(t)$ be those of $\mathsf{Q}(t)$. The scalar field in eq. (12.11) has

$$f_p = -2 - \frac{2\kappa\bar{\alpha}}{p^2} - \frac{\kappa^2\alpha_0\bar{\alpha}}{(p+\kappa\bar{\alpha})^2} < 0. \tag{12.26}$$

Its solution map $\Phi_t(p_0)$ satisfies $\partial_{p_0}\Phi_t = \exp(\int_0^t f_p du) > 0$ for finite t. Hence $p_+ = \Phi_t(e^a) > p_- = \Phi_t(e^{-a}) > 0$, $\partial_a p_+ > 0$, and $\partial_a p_- < 0$. The model eigenvalues also remain distinct and have no a dependence. Define

$$k_1 = 2A_I\frac{(p_+ - p_-)^2}{p_+p_-}, \qquad k_2 = 2B_I\frac{(q_+ - q_-)^2}{q_+q_-} > 0. \tag{12.27}$$

Writing $r = p_+/p_- > 1$ gives $k_1 = 2A_I(r + r^{-1} - 2)$ and $\partial_a k_1 > 0$. Radial precision variations are diagonal in an eigenframe, while orbit variations are off-diagonal, so their Fisher pairing vanishes. The collective normal equation then gives

$$A^F = -\frac{k_1}{k_1+k_2}d\chi, \qquad h^{\min} = g_{aa}da^2 + \frac{k_1k_2}{k_1+k_2}d\chi^2, \tag{12.28}$$
$$g_{aa} = A_I\left[\left(\frac{\partial_a p_+}{p_+}\right)^2 + \left(\frac{\partial_a p_-}{p_-}\right)^2\right] > 0.$$

Identifying the one-dimensional Lie algebra with its rotation generator, the curvature is

$$F^F = dA^F = -\frac{(\partial_a k_1)k_2}{(k_1+k_2)^2}da \wedge d\chi \neq 0. \tag{12.29}$$

These computations hold for every retained block and its fine constituents with the same totals. Under a passive local rotation by $\vartheta(c)$, the angular momentum one-form gains $(k_1+k_2)d\vartheta$, so $A^{F\prime} = A^F - d\vartheta$. The tensor is invariant, with exactly the affine connection law of the general construction. □

The context-constant allocation assumption is substantive. If $d\eta_1 = 1, d\eta_2 = -1$ while precisions and all other shapes are fixed, the parent model sum has zero derivative, whereas the fine score is the centered difference of two log determinants with positive variance. It is orthogonal to rotation scores because determinant is conjugation invariant. Its fine minimized length is positive and its retained length is zero. Such a contextual direction violates exact tensor inheritance.

Chapter 13

Formation Beyond the Prepared Examples

The formation obligation in definition 6.17 combines selected persistent membership with inherited evolution and responses. Chapter 12 realizes it conditionally under the original smooth KL potential. Four matrix-law agents form two parents and then their union through one retained-state rule. The full channel derivative, additive parent laws, original-time Fisher evolution, admitted adaptive response histories, and collective contextual geometry are proved together.

Its assumptions identify what remains to be derived. Common precision is prepared, and unequal precisions do not first agree at finite time in this model. The normalized channels read a global moment; they do not create preferential source groups. The threshold is fixed constitutive data, and every additive quotient exists before its selection event. The construction therefore establishes state-selected use of sufficient parent descriptions, not the first appearance of new operational redundancy.

The response class is also substantive. Fresh extensive queries and readouts of their full joint vector descend. Child-specific probes, reused hidden children, or Bayesian interventions can require distinctions that the parent discards. A broader formation mechanism must specify its actual interface and show that the resulting conditional kernels and updates close, rather than infer this from a static sum law.

The geometry in theorem 12.6 is inherited because all contextual and orbit scores are measurable from the sums, and the infinitesimal stabilizers agree. Discarded shape allocations are constant over context. Allowing those allocations to vary can give positive fine contextual length with zero parent length. Formation with inherited geometry from less restricted contextual families requires satisfying the corresponding score and connection conditions, not merely Fisher contraction.

The preceding constructions retain their separate scopes. The profile hierarchy in chapter 11 uses identity channels and a supplied tree, with a different response protocol and no Fisher tangent for its discarded phases. The phase reduction in chapter 10 loses ordinary child sampling. The erasure example in chapter 9 keeps every public child state, and chapter 7 changes the interaction to a nonsmooth one. Their conclusions do not transfer to one another without checking the specified laws, channels, and retentions.

The outstanding problem is to obtain persistent sufficient organization from less structured preparations, with a justified membership criterion and an appropriate interaction architecture. Full permutation symmetry still constrains law-only deterministic selection as in proposition 6.16. None of the finite constructions establishes an indefinite spacetime metric, a thermodynamic limit, or unbounded recursive formation.

Appendix A

A Normalized Internal Record Process

This appendix constructs a finite-history law from an internal interaction rule. The rule is supplied; normalization does not select it. Let S and O be standard Borel spaces. A state x includes an accessible record register. Write $\rho : \mathsf{S} \to \mathsf{O}$ for its readout.

Definition A.1 (Event and registration). Choose a normalized kernel $R : \mathsf{S} \rightsquigarrow \mathsf{O}$ and a measurable update $A : \mathsf{S} \times \mathsf{O} \to \mathsf{S}$ satisfying

$$\rho(A(x, o)) = o, \tag{A.1}$$

for $R(x, \cdot)$-almost every o, for every admitted x. Define

$$J(x, dx', do) = R(x, do)\delta_{A(x,o)}(dx'). \tag{A.2}$$

If necessary, the state can be explicitly augmented by a register so that eq. (A.1) holds.

Proposition A.2 (Consistent finite histories). For an initial law μ_0 on S, the iterated kernel product

$$\mathbb{P}_n(dx_0, do_1, dx_1, \ldots, do_n, dx_n) = \mu_0(dx_0) \prod_{t=1}^{n} J(x_{t-1}, dx_t, do_t), \tag{A.3}$$

is a probability law. Marginalizing its last pair gives $\mathbb{P}_{n-1}$, and every record agrees with the readout of the resulting state almost surely.

Proof. The product denotes successive kernel integration, not independent factors. Integrating the last pair gives one because $J(x, \mathsf{S} \times \mathsf{O}) = 1$. Repeating this argument gives total mass one and the stated consistency. Each transition is supported on $x' = A(x, o)$ and hence on $\rho(x') = o$. For a finite history, the intersection of these probability-one events still has probability one. □

Nothing in this construction requires a variable external to the agents. It does require a declared accessible register and an update. A stochastic A is handled by replacing its Dirac kernel with a normalized kernel supported on the registration condition. A deterministic transition is recovered when R is a Dirac kernel as well.

A.1 Prediction followed by assimilation

One possible internal state contains a model law s and a record register. Fix a normalized density $k(o \mid m)$ with respect to a reference measure on O. Let

$$R(s, do) = \left[\int k(o \mid m)s(dm) \right] do, \qquad s^o(dm) = \frac{k(o \mid m)s(dm)}{\int k(o \mid m')s(dm')}. \tag{A.4}$$

The update replaces s by s^o and stores o. The denominator is positive and finite for predictive-almost every record. Defining an arbitrary probability law on the remaining null set completes the measurable rule under the usual conditional-law hypotheses.

Proposition A.3 (The Bayesian history identity). For a fixed latent model m, initial law s_0, and fixed normalized observation density k, sequential prediction and Bayesian assimilation give the record density

$$p(o_1, \dots, o_n) = \int \prod_{t=1}^{n} k(o_t \mid m) s_0(dm). \tag{A.5}$$

Proof. After $t-1$ records, the posterior is proportional to $s_0(dm) \prod_{r<t} k(o_r \mid m)$. The next predictive density is the integral with the product extended to t, divided by the integral up to $t-1$. Multiplying these conditional densities cancels every intermediate normalizer and leaves eq. (A.5). □

This identity applies to the stated fixed observation model. If the likelihood itself is changed using the current state, the product of the actual predictive kernels still defines a normalized process, but it need not equal the fixed-latent expression. Likewise, replacing Bayes' rule by a finite natural-gradient step defines a different assimilation mechanism unless the updates coincide. The normalization of a process and its interpretation as Bayesian conditioning are separate statements.

Appendix B

From Directional Interaction to Base Transport

Links between agents at a single context do not determine transport between neighboring contexts. A connection requires a smooth dependence on the direction and size of a base displacement. This appendix gives a matrix realization of the connection used in chapter 3. It can occur within one agent; collective least-squares selection is a separate construction.

Fix one sector and one agent, suppressing the sector superscript. On a coordinate neighborhood of $\mathcal{C}$, let $U_i(c)$ be a twice continuously differentiable invertible presentation and let $a_i = a_{i,\mu}(c)dc^\mu$ be a continuously differentiable matrix-valued one-form. For a principal G-connection its values must lie in the represented Lie algebra. General endomorphism-valued forms instead describe connections for the extended general linear frame bundle.

Definition B.1 (Infinitesimal directional transport). The proposed transport family has first-order behavior

$$\Theta_i(c+\delta c, c) = U_i(c+\delta c)\exp[-a_{i,\mu}(c)\delta c^\mu]U_i(c)^{-1} + O(|\delta c|^2). \tag{B.1}$$

The first-order generator is linear in the tangent displacement. A single finite matrix at c supplies neither this linear dependence nor the identity value at zero displacement.

Proposition B.2 (Connection and curvature). With the convention $D_i = d + B_i$, eq. (B.1) has connection and curvature

$$B_i = U_i a_i U_i^{-1} - dU_i U_i^{-1}, \qquad F_i = U_i(da_i + a_i \wedge a_i)U_i^{-1}. \tag{B.2}$$

Proof. Expanding the two factors at c gives $\Theta_i = I + (dU_iU_i^{-1} - U_ia_iU_i^{-1})(\delta c) + O(|\delta c|^2)$, which identifies B_i. Acting on a vector-valued function gives the operator identity $d + B_i = U_i(d + a_i)U_i^{-1}$. Squaring it yields $(d + B_i)^2 = U_i(d + a_i)^2U_i^{-1}$, and the squares act by the curvature two-forms in eq. (B.2). □

For an explicit curved example, take $U_i = I$ over an open subset of $\mathbb{R}^2$ and set

$$a_{i,x} = E_{12}, \qquad a_{i,y} = E_{21}. \tag{B.3}$$

Both are rank-one receiving–sending products, while their derivatives vanish. Nevertheless,

$$F_{i,xy} = [E_{12}, E_{21}] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0. \tag{B.4}$$

Thus even constant directional interfaces can give curvature through noncommutativity. By contrast, replacing a finite pointwise map by a new frame $V(c)$ and setting $B = -dVV^{-1}$ gives zero curvature: it is the case $a = 0$ of eq. (B.2).

Finite paths use the solution of

$$\frac{d}{dt}T_i(t) = -B_i(\dot{\gamma}(t))T_i(t), \qquad T_i(0) = I. \tag{B.5}$$

Uniqueness of this linear equation gives composition under concatenation and inversion under path reversal. The solution is the path-ordered exponential; the usual link and Wilson-line constructions use this ordering [23]. Independently declared opposite attending edges are not two orientations of this same path and need not be inverses.

Proposition B.3 (Compatibility under re-expression). Under an agent-coordinate change $U_i' = g_i U_i$ with the same reference form,

$$B_i' = g_i B_i g_i^{-1} - dg_i g_i^{-1}. \tag{B.6}$$

Under a change of reference presentation $U_i' = U_i h$, keeping the same intrinsic transport instead requires

$$a_i' = h^{-1} a_i h + h^{-1} dh. \tag{B.7}$$

Proof. Insert $g_i U_i$ into eq. (B.2) and apply the product rule. Inserting $U_i h$ and eq. (B.7) cancels the two terms containing dh, leaving the original B_i. □

The same affine law must hold between overlapping coordinate patches. A directional interface product transforming only by conjugation is a tensorial one-form. It can describe the difference between two connections, but it is not by itself the full affine connection data. Constructing displacement links with the endpoint transformation law and then differentiating them is one way to obtain the required inhomogeneous term.

If patch coordinates satisfy $v_\alpha = g_{\alpha\beta} v_\beta$, their transition functions must obey $g_{\alpha\beta} g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps. The displayed affine law then makes the local connections compatible. A nontrivial bundle need not admit a single global presentation U_i. Even on a trivial bundle, zero curvature does not imply path independence: the scalar connection $B = b\,d\theta$ on the circle is flat but has holonomy $e^{-2\pi b}$, which differs from one for real $b \neq 0$. Global path independence requires trivial holonomy as well as local flatness.

Different agents may carry different connections on the same bundle. They represent one common connection only when their local forms satisfy its overlap relations. Neither case requires an averaging construction.

Open problem B.4 (Deriving directional transport from agent interactions). Specify a constitutive family of displacement links from the agents' laws and interfaces that has identity at zero displacement, a smooth first derivative linear in displacement, and compatible endpoint laws on overlaps. Its derivative then determines the connection by proposition B.2. The unresolved step is the selection and construction of that family; pointwise covariance and the possibility of curvature alone do not determine it.

Appendix C

A Conditional Determinant Geometry

The collective construction of proposition 3.8 does not require a Lorentzian target. This appendix gives one additional choice for a complex rank-two representation. Its indefinite form is supplied by a determinant; the Fisher metric used to choose the connection remains positive.

Let $W \to \mathcal{U}$ be a complex rank-two vector bundle with structure group $\mathrm{GL}(2,\mathbb{C})$. In a frame, identify the real rank-four bundle $\mathrm{Herm}(W)$ with Hermitian matrices transforming by $H' = gHg^\dagger$. Suppose the collective target laws p_i on W have finite second moments and that their moment maps are smooth on the admitted regular families. With the same declared positive weights as in the collective metric, set

$$M = \mathcal{M}(\boldsymbol{p}) = \sum_i w_i \int zz^\dagger p_i(dz), \qquad e(X) = d\mathcal{M}_{\boldsymbol{p}} r_X. \tag{C.1}$$

Equivariance of the moment map gives

$$e = dM + A^F M + M(A^F)^\dagger, \qquad e' = geg^\dagger. \tag{C.2}$$

The first expression is independent of the stabilizer-valued representative of A^F by proposition 3.9.

Proposition C.1 (Determinant metric). Assume $\mathcal{U}$ is four dimensional and the actual one-form $e : T\mathcal{U} \to \mathrm{Herm}(W)$ is a bundle isomorphism. Supply a smooth positive determinant-scale section $\kappa_{\det}$ with local transformation $\kappa'_{\det} = |\det g|^2 \kappa_{\det}$. Then

$$\eta_{\det}(H,K) = \tfrac{1}{2}[\mathrm{tr}(HK) - \mathrm{tr}(H)\,\mathrm{tr}(K)], \qquad \gamma(X,Y) = \kappa_{\det}^{-1}\eta_{\det}(eX,eY). \tag{C.3}$$

defines a gauge-invariant metric with one negative and three positive directions. The positive-definite Hermitian cone determines a time orientation through e.

Proof. For a Hermitian matrix $H = \begin{pmatrix} t+z & x-iy \\ x+iy & t-z \end{pmatrix}$, direct evaluation gives

$$\eta_{\det}(H,H) = -\det H = -t^2 + x^2 + y^2 + z^2. \tag{C.4}$$

Multiplicativity of the determinant under congruence gives $-\det(gHg^\dagger) = |\det g|^2(-\det H)$. Polarizing proves the same scaling law for $\eta_{\det}(H,K)$. The compensator cancels that factor in eq. (C.3), and the isomorphism e preserves its inertia. Positive Hermitian matrices have $t > 0$ and $t^2 > x^2 + y^2 + z^2$, so they form one timelike cone, preserved by congruence. □

The rank hypothesis is imposed on the actual contextual first jet. Surjectivity of the horizontal differential of $\mathcal{M}$ establishes the availability of such a jet at a law tuple, but does not select a section or prove persistence under inference. If the base has dimension greater than four and

e is surjective, the same formula gives a nondegenerate form on the quotient vector bundle by $\ker e$. A quotient-manifold interpretation still requires a regular leaf quotient and basicness, as in proposition 3.6.

Proposition C.2 (Circle completion). Let $\mathscr{Q}_{\det} \to \mathcal{U}$ be the unit circle bundle of $\det W$ for the line metric whose local coefficient is $\kappa_{\det}^{-1}$. Write $\pi_{\det}$ for its projection. Supply a real principal circle connection $\omega_{\det}$ normalized to one on the vertical generator of period 2π, and a smooth positive radius $r_{\det} : \mathcal{U} \to (0, \infty)$. Then

$$G_{\mathrm{KK}} = \pi_{\det}^{*}\gamma + (r_{\det} \circ \pi_{\det})^{2}\omega_{\det} \otimes \omega_{\det} \tag{C.5}$$

is circle invariant and has one negative and four positive directions.

Proof. The connection splits each total tangent space into its horizontal subspace and the one-dimensional vertical subspace. On the first, eq. (C.5) is the pullback of γ; on the second, it is positive because $r_{\det} > 0$. The two subspaces are orthogonal. The base pullback, radius, and principal connection are invariant under the circle action, proving invariance of the total metric. □

A supplied global $\mathrm{GL}(2, \mathbb{C})$ connection A gives the local expression $\omega_{\det} = d\theta_{\det} + \operatorname{Im} \operatorname{tr} A$ on this circle bundle. The coordinate transformation $\theta'_{\det} = \theta_{\det} + \arg \det g$ cancels $\operatorname{Im} \operatorname{tr} A' = \operatorname{Im} \operatorname{tr} A - d \arg \det g$. When the collective minimization determines only a class, its lift to a full connection is further data. For instance, central phase rotations act trivially on centered proper complex Gaussian laws and their covariance matrices, so that family cannot fix the corresponding phase connection from its Fisher normal equations. Neither the positive circle radius nor equations of motion for γ follow from the construction. The determinant and circle choices give this conditional geometric example without identifying it with physical spacetime dynamics.

Appendix D

Notation

Symbol	Meaning
$\mathcal{C}, c_*$	Context base and a selected context.
P, G	Principal bundle and its structure group.
K, M	Represented state contents and model presentations.
$\mathcal{P}(X)$	Probability measures on a measurable space X.
ζ_i, q_i, s_i	Joint agent law and its state and model marginals.
g^F, ℓ_v	Fisher metric of an admitted regular family and the score in tangent direction v.
$D^\omega\zeta, h^\omega_\zeta$	Covariant statistical derivative and its Fisher pullback on the context base.
$J_{\boldsymbol{p}}, H_{\boldsymbol{p}}$	Infinitesimal collective group action and its Fisher-orthogonal complement.
$\mathfrak{k}, A^F$	Collective orbit metric and a representative of the minimizing connection class.
U^q_i, U^s_i	Frame representatives in the two sectors.
$\bar{q}_i, \bar{s}_i$	Laws in common reference coordinates on a trivializing patch.
L^q_{ij}, L^s_{ij}	Law-dependent interaction generators in the vector realization.
$\Theta^q_{ij}, \Theta^s_{ij}$	Invertible source-to-receiver interaction maps.
$\mathcal{K}^q_{ij}, \mathcal{K}^s_{ij}$	Normalized source-to-receiver comparison channels.
T^a_j, O^a_i	Sending and receiving maps in sector $a \in \{q, s\}$.
β_i, γ_i	Belief-sector and model-sector source rows.
G_i, H_i	Local model-to-state kernel and state-to-record kernel.
K^{rec}_i	Normalized model-to-record predictive evaluator.
S, x	Complete agent-configuration space and a configuration.
O_i, ρ_i	Record alphabet and internal record readout.
$O_{ij,t}, o_{ij,t}$	Record random variable and one realized value.
Γ_{ij}	Interaction transition on configurations.
$\mathcal{E}$	Declared law-level interaction potential.
$W_2(\nu, \mathsf{P})$	Matrix Wishart law with shape $\nu > 1/2$ and precision P in the conditional formation example.
n_I, A_I, B_I	Primitive count and belief/model shape sums of a retained block in that example.
Λ_q, C_q	Moment-dependent comparison precision and its complete force coefficient; the model sector uses Λ_s, C_s.
$\mathcal{F}$	Variational free energy for a specified normalized probabilistic model.
F, R, F^c	Fine deterministic update, retention map, and inherited update.
a_i, B_i, F_i	Reference connection form, its local representative, and curvature.
X, U, Z, Y	Source, encoded representation, decoder information, and prediction target in the coding construction.

The symbol F for a discrete update and F_i for curvature are used in separate constructions. A frame U^a_i is an invertible map; the coding variable U is a random representation. A Markov kernel returns a probability measure, whereas an ordinary map returns one value. Kernel composition is written in the order in which a law traverses the kernels.

Bibliography

[1] Shun-ichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10(2):251–276, 1998. doi: 10.1162/089976698300017746.

[2] Nihat Ay, Jürgen Jost, Hông Vân Lê, and Lorenz Schwachhöfer. Information geometry and sufficient statistics. *Probability Theory and Related Fields*, 162:327–364, 2015. doi: 10.1007/s00440-014-0574-8.

[3] Nihat Ay, Jürgen Jost, Hông Vân Lê, and Lorenz Schwachhöfer. Parametrized measure models. *Bernoulli*, 24(3):1692–1725, 2018. doi: 10.3150/16-BEJ910. URL https://arxiv.org/abs/1510.07305.

[4] Lionel Barnett and Anil K. Seth. Dynamical independence: Discovering emergent macroscopic processes in complex dynamical systems. *Physical Review E*, 108:014304, 2023. doi: 10.1103/PhysRevE.108.014304.

[5] Chung Chan, Ali Al-Bashabsheh, Qiaoqiao Zhou, Tarik Kaced, and Tie Liu. Info-clustering: A mathematical theory for data clustering, 2016. URL https://arxiv.org/abs/1605.01233.

[6] Jorge Cortés. Finite-time convergent gradient flows with applications to network consensus. *Automatica*, 42(11):1993–2000, 2006. doi: 10.1016/j.automatica.2006.06.015.

[7] Jaap Eldering, Matthew Kvalheim, and Shai Revzen. Global linearization and fiber bundle structure of invariant manifolds. *Nonlinearity*, 31(9):4202–4245, 2018. doi: 10.1088/1361-6544/aaca8d. URL https://arxiv.org/abs/1711.03646.

[8] Robert M. Gray. *Entropy and Information Theory*. Springer-Verlag, New York, 1990. URL https://www-ee.stanford.edu/~gray/it.pdf. Corrected first edition, revised June 26, 2023.

[9] A. Guichardet. On rotation and vibration motions of molecules. *Annales de l'I.H.P. Physique théorique*, 40(3):329–342, 1984. URL https://www.numdam.org/item/AIHPA_1984__40_3_329_0/.

[10] Robert Hermann and Arthur J. Krener. Nonlinear controllability and observability. *IEEE Transactions on Automatic Control*, 22(5):728–740, 1977. doi: 10.1109/TAC.1977.1101601.

[11] Toby Dylan Hocking, Armand Joulin, Francis Bach, and Jean-Philippe Vert. Clusterpath: An algorithm for clustering using convex fusion penalties. In *Proceedings of the 28th International Conference on Machine Learning*, pages 745–752, 2011. URL https://icml.cc/2011/papers/419_icmlpaper.pdf.

[12] Leonhard Horstmeyer and Fatihcan M. Atay. Characterization of exact lumpability for vector fields on smooth manifolds. *Differential Geometry and its Applications*, 48:46–60, 2016. doi: 10.1016/j.difgeo.2016.06.001.

[13] Olav Kallenberg. *Foundations of Modern Probability*. Springer, Cham, 3 edition, 2021. doi: 10.1007/978-3-030-61871-1.

[14] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Springer, New York, 1976. ISBN 9780387901923. Reprint of the 1960 edition, with a new appendix.

[15] Solomon Kullback and Richard A. Leibler. On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79–86, 1951. doi: 10.1214/aoms/1177729694.

[16] Gérard Letac and Hélène Massam. Existence and non-existence of the non-central Wishart distributions, 2017. URL https://arxiv.org/abs/1108.2849v2. Version 2.

[17] Debra Lewis, Nilima Nigam, and Peter Olver. Connections for general group actions, 2003. URL https://arxiv.org/abs/math/0309234.

[18] David Pollard. A note on insufficiency and the preservation of Fisher information. In *From Probability to Statistics and Back: High-Dimensional Models and Processes*, volume 9 of *IMS Collections*, pages 266–275. Institute of Mathematical Statistics, 2013. doi: 10.1214/12-IMSCOLL919. URL https://arxiv.org/abs/1107.3797.

[19] Daniil Ryabko. Independence clustering (without a matrix), 2017. URL https://arxiv.org/abs/1703.06700.

[20] Cosma Rohilla Shalizi and James P. Crutchfield. Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104:817–879, 2001. URL https://arxiv.org/abs/cond-mat/9907176.

[21] Claude E. Shannon. Coding theorems for a discrete source with a fidelity criterion. *IRE International Convention Record*, 7(4):142–163, 1959. URL https://gwern.net/doc/cs/algorithm/information/1959-shannon.pdf.

[22] Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method, 1999. URL https://arxiv.org/abs/physics/0004057. Manuscript dated September 30, 1999; arXiv:physics/0004057, posted 2000.

[23] David Tong. Gauge theory. University of Cambridge lecture notes, 2018. URL https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf.

[24] Kenneth G. Wilson. Confinement of quarks. *Physical Review D*, 10(8):2445–2459, 1974. doi: 10.1103/PhysRevD.10.2445.

[25] Aaron D. Wyner and Jacob Ziv. The rate-distortion function for source coding with side information at the decoder. *IEEE Transactions on Information Theory*, 22(1):1–10, 1976. doi: 10.1109/TIT.1976.1055508.